

Exercises for Natural Log Functions

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Chapter 1

Linear Functions

1.1 Linear relation between variables

Definition 1.1.1: Linear Relation

Two real variables x and y are said to satisfy a **linear relation** if their relationship can be expressed in the general first-degree polynomial equation form:

$$Ax + By + C = 0$$

where $A, B, C \in \mathbb{R}$ and A, B are not both zero.

When $B \neq 0$, y can be expressed as the subject of the relation:

$$y = mx + c$$

where $m = -\frac{A}{B}$ represents the **slope** (or gradient) of the straight line, and $c = -\frac{C}{B}$ represents the **vertical intercept** (y -intercept).

Fundamental Properties of Linear Functions:

1. **Constant Rate of Change:** For any two distinct points (x_1, y_1) and (x_2, y_2) satisfying the relation with $x_1 \neq x_2$:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \text{constant}$$

2. **Intercepts:**

- Vertical intercept (y -intercept): Point $(0, c)$ obtained by setting $x = 0$.
- Horizontal intercept (x -intercept): Point $(-\frac{c}{m}, 0)$ obtained by setting $y = 0$ (for $m \neq 0$).

3. **Domain and Range:** For $m \neq 0$, the domain is $\mathbb{R}$ and the range is $\mathbb{R}$.

Remark

In the HKDSE syllabus, if x and y are of linear relation, and y is the subject of the relation between x and y , we say that y is a linear function of x .

1.1.1 Level 1 Exercise

1. Find the slope, x -intercept, and y -intercept of each of the following straight lines:
 - (a) $2x - 3y + 12 = 0$
 - (b) $4x + 5y - 20 = 0$
 - (c) $y = -3x + 7$
2. Find the equation of the straight line passing through each given pair of points in the form $y = mx + c$:
 - (a) $(1, 4)$ and $(3, 10)$
 - (b) $(-2, 5)$ and $(4, -1)$
 - (c) $(0, -3)$ and $(5, 2)$
3. Given that y is a linear function of x such that the graph passes through $(2, 7)$ and has a slope of -3 , express y in terms of x .

4. Determine whether each of the following tables of values represents a linear function $y = f(x)$ by examining the rate of change $\frac{\Delta y}{\Delta x}$:

(a)

x	1	2	4	7
y	3	5	9	15

(b)

x	0	1	3	6
y	1	2	5	10

5. A straight line passes through the points $P(a, 3)$ and $Q(5, 11)$ with a slope of 2. Find the value of a .
6. Find the equation of the straight line L in the form $Ax + By + C = 0$ that:
- (a) passes through $(-1, 4)$ and is parallel to $y = 3x - 5$.
 - (b) passes through $(2, -3)$ and is perpendicular to $2x - y + 4 = 0$.
7. Determine the point of intersection of the two linear equations $y = 2x + 3$ and $3x - 2y + 1 = 0$.
8. The line L_1 has x -intercept 4 and y -intercept -2 . Write down the equation of L_1 in slope-intercept form.
9. For the linear function $f(x) = 5 - 2x$:
- (a) Evaluate $f(-3)$ and $f(4)$.
 - (b) Let $y = f(x)$. Express x as a linear function of y .

1.1.2 Level 2 Exercise

1. Let y be a linear function of x^2 . It is given that the linear function passes through $(1, 4)$ and $(3, 10)$. Express y explicitly in terms of x .
2. Given that $\frac{1}{y}$ is a linear function of x . The slope and the vertical intercept of the linear function are 3 and 2 respectively. Express y in terms of x as a single fraction.
3. The straight line $L : 3x + ky - 12 = 0$ passes through the point $(2, 3)$.
 - (a) Find the value of k .
 - (b) Calculate the area of the triangle bounded by L , the x -axis, and the y -axis.
4. The three points $A(-1, 2)$, $B(2, k)$, and $C(5, 14)$ are collinear. Find the value of k .
5. A straight line L has non-zero x -intercept a and y -intercept b .
 - (a) Show that the equation of L can be written in two-intercept form: $\frac{x}{a} + \frac{y}{b} = 1$.
 - (b) If L passes through $(2, 3)$ and $a = 2b$, find the values of a and b .
6. The graph of y^2 against $\sqrt{x}$ is a straight line with slope 5 and y^2 -intercept -10 . Express y^2 in terms of x , and determine the domain of x for which y is real.
7. Two straight lines $L_1 : y = m_1x + c_1$ and $L_2 : y = m_2x + c_2$ intersect at $P(3, 7)$. If $L_1 \perp L_2$ and L_1 passes through the origin $(0, 0)$:
 - (a) Find the values of m_1, c_1, m_2 , and c_2 .
 - (b) Find the x -intercept of L_2 .

8. Suppose that W is a linear function of t . It is given that $W = 45$ when $t = 2$ and $W = 75$ when $t = 8$.
 - (a) Express W in terms of t .
 - (b) Find the value of t when $W = 100$.
9. The line $L_1 : kx + (k + 2)y - 8 = 0$ is perpendicular to $L_2 : 3x - 2y + 1 = 0$. Find the value of k .
10. Let $f(x)$ be a linear function such that $f(f(x)) = 9x + 8$. Find the two possible algebraic expressions for $f(x)$.

1.1.3 Exam-type Problem

Question 1 (8 marks) A straight line L passes through the points $A(2, 5)$ and $B(6, 13)$.

- (a) Find the equation of L in the form $y = mx + c$. (2 marks)
- (b) L cuts the x -axis at point P and the y -axis at point Q . Find the coordinates of P and Q . (3 marks)
- (c) Calculate the area of $\triangle OPQ$, where O is the origin. (3 marks)

Question 2 (9 marks) The variables x and y satisfy a transformed linear relationship. The graph of $(y - 2x)$ against $(x + 1)$ is a straight line with slope 3 and vertical intercept -4 .

- (a) Express $(y - 2x)$ as a linear function of $(x + 1)$. (2 marks)
- (b) Hence, express y explicitly in terms of x . (3 marks)
- (c) Find the x -intercept and y -intercept of the relation between y and x . (4 marks)

Question 3 (10 marks) The temperature measured in degrees Fahrenheit (F) and degrees Celsius (C) satisfy a linear relationship. Water freezes at 0°C (32°F) and boils at 100°C (212°F).

- (a) Express F as a linear function of C . (3 marks)
- (b) Find the temperature at which the value in degrees Fahrenheit equals the value in degrees Celsius. (3 marks)
- (c) Find the inverse function $C(F)$ expressing Celsius in terms of Fahrenheit. (4 marks)

Question 4 (9 marks) Consider a family of straight lines given by $L_k : (2k + 1)x + (k - 1)y - (5k + 1) = 0$, where k is a real parameter.

- (a) Show that all lines L_k pass through a fixed point P for any value of k . Find the coordinates of P . (5 marks)
- (b) Find the value of k such that L_k is parallel to the line $4x + y - 3 = 0$. (4 marks)

Question 5 (10 marks) The variables x and y satisfy a reciprocal linear relation. The graph of $\frac{1}{y}$ against $\frac{1}{x}$ is a straight line passing through $(1, 3)$ and $(4, 9)$.

- (a) Express $\frac{1}{y}$ as a linear function of $\frac{1}{x}$. (3 marks)

- (b) Express y in terms of x as a single simplified rational function $y = \frac{Ax}{Bx+C}$. (4 marks)
- (c) Find the value of y as x becomes extremely large. (3 marks)

Question 6 (10 marks) A company's total production cost $C(x)$ (in dollars) for manufacturing x units of a product is a linear function $C(x) = mx + c$. Producing 100 units costs \$5,000 and producing 300 units costs \$11,000.

- (a) Find the values of m and c , and explain their physical meanings. (4 marks)
- (b) Each unit is sold at a fixed price of $p = \$45$. Express the revenue $R(x)$ and profit $P(x)$ as linear functions of x . (3 marks)
- (c) Find the break-even quantity x_0 (minimum number of units needed to avoid a net loss). (3 marks)

Question 7 (12 marks) Let $f(x) = ax + b$ and $g(x) = cx + d$ be two non-constant linear functions ($a, c \neq 0$).

- (a) Show that the composite function $(f \circ g)(x) = f(g(x))$ is also a linear function, and state its slope. (3 marks)
- (b) Prove that $f \circ g = g \circ f$ if and only if $b(c - 1) = d(a - 1)$. (4 marks)
- (c) Given $f(x) = 2x + 3$ and $g(x) = 5x - 1$, evaluate the limit:

$$\lim_{x \rightarrow \infty} \frac{(f \circ g)(x)}{x}$$

(5 marks)

1.2 Linear Regression

Definition 1.2.1: Method of Linear Regression

For a set of n bivariate data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the **least-squares regression line of y on x** is the straight line:

$$\hat{y} = a + bx \quad (\text{or } y = mx + c)$$

that minimizes the sum of squared vertical residuals:

$$S = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (a + bx_i))^2$$

Fundamental Properties of the Regression Line of y on x :

1. **Mean Point (Centroid):** The regression line always passes through the mean point $(\bar{x}, \bar{y})$, where:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Consequently, $\bar{y} = a + b\bar{x}$.

2. **Regression Coefficient (Slope b):**

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

3. **Vertical Intercept a :**

$$a = \bar{y} - b\bar{x}$$

4. **Direction of Prediction:** The regression line of y on x is used exclusively to estimate or predict the dependent variable y given a known value of the independent variable x .

Remark

We can easily perform linear regression using calculator in Mode 5 (Reg). Select the option 1 (Lin) to use linear regression mode, and input the coordinates as X,Y .

1.2.1 Level 1 Exercise

1. A set of 5 data points has $\bar{x} = 4$ and $\bar{y} = 11$. The least-squares regression line of y on x is given by $y = 1.8x + c$. Find the value of c .
2. The linear regression line of y on x for a dataset is $\hat{y} = -2.5x + 18.4$.
 - (a) Predict the value of y when $x = 4$.
 - (b) Predict the value of y when $x = 6.2$.
3. Given the summary statistics for 6 pairs of data (x, y) :

$$\sum x = 30, \quad \sum y = 48, \quad \sum x^2 = 190, \quad \sum xy = 282$$

- (a) Find the mean values $\bar{x}$ and $\bar{y}$.

- (b) Find the equation of the regression line of y on x in the form $y = a + bx$.
4. A student obtains a regression line of y on x as $y = 0.75x + 3.2$ to model student exam scores (y) against study hours (x).
- (a) Interpret the physical meaning of the slope 0.75.
- (b) State whether it is appropriate to use this equation to predict study hours x if an exam score $y = 80$ is given. Explain briefly.
5. For a set of bivariate data, the regression equation of y on x is $y = 3.4x - 5.1$, and $\bar{x} = 5$. Find $\bar{y}$.
6. The following table shows 4 data points:

x	1	2	3	4
y	3	5	6	8

- (a) Calculate $\bar{x}$ and $\bar{y}$.
- (b) Find the equation of the regression line of y on x .

1.2.2 Level 2 Exercise

1. Non-linear data (x, y) are transformed using $Y = \ln y$ and $X = x$. The least-squares regression line of Y on X is calculated to be $Y = 0.45X + 1.30$. Express y in the form $y = Ae^{kx}$, giving A and k correct to 3 significant figures.
2. A set of data (x_i, y_i) yields $\bar{x} = 10$, $\bar{y} = 25$, and the regression line of y on x is $y = 2x + 5$. A new data point $(10, 25)$ is added to the dataset.
- (a) Does the mean point $(\bar{x}, \bar{y})$ change?
- (b) Does the slope of the regression line change? Explain briefly.
3. Data variables x and y satisfy $y = ab^x$. Taking natural logarithms gives $\ln y = \ln a + x \ln b$. If $\bar{x} = 3.0$, $\ln \bar{y} = 2.4$, and the slope of the regression line of $\ln y$ on x is 0.5, find the exact values of a and b .
4. A dataset is transformed using $X = \frac{1}{x}$ and $Y = \frac{1}{y}$. The regression line of Y on X is $Y = 2.5X + 0.4$. Express y explicitly in terms of x as a single rational function.
5. For $n = 10$ pairs of data, $\sum(x_i - \bar{x})^2 = 40$, $\sum(y_i - \bar{y})^2 = 90$, and $\sum(x_i - \bar{x})(y_i - \bar{y}) = 48$. Calculate the slope b of the regression line of y on x .
6. The residuals of a linear regression model are defined as $e_i = y_i - \hat{y}_i$. Given the dataset $(1, 2), (2, 4), (3, 5)$, find the regression line $\hat{y} = a + bx$ and verify that $\sum_{i=1}^3 e_i = 0$.
7. Data (x, y) are coded using $u = x - 100$ and $v = y - 500$. The regression line of v on u is $v = 3.2u + 15$.
- (a) Express y as a linear function of x .
- (b) Predict y when $x = 110$.
8. Show that if $y = Ax^b$, the slope of the regression line of $\ln y$ on $\ln x$ equals the power exponent b .

1.2.3 Exam-type Problem

Question 1 (11 marks) The population P (in thousands) of a city at time t (in years) is modeled by $P = Ae^{kt}$, where A and k are positive constants. The values of t and P are recorded, and a transformation $Y = \ln P$ is performed. The regression line of Y on t is $Y = 0.04t + 1.25$.

- (a) Express $\ln P$ as a linear function of t . (2 marks)
- (b) Find the value of k and the exact value of A . (3 marks)
- (c) Estimate the population when $t = 10$, correct to 3 significant figures. (3 marks)
- (d) Using the limit $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$, evaluate the instantaneous rate of change at $t = 0$:

$$\lim_{h \rightarrow 0} \frac{P(h) - P(0)}{h}$$

(3 marks)

Question 2 (10 marks) An experiment measures the pressure P and volume V of a gas. A power law model $P = CV^{-\gamma}$ is proposed. Taking natural logarithms gives $\ln P = \ln C - \gamma \ln V$. Let $Y = \ln P$ and $X = \ln V$. The summary statistics for 8 data pairs (X, Y) are:

$$\bar{X} = 1.5, \quad \bar{Y} = 4.2, \quad \text{slope of regression line of } Y \text{ on } X \text{ is } -1.4$$

- (a) Find the equation of the regression line of Y on X in the form $Y = a + bX$. (3 marks)
- (b) Calculate the exact value of γ and the value of C correct to 3 significant figures. (4 marks)
- (c) Estimate the pressure P when the volume $V = 5.0$. (3 marks)

Question 3 (6 marks) The heights x (in cm) and weights y (in kg) of 10 students are recorded. It is given that:

$$\sum x = 1650, \quad \sum y = 580, \quad \sum x^2 = 273000, \quad \sum y^2 = 34200, \quad \sum xy = 96400$$

- (a) Find the mean point $(\bar{x}, \bar{y})$. (2 marks)
- (b) Find the equation of the least-squares regression line of y on x . (4 marks)

Question 4 (10 marks) A scientist collects bivariate data (x, y) and tests two models using linear regression:

- **Linear Model:** y on $x \implies y = 2.1x + 3.4$, with $r = 0.72$.
- **Exponential Model:** $\ln y$ on $x \implies \ln y = 0.35x + 0.80$, with $r = 0.98$.

- (a) Which model is more appropriate for predicting y ? Justify using the correlation coefficients. (2 marks)
- (b) Using the preferred model, express y in the form $y = Ab^x$. (4 marks)

- (c) Predict the value of y when $x = 4$ using both models, and calculate the absolute difference between the two predictions. (4 marks)

Question 5 (9 marks) Let $(x_1, y_1), \dots, (x_n, y_n)$ be n data points with regression line $y = a + bx$. A linear transformation is defined by $u_i = 3x_i - 2$ and $v_i = 2y_i + 5$.

- (a) Express $\bar{u}$ and $\bar{v}$ in terms of $\bar{x}$ and $\bar{y}$. (3 marks)
- (b) Show that the slope b' of the regression line of v on u is given by $b' = \frac{2}{3}b$. (3 marks)
- (c) If the original regression line is $y = 1.5x + 4.0$, find the transformed regression line of v on u . (3 marks)

Question 6 (10 marks) A logarithmic regression model $y = a + b \ln x$ is fitted to data. Let $X = \ln x$. The regression line of y on X passes through $(0, 5)$ and $(2, 11)$.

- (a) Find the values of a and b . (3 marks)
- (b) Express y explicitly as a function of x . (2 marks)
- (c) Find y when $x = e^3$. (2 marks)
- (d) Using the limit $\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$, evaluate:

$$\lim_{x \rightarrow 1} \frac{y(x) - a}{x - 1}$$

(3 marks)

Question 7 (11 marks) A cooling body satisfies Newton's Law of Cooling $T(t) - 20 = Ae^{-kt}$, where T is temperature in $^{\circ}\text{C}$ and t is time in minutes. Data of $Y = \ln(T - 20)$ against t are analyzed, giving a linear regression line $Y = -0.05t + 3.80$.

- (a) Express $\ln(T - 20)$ as a linear function of t . (2 marks)
- (b) Find the exact value of k and the value of A correct to 3 significant figures. (3 marks)
- (c) Find the initial temperature of the body at $t = 0$. (3 marks)
- (d) Evaluate $\lim_{t \rightarrow \infty} T(t)$ and state its physical meaning. (3 marks)

Question 8 (10 marks) A reciprocal relation $y = \frac{x}{ax+b}$ is modeled by rewriting as $\frac{1}{y} = a + b\left(\frac{1}{x}\right)$. Let $Y = \frac{1}{y}$ and $X = \frac{1}{x}$. The linear regression line of Y on X is $Y = 0.5X + 0.2$.

- (a) Find the values of a and b . (3 marks)
- (b) Express y as a function of x . (3 marks)
- (c) Evaluate the limit $\lim_{x \rightarrow \infty} y(x)$ and interpret its physical meaning. (4 marks)

Question 9 (11 marks) A saturation growth model $y = \frac{100}{1 + Ae^{-kx}}$ is linearized by $Y = \ln\left(\frac{100}{y} - 1\right) = \ln A - kx$. The regression line of Y on x is $Y = -0.2x + 1.6$.

- (a) Find the value of k and the value of A correct to 3 significant figures. (4 marks)
- (b) Predict y when $x = 5$. (3 marks)
- (c) Evaluate $\lim_{x \rightarrow \infty} y(x)$. (4 marks)

Question 10 (12 marks) Consider a set of bivariate data $(x_1, y_1), \dots, (x_n, y_n)$.

- (a) Define the residual sum of squares $S(a, b) = \sum_{i=1}^n (y_i - a - bx_i)^2$. Show that the condition $\frac{\partial S}{\partial a} = 0$ leads directly to the mean point property $\bar{y} = a + b\bar{x}$. (4 marks)
- (b) Given $\bar{x} = 2$, $\bar{y} = 7$, and slope $b = 3$, write down the regression equation. (3 marks)
- (c) Using the limit $\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$, evaluate:

$$\lim_{x \rightarrow 0} \frac{\ln(1 + \hat{y}(x) - a)}{\hat{y}(x) - a}$$

where $\hat{y}(x) = a + bx$. (5 marks)

Chapter 2

Natural Logarithmic function

2.1 Logarithmic function as inverse function of exponential function

Definition 2.1.1: Logarithmic function

For a real constant $a > 0$ and $a \neq 1$, the exponential function $f : \mathbb{R} \rightarrow (0, \infty)$ defined by $f(x) = a^x$ is a bijective (one-to-one and onto) function.

The **logarithmic function with base a** , denoted by $f^{-1}(x) = \log_a x$, is defined as the inverse function of $f(x)$.

Consequently, for any $x > 0$ and $y \in \mathbb{R}$:

$$y = \log_a x \iff x = a^y$$

Fundamental Properties derived from Inverse Relationship:

1. Domain and Range:

$$\text{Domain of } \log_a x = \text{Range of } a^x = (0, \infty)$$

$$\text{Range of } \log_a x = \text{Domain of } a^x = (-\infty, \infty)$$

2. Cancellation Identities:

$$\log_a(a^x) = x \quad \text{for all } x \in \mathbb{R}$$

$$a^{\log_a x} = x \quad \text{for all } x > 0$$

3. Geometric Reflection: The graph of $y = \log_a x$ is obtained by reflecting the graph of $y = a^x$ in the straight line $y = x$.

2.1.1 Level 1 Exercise

1. Convert each of the following exponential statements into logarithmic statements:

(a) $3^4 = 81$

(b) $e^{2k} = 5$

(c) $10^{-3} = 0.001$

2. Convert each of the following logarithmic statements into exponential statements:

(a) $\log_2 64 = 6$

(b) $\ln(x+1) = 3$

(c) $\log_{10} y = -2$

3. Without using a calculator, evaluate the exact value of each of the following expressions:

(a) $\log_5 125$

(b) $\ln(e^{-7})$

(c) $10^{\log_{10} 42}$

(d) $e^{\ln 12 - \ln 4}$

4. State the domain and range of each of the following functions:

(a) $f(x) = \log_3(x-4)$

- (b) $g(x) = \ln(2x + 6)$
 (c) $h(x) = 5 - \log_2 x$
5. Simplify each of the following expressions using cancellation identities:
- (a) $\log_7(7^{3x+1})$
 (b) $e^{\ln(x^2+1)}$
 (c) $10^{\log_{10}(5x)}$
6. Find the inverse function $f^{-1}(x)$ for each of the following exponential functions:
- (a) $f(x) = 2^x - 3$
 (b) $f(x) = e^{x+1}$
 (c) $f(x) = 4(10^{2x})$
7. Find the inverse function $f^{-1}(x)$ for each of the following logarithmic functions, and state the domain of $f^{-1}(x)$:
- (a) $f(x) = \log_3(x + 2)$
 (b) $f(x) = \ln x - 4$
 (c) $f(x) = 2\log_5(x - 1)$
8. Determine whether each of the following points lies on the graph of $y = \ln x$:
- (a) $A(1, 0)$
 (b) $B(e^2, 2)$
 (c) $C\left(\frac{1}{e}, -1\right)$
9. Solve each of the following simple equations directly using the definition of logarithms:
- (a) $\log_2(x - 3) = 4$
 (b) $\ln(2x + 1) = 0$
 (c) $\log_{10}(x^2 - 21) = 2$

2.1.2 Level 2 Exercise

1. Consider the function $f(x) = \ln\left(\frac{1+x}{1-x}\right)$ defined on the domain $-1 < x < 1$.
- (a) Show that $f(x)$ is an odd function, i.e., $f(-x) = -f(x)$.
 (b) Find the inverse function $f^{-1}(x)$ and state its domain and range.
2. The function $g(x) = 3 - 2e^{1-x}$ is given.
- (a) Find the range of $g(x)$.
 (b) Express $g^{-1}(x)$ in terms of x , and state its domain.
 (c) Find the coordinates of the x -intercept and y -intercept of the graph of $y = g^{-1}(x)$.
3. Let $f(x) = e^x$ and $g(x) = \ln(x^2 + 1)$.
- (a) Find explicit expressions for the composite functions $(f \circ g)(x)$ and $(g \circ f)(x)$.
 (b) State the domain and range of $(f \circ g)(x)$.
4. Consider the function $f(x) = 2 - \ln(3x + 1)$.

- (a) Find the exact domain and range of $f(x)$.
 - (b) Find the inverse function $f^{-1}(x)$ and state its domain.
5. Solve the quadratic-in-form exponential equation $e^{2x} - 5e^x + 6 = 0$ for x by expressing the roots in terms of natural logarithms.
 6. The hyperbolic sine function is defined as $\sinh(x) = \frac{e^x - e^{-x}}{2}$ for $x \in \mathbb{R}$.
 - (a) By letting $y = \sinh(x)$ and setting up a quadratic equation in e^x , show that $\sinh(x)$ is bijective.
 - (b) Express the inverse function $\sinh^{-1}(x)$ in terms of natural logarithms.
 7. Determine the complete domain and range of the function $f(x) = \ln(x^2 - 4x + 3)$.
 8. It is given that $f(x) = e^{x-1} + 1$.
 - (a) Find $f^{-1}(x)$.
 - (b) Prove that if the graphs of $y = f(x)$ and $y = f^{-1}(x)$ intersect at a point (a, b) , then $a = b$.
 - (c) Hence, solve the equation $e^{x-1} + 1 = \ln(x - 1) + 1$.
 9. Solve the logarithmic inequality $\log_2(x^2 - 3x) \leq 2$ and express the solution set in interval notation.

2.1.3 Exam-type Problem

Question 1 (11 marks) Let $f : \mathbb{R} \rightarrow (-\infty, 5)$ be defined by $f(x) = 5 - 2e^{3x}$.

- (a) Find $f^{-1}(x)$ and state the domain of $f^{-1}(x)$. (3 marks)
- (b) The graph of $y = f(x)$ cuts the y -axis at point P , and the graph of $y = f^{-1}(x)$ cuts the x -axis at point Q .
 - (i) Find the coordinates of P and Q .
 - (ii) Explain, from geometric transformation principles, why P and Q are symmetric about the line $y = x$. (3 marks)
- (c) Show that if the curves $y = f(x)$ and $y = f^{-1}(x)$ intersect at a point (k, k) , then k satisfies the equation $k + 2e^{3k} - 5 = 0$. (2 marks)
- (d) Using the limit $\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = 1$, evaluate the limit $\lim_{x \rightarrow 3} \frac{f^{-1}(x) - 1}{x - 3}$. (3 marks)

Question 2 (7 marks) Let $g(x) = \ln(x + \sqrt{x^2 + 1})$ for $x \in \mathbb{R}$.

- (a) Show that $x + \sqrt{x^2 + 1} > 0$ for all real numbers x . (2 marks)
- (c) Find $g^{-1}(x)$ and express it in terms of exponential functions. (3 marks)
- (d) Deduce the range of $g(x)$. (2 marks)

Question 3 (8 marks) The graph of $y = a^x + b$ passes through $(0, 3)$ and $(2, 11)$, where $a > 1$ and b are constants.

- (a) Find the values of a and b . (3 marks)

- (b) Find the inverse function $f^{-1}(x)$ of $f(x) = a^x + b$. (2 marks)
- (c) State the domain and range of $f^{-1}(x)$, and find the x -intercept of its graph. (3 marks)

Question 4 (9 marks) Let $f(x) = \ln(k - x)$, where k is a real constant.

- (a) State the domain of $f(x)$ in terms of k . (1 mark)
- (b) Find $f^{-1}(x)$ and state its range. (3 marks)
- (c) If the graph of $y = f^{-1}(x)$ passes through $(\ln 4, -1)$, find the exact value of k . (2 marks)
- (d) For the value of k found in (c), solve the equation $f(x) = f^{-1}(x)$. (3 marks)

Question 5 (10 marks) Consider the function $h(x) = \frac{e^x - 1}{e^x + 1}$ defined for $x \in \mathbb{R}$.

- (a) Show that $-1 < h(x) < 1$ for all $x \in \mathbb{R}$. (3 marks)
- (b) Find $h^{-1}(x)$ and state its domain. (4 marks)
- (c) Solve $h^{-1}(x) = \ln 3$. (3 marks)

Question 6 (8 marks) It is given that $f(x) = 2^{3x-1} + 4$.

- (a) Express $f^{-1}(x)$ in terms of base-2 logarithms. (3 marks)
- (b) Express $f^{-1}(x)$ in terms of natural logarithms ($\ln$). (2 marks)
- (c) Find all axis intercepts and the line of symmetry. (3 marks)

Question 7 (11 marks) Let $f(x) = \ln(x^2 - 2x + 2)$ for $x \geq 1$.

- (a) By completing the square for $x^2 - 2x + 2$, show that $f(x)$ is well-defined. (3 marks)
- (b) Find $f^{-1}(x)$ and state its domain. (4 marks)
- (c) Find the coordinates of the intersection point of $y = f(x)$ and $y = f^{-1}(x)$. (4 marks)

Question 8 (9 marks) A curve C_1 has equation $y = e^{2x} - 4e^x + 3$.

- (a) Find the x -intercepts and y -intercept of C_1 . (3 marks)
- (b) Let C_2 be the reflection of C_1 in the line $y = x$.
- (i) Write down the x -intercepts and y -intercept of C_2 .
- (ii) Find the equation of C_2 for $y \geq \ln 2$. (6 marks)

Question 9 (10 marks) Let $p(x) = \log_a(x + b)$, where $a > 1$ and $b > 0$. The graph of $y = p(x)$ has an x -intercept at $(3, 0)$ and its inverse function graph $y = p^{-1}(x)$ passes through $(2, 19)$.

- (a) Find the values of a and b . (4 marks)
- (b) Find $p^{-1}(x)$ and state its domain and range. (3 marks)

- (c) Solve $p(x) < 2$. (3 marks)

Question 10 (12 marks) Let $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ for $x \in \mathbb{R}$.

- (a) Show that $f(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$. (2 marks)
- (b) Find $f^{-1}(x)$ and state its domain. (4 marks)
- (c) Using the limit $\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$, evaluate $\lim_{x \rightarrow 0} \frac{f^{-1}(x)}{x}$. (3 marks)

2.2 Natural Logarithm

Definition 2.2.1: Natural Logarithm

The **natural logarithm**, denoted by $\ln x$ (or $\log_e x$), is the logarithmic function with base e , where $e \approx 2.71828$ is the natural number defined by the limit:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}}$$

Since $f(x) = e^x$ is a bijective function from $\mathbb{R}$ to $(0, \infty)$, its inverse function is $f^{-1}(x) = \ln x$.

Consequently, for any $x > 0$ and $y \in \mathbb{R}$:

$$y = \ln x \iff x = e^y$$

Fundamental Properties of Natural Logarithms:

1. Domain and Range:

$$\text{Domain of } \ln x = (0, \infty), \quad \text{Range of } \ln x = (-\infty, \infty)$$

2. Cancellation Identities:

$$\ln(e^x) = x \quad \text{for all } x \in \mathbb{R}$$

$$e^{\ln x} = x \quad \text{for all } x > 0$$

3. Special Values:

$$\ln 1 = 0, \quad \ln e = 1$$

4. Fundamental Limit Formula:

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

2.2.1 Level 1 Exercise

- Convert each of the following exponential statements into equivalent logarithmic statements using $\ln$:

(a) $e^x = 7$

(b) $e^{3-x} = 2$

(c) $e^{x^2} = 10$

- Without using a calculator, evaluate the exact value of each of the following expressions:

(a) $\ln(\sqrt{e})$

(b) $\ln(e^5) - \ln(e^2)$

(c) $e^{2 \ln 3}$

(d) $e^{-\ln 4}$

- State the domain and range of each of the following functions:

(a) $f(x) = \ln(x+5)$

(b) $g(x) = \ln(3 - 2x)$

(c) $h(x) = 4 - 2 \ln x$

4. Simplify each of the following expressions using cancellation identities:

(a) $e^{\ln(2x-1)}$

(b) $\ln(e^{x^2+3x})$

(c) $e^{3 \ln x + 1}$

5. Find the inverse function $f^{-1}(x)$ for each of the following functions, and state the domain of $f^{-1}(x)$:

(a) $f(x) = 2e^x - 1$

(b) $f(x) = \ln(x - 3) + 2$

(c) $f(x) = \frac{1}{2}e^{4x}$

6. Solve each of the following equations directly using the definition of natural logarithms:

(a) $\ln(3x - 2) = 0$

(b) $e^{2x+1} = 9$

(c) $\ln(x^2 - 3) = \ln 6$

7. Given $a = \ln 2$ and $b = \ln 3$, express each of the following in terms of a and b :

(a) $\ln 6$

(b) $\ln\left(\frac{8}{9}\right)$

(c) $\ln(\sqrt{12})$

8. Determine whether $x = e^2$ is a solution to the equation:

$$\ln x + \ln(x - e) = 2 + \ln(e - 1)$$

9. Find the exact coordinates of the x -intercept and y -intercept (if any) of the graph of $y = 3 \ln(x + 1) - 2$.

10. Solve for x given that $e^{\ln(x+2)} = 2x - 5$.

2.2.2 Level 2 Exercise

1. Solve the quadratic-in-form equation $(\ln x)^2 - 3 \ln x + 2 = 0$ for x .

2. Solve the exponential equation $e^x - 6e^{-x} = 1$ for $x \in \mathbb{R}$, leaving your answer in terms of natural logarithms.

3. Let $g(x) = \frac{e^x + e^{-x}}{2}$. Solve the equation $g(x) = 2$ for $x \in \mathbb{R}$, expressing the solutions in terms of natural logarithms.

4. Find the complete domain and range of the function $f(x) = \ln(9 - x^2)$.

5. Solve the logarithmic inequality $\ln(x^2 - 5x + 6) \leq \ln 2$ and write the solution in interval notation.

2.2.3 Exam-type Problem

Question 1 (11 marks)

- (a) Using the exponential series expansion $e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \dots$, expand e^{2x} up to the term in x^3 . (2 marks)
- (b) Hence, estimate the value of $\frac{e^{0.2}-1}{0.1}$ using the first 4 terms of the series expansion. (3 marks)
- (c) Using the fundamental limit $\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = 1$ and $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$, evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{\ln(1 + 2x)}$$

(6 marks)

Question 2 (7 marks) Let $f(x) = \ln\left(\frac{e^x-1}{e^x+1}\right)$ for $x > 0$.

- (a) Show that $f(x) = \ln\left(\frac{1-e^{-x}}{1+e^{-x}}\right)$. (2 marks)
- (b) Find the inverse function $f^{-1}(x)$, and state its domain and range. (5 marks)

Question 3 (9 marks) The population $P(t)$ of a species at time t (in years) is modeled by $P(t) = P_0 e^{kt}$, where P_0 and k are positive constants.

- (a) If the population doubles in 5 years, find the exact value of k in terms of $\ln 2$. (3 marks)
- (b) Find the time required for the population to triple. Express your answer in terms of natural logarithms, and evaluate correct to 3 significant figures. (3 marks)
- (c) Using the approximation $e^x \approx 1 + x$ for small x , estimate the percentage increase in population over a short time period of $t = 0.1$ years when $k = 0.14$. (3 marks)

Question 4 (9 marks) Let $g(x) = \ln(x^2 + 2x + 2)$ for $x \in \mathbb{R}$.

- (a) Write down the domain and range of $g(x)$. (2 marks)
- (b) Solve $g(x) = \ln 5$. (3 marks)
- (c) Using the limit $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$, evaluate:

$$\lim_{x \rightarrow 0} \frac{g(x) - \ln 2}{x}$$

(4 marks)

Question 5 (10 marks) Let $f(x) = e^{2x} - 4e^x + 3$ for $x \in \mathbb{R}$.

- (a) Find all x -intercepts of $y = f(x)$. (3 marks)
- (b) For $x \geq \ln 2$, find $f^{-1}(x)$. State the domain of $f^{-1}(x)$. (3 marks)

- (c) Using the limit $\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = 1$, evaluate:

$$\lim_{x \rightarrow 3} \frac{f^{-1}(x) - \ln 2}{x - 3}$$

(4 marks)

Question 6 (5 marks) Let $h(x) = \ln\left(\frac{x+a}{x-a}\right)$, where $a > 0$ is a constant.

- (a) State the domain of $h(x)$ in terms of a . (2 marks)
- (c) Find $h^{-1}(x)$ in terms of a and e^x . (3 marks)

Question 7 (10 marks) Let $f(x) = \ln(e^x + k)$, where $k > 0$ is a constant.

- (a) State the domain and range of $f(x)$. (2 marks)
- (b) Find $f^{-1}(x)$ and state its domain. (4 marks)
- (c) If the graph of $y = f^{-1}(x)$ passes through $(\ln 5, \ln 2)$, find the value of k . (2 marks)
- (d) For this value of k , solve $f(x) = 2x$. (2 marks)

Question 8 (9 marks)

- (a) Show that $\lim_{x \rightarrow 0} \frac{\ln(1+3x)}{x} = 3$ using the standard limit $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$. (3 marks)
- (b) Hence, evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+3x) - \ln(1-2x)}{x}$. (3 marks)
- (c) Find the value of the constant c such that:

$$\lim_{x \rightarrow 0} \frac{e^{cx} - 1}{\ln(1+3x)} = 2$$

(3 marks)

Question 9 (10 marks) Consider the natural logarithm function expansion $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ for $|x| < 1$.

- (a) Find the exact value of $\ln(x+1)$ when $x = e^2 - 1$. (1 mark)
- (b) (i) Find the first 3 non-zero terms of the series expansion of $\ln\left(\frac{1+x}{1-x}\right)$ for $|x| < 1$. (4 marks)
- (ii) By substituting $x = \frac{1}{3}$, estimate the value of $\ln 2$. Give your answer as an exact fraction. (5 marks)

2.3 Properties of Logarithmic Function

Theorem 2.3.1: Properties of natural log

Let $x, y > 0$.

- $\ln(xy) = \ln(x) + \ln(y)$
- $\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$
- $\ln(x^y) = y \ln(x)$

Remark

The third identity is only defined directly for $x > 0$ (base > 0). If $x < 0$, we require y to be an even integer, and the result becomes $\ln(x^y) = y \ln|x|$, where $|\cdot|$ is the absolute-value function.

2.3.1 Level 1 Exercise

1. Expand each of the following expressions into sums, differences, or multiples of logarithms (assume all variables represent positive real numbers):

(a) $\ln(x^3y^2)$

(b) $\ln\left(\frac{\sqrt{x}}{y^4}\right)$

(c) $\ln\left(\frac{x^2y^3}{z^5}\right)$

2. Express each of the following as a single logarithm in the form $\ln(K)$:

(a) $2 \ln 3 + 3 \ln 2$

(b) $\frac{1}{2} \ln 16 - 2 \ln 2 + \ln 5$

(c) $3 \ln x - 2 \ln y + \frac{1}{3} \ln z \quad (x, y, z > 0)$

3. Without using a calculator, evaluate the exact value of each expression:

(a) $\ln(e^4 \cdot \sqrt{e})$

(b) $\ln 18 - 2 \ln 3 + \ln 5 - \ln 10$

(c) $e^{3 \ln 2 - \ln 4}$

4. Using the Remark above, simplify each of the following expressions:

(a) $\ln((-4)^2)$

(b) $\ln((-x)^6)$ for $x < 0$

(c) $\ln(x^4) - 2 \ln|x|$ for $x \neq 0$

5. Given $p = \ln 2$, $q = \ln 3$, and $r = \ln 5$, express each of the following in terms of p , q , and r :

(a) $\ln 30$

(b) $\ln\left(\frac{12}{25}\right)$

(c) $\ln(0.072)$

6. Given that $\ln a = 3$ and $\ln b = 5$, evaluate the exact value of:

(a) $\ln(a^2b^3)$

(b) $\ln\left(\frac{\sqrt{a}}{b^2}\right)$

(c) $\ln\left(\frac{e^2 \cdot a^4}{\sqrt{b}}\right)$

7. Solve each of the following simple logarithmic equations for x :

(a) $\ln x + \ln 4 = \ln 12$

(b) $\ln(x+3) - \ln x = \ln 2$

(c) $2 \ln x = \ln(3x+4)$

8. Using the Change of Base Formula $\log_a b = \frac{\ln b}{\ln a}$, evaluate the exact value of:

(a) $\log_2 32$

(b) $\log_9 27$

(c) $(\log_3 5)(\log_5 81)$

9. Solve the following exponential equations by taking natural logarithms on both sides:

(a) $3^x = 7$

(b) $2^{x+1} = 5^x$

(c) $e^{2x-1} = 4^x$

10. Determine whether the following statements are True or False. If false, provide a counterexample or correct expression:

(a) $\frac{\ln a}{\ln b} = \ln a - \ln b$ for $a, b > 1$

(b) $\ln(a+b) = \ln a + \ln b$ for $a, b > 0$

(c) $(\ln x)^2 = 2 \ln x$ for $x > 0$

2.3.2 Level 2 Exercise

1. Solve the logarithmic equation for x :

$$\ln(x-2) + \ln(x+1) = \ln(x+7)$$

Make sure to verify and state all domain restrictions.

2. Solve the equation for x :

$$(\ln x)^2 - \ln(x^3) + 2 = 0$$

3. Solve the system of simultaneous equations for x and y :

$$\begin{cases} \ln x + 2 \ln y = 7 \\ 2 \ln x - \ln y = 1 \end{cases}$$

4. Prove the following identity for $x > 0$:

$$\ln(\sqrt{x+1} - \sqrt{x}) + \ln(\sqrt{x+1} + \sqrt{x}) = 0$$

5. Let $x < 0$. Simplify the function $f(x) = \ln(x^6) - 3 \ln(x^2) + \ln(-x)$ into a single logarithm term in terms of $-x$.

6. Solve the equation involving change of base for $x > 0$ and $x \neq 1$:

$$\log_x e + 2 \ln x = 3$$

7. Solve the equation for x :

$$x^{\ln x} = e^4 x^3$$

8. Solve the logarithmic inequality:

$$\ln(x^2 - 4) - \ln(x - 2) \leq \ln 5$$

and express the solution set in interval notation.

9. Let $f(x) = \ln\left(\frac{2+x}{2-x}\right)$ defined for $-2 < x < 2$.

- (a) Show that $f(x) + f(y) = f\left(\frac{4x+4y}{4+xy}\right)$.
 (b) Simplify $f(x) + f(y) + f(-x - y)$.

2.3.3 Exam-type Problem

Question 1 (9 marks) Consider the logarithmic equation $\ln(x^2 - kx + 4) - \ln(x - 1) = \ln 3$, where k is a real constant.

- (a) Express x in terms of k . (3 marks)
 (a) Find the range of values of k such that the equation has exactly one real root. (6 marks)

Question 2 (10 marks) The variables x and y satisfy a non-linear relationship. The graph of $\ln y$ against $\ln x$ is a straight line passing through $(2, 1)$ and $(5, 7)$.

- (a) Express $\ln y$ as a linear function of $\ln x$. (2 marks)
 (b) Express y in the form $y = Ax^p$, where A and p are exact constants. (4 marks)
 (c) Using the result in (b), evaluate y when $x = e^2$, in terms of e . (4 marks)

Question 3 (10 marks) Let $S_n = \sum_{k=2}^n \ln\left(1 - \frac{1}{k^2}\right)$ for integers $n \geq 2$.

- (a) Show that $1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2}$. (1 mark)
 (b) Using logarithmic properties, show that $S_n = \ln\left(\frac{n+1}{2n}\right)$. (5 marks)
 (c) Evaluate the limit $\lim_{n \rightarrow \infty} S_n$. (4 marks)

Question 4 (9 marks) Solve the simultaneous system for $x, y > 0$:

$$\begin{cases} xy = e^5 \\ \ln x \cdot \ln y = 6 \end{cases}$$

- (a) Show that $\ln x$ and $\ln y$ are the roots of the quadratic equation $t^2 - 5t + 6 = 0$. (4 marks)
 (b) Hence, find all ordered pairs (x, y) satisfying the system. Express your answers in exact form in terms of e . (5 marks)

Question 5 (10 marks) A scientist models the radioactive decay of a substance by $N(t) = N_0 e^{-\lambda t}$, where $N(t)$ is the mass remaining at time t (in years), N_0 is the initial mass, and $\lambda > 0$ is the decay constant.

- (a) Express $\ln N(t)$ as a linear function of t . (2 marks)
- (b) The half-life T is defined as the time required for the mass to decay to half of its initial value. Show that $T = \frac{\ln 2}{\lambda}$. (3 marks)
- (c) If the mass decays to 25% of its initial value in 12 years, find the exact value of T , and determine the time required for the mass to decay to 10% of its initial value (correct to 3 significant figures). (5 marks)

Question 6 (8 marks)

- (a) Using the fundamental limit $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$, evaluate:

$$\lim_{x \rightarrow 0} \frac{\ln(1 + 3x + 2x^2)}{x}$$

(4 marks)

- (b) Using the binomial expansion $(1+y)^4 = 1 + 4y + 6y^2 + 4y^3 + y^4$, expand $\ln((1+x)^4)$ using logarithmic properties and find:

$$\lim_{x \rightarrow 0} \frac{\ln((1+x)^4)}{2x}$$

(4 marks)

Question 7 (11 marks) Let $f(x) = \ln(x^4 - 16)$ defined on its maximal domain D .

- (a) Find the domain D of $f(x)$. (3 marks)
- (b) Using the Remark on $\ln(x^y) = y \ln|x|$ for negative bases, simplify $f(x) - \ln(x^2 + 4)$ for $x \in D$. (4 marks)
- (c) Solve the equation $f(x) - \ln(x^2 + 4) = \ln 5$. (4 marks)

Question 8 (12 marks) Let $P_n = \prod_{k=2}^n \left(1 - \frac{2}{k(k+1)}\right)$ for $n \geq 2$.

- (a) Show that $1 - \frac{2}{k(k+1)} = \frac{(k-1)(k+2)}{k(k+1)}$. (2 marks)
- (b) Taking natural logarithms on both sides, express $\ln P_n$ as a telescoping sum and show that:

$$\ln P_n = \ln \left(\frac{n+2}{3n} \right)$$

(6 marks)

- (c) Deduce an explicit expression for P_n in terms of n , and evaluate $\lim_{n \rightarrow \infty} P_n$. (4 marks)

Question 9 (11 marks) Consider the function $g(x) = \ln \left(\frac{e^x + 1}{e^x - 1} \right)$ for $x > 0$.

- (a) Express $g(x)$ in terms of $\ln(1 + e^{-x})$ and $\ln(1 - e^{-x})$. (3 marks)

(b) Show that $g^{-1}(x) = g(x)$ for all $x > 0$. *(5 marks)*

(c) Using the limit $\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$, evaluate:

$$\lim_{x \rightarrow \infty} e^x g(x)$$

(3 marks)

2.4 Change of Base

Theorem 2.4.1: Change of Base

Let $x, y > 0$ with $x \neq 1$.

$$\log_x(y) = \frac{\ln(y)}{\ln(x)}$$

Proof. Let $z = \log_x(y)$.

By the definition of logarithmic functions, this is equivalent to:

$$x^z = y$$

Taking the natural logarithm ($\ln$) on both sides of the equation:

$$\ln(x^z) = \ln(y)$$

Applying the power property of logarithms ($\ln(a^b) = b \ln a$):

$$z \ln(x) = \ln(y)$$

Since $x > 0$ and $x \neq 1$, we have $\ln(x) \neq 0$. Dividing both sides by $\ln(x)$:

$$z = \frac{\ln(y)}{\ln(x)}$$

Substituting $z = \log_x(y)$ back yields:

$$\log_x(y) = \frac{\ln(y)}{\ln(x)}$$

This completes the proof. □

Remark

In general, we write

$$\log_x(y) = \frac{\log_a(y)}{\log_a(x)}$$

for an arbitrary positive real number $a \neq 1$. However, due to consistency and convenience, I prefer writing without other bases. Let the arithmetic follow whenever we need that extra base.

2.4.1 Level 1 Exercise

1. Express each of the following logarithms in terms of natural logarithms ($\ln$):

(a) $\log_2 7$

(b) $\log_5 100$

(c) $\log_{10} e$

2. Without using a calculator, evaluate the exact value of each of the following:

(a) $\log_2 8$

(b) $\log_9 27$

(c) $\log_{16} 64$

3. Simplify each of the following logarithmic products:

(a) $\log_3 5 \cdot \log_5 9$

(b) $\log_2 3 \cdot \log_3 4 \cdot \log_4 8$

(c) $\frac{\log_5 16}{\log_5 2}$

4. Given $a = \ln 2$ and $b = \ln 3$, express each of the following in terms of a and b :

(a) $\log_2 3$

(b) $\log_3 12$

(c) $\log_6 18$

5. Find the exact value of x in each of the following simple equations:

(a) $\log_x 81 = 4$

(b) $\log_x 5 = \frac{1}{2}$

(c) $\log_x e = 2$

6. Prove that for any $A, B > 0$ with $B \neq 1$:

$$\frac{\log_{10} A}{\log_{10} B} = \frac{\ln A}{\ln B}$$

7. Evaluate the exact value of each of the following exponential expressions:

(a) $2^{\log_2 5}$

(b) $3^{\log_9 16}$

(c) $4^{\log_2 3}$

8. Solve the equation $\log_4 x = \log_2 3$ for x .

9. Simplify the expression $\log_a \sqrt{b} \cdot \log_b (a^3)$ for $a, b > 0$ and $a, b \neq 1$.

10. Show that $\log_a b = \frac{1}{\log_b a}$ for all $a, b > 0$ and $a, b \neq 1$.

2.4.2 Level 2 Exercise

1. Solve the logarithmic equation for x :

$$\log_2 x + \log_4 x = 6$$

2. Solve the equation involving a variable base for $x > 0$ and $x \neq 1$:

$$\log_x 4 + \log_4 x = \frac{5}{2}$$

3. Solve the simultaneous system for $x, y > 0$ and $x, y \neq 1$:

$$\begin{cases} \log_2 x + \log_4 y = 3 \\ \log_4 x + \log_2 y = 3 \end{cases}$$

4. Solve the equation for $x > 0$:

$$x^{\log_2 x} = 16x$$

5. Given $a, b, c > 1$, prove that:

$$\frac{1}{\log_a(abc)} + \frac{1}{\log_b(abc)} + \frac{1}{\log_c(abc)} = 1$$

6. Solve the equation for x :

$$\log_2(x+1) - \log_8(x+1) = \frac{2}{3}$$

7. Given $a^x = b^y = c^z = d$, where $a, b, c, d > 1$. Show that:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \log_d(abc)$$

8. Solve the logarithmic inequality for $x > 0$ and $x \neq 1$:

$$\log_x 2 < 1$$

9. Given that $\log_a b = k$, express each of the following in terms of k :

(a) $\log_{a^2}(b^3)$

(b) $\log_{\sqrt{a}}\left(\frac{1}{b}\right)$

10. Solve the equation for $x > 0$ and $x \neq 1$:

$$\log_3 x - \log_x 9 + 1 = 0$$

2.4.3 Exam-type Problem

Question 1 (8 marks) Solve the logarithmic equation:

$$\log_3 x + \log_9 x + \log_{27} x = \frac{11}{2}$$

(8 marks)

Question 2 (9 marks) Consider the equation $\log_x 16 - \log_4 x = 3$.

(a) By using the Change of Base Theorem, show that the equation can be rewritten as:

$$(\log_4 x)^2 + 3 \log_4 x - 2 = 0$$

(4 marks)

(b) Hence, find the exact values of x .

(5 marks)

Question 3 (10 marks) The graph of $\log_2 y$ against $\log_4 x$ is a straight line passing through the points $(1, 3)$ and $(3, 7)$.

(a) Express $\log_2 y$ as a linear function of $\log_4 x$.

(2 marks)

(b) Express y in the form $y = Ax^p$, where A and p are exact constants.

(5 marks)

(c) Find the value of y when $x = 16$.

(3 marks)

Question 4 (10 marks) Let $a, b, c > 1$. It is given that $\log_a x, \log_b x, \log_c x$ form an arithmetic progression for any $x > 0$ ($x \neq 1$).

- (a) Show that $\frac{2}{\ln b} = \frac{1}{\ln a} + \frac{1}{\ln c}$. (4 marks)
- (b) Express b in terms of a and c . (3 marks)
- (c) If $a = 2$ and $c = 8$, find the exact value of b . (3 marks)

Question 5 (10 marks) Solve the system of non-linear equations for $x, y > 0$:

$$\begin{cases} xy = 64 \\ \log_x y + \log_y x = \frac{5}{2} \end{cases}$$

(10 marks)

Question 6 (9 marks) Let $a > 0$ and $a \neq 1$.

- (a) Express $\log_a x$ in terms of natural logarithms. (1 mark)
- (b) Using the limit $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$, evaluate:

$$\lim_{x \rightarrow 1} \frac{\log_a x}{x - 1}$$

(5 marks)

- (c) Deduce the value of $\lim_{x \rightarrow 1} \frac{\log_2 x}{x - 1}$. (3 marks)

Question 7 (11 marks) For any integer $n \geq 2$, define $P_n = \sum_{k=2}^n \frac{1}{\log_k(n!)}$.

- (a) Using the Change of Base Theorem, prove that $\frac{1}{\log_k(n!)} = \frac{\ln k}{\ln(n!)}$. (3 marks)
- (b) Show that $P_n = 1$ for all integers $n \geq 2$. (5 marks)
- (c) Evaluate $\lim_{n \rightarrow \infty} P_n$. (3 marks)

Question 8 (9 marks) Consider the exponential equation $2^{2x+1} \cdot 3^x = 5^{1-x}$.

- (a) By taking natural logarithms on both sides and applying base conversion properties, show that:

$$x = \frac{\ln 5 - \ln 2}{\ln 12 + \ln 5}$$

(5 marks)

- (b) Express x in the form $\log_A B$, where A and B are integers. (4 marks)

Question 9 (10 marks) Consider the logarithmic inequality $\log_x(3x - 2) > 2$.

- (a) Find the domain of the expression $\log_x(3x - 2)$. (3 marks)
- (b) By considering two cases ($0 < x < 1$ and $x > 1$), solve the inequality completely. Give your answer in interval notation. (7 marks)

Question 10 (10 marks)

(a) Express $\log_2(1 + 4x)$ and $\log_3(1 + 2x)$ in terms of natural logarithms. *(2 marks)*

(b) Using the fundamental limit $\lim_{u \rightarrow 0} \frac{\ln(1 + u)}{u} = 1$, evaluate the following limits:

(i) $\lim_{x \rightarrow 0} \frac{\log_2(1 + 4x)}{x}$ *(3 marks)*

(ii) $\lim_{x \rightarrow 0} \frac{\log_2(1 + 4x)}{\log_3(1 + 2x)}$ *(5 marks)*

2.5 Application Problems of Logarithmic Function

Definition 2.5.1: Applications of Logarithmic Functions

Logarithmic functions are extensively used in modeling real-world phenomena involving rapidly growing or decaying quantities, as well as in linearizing non-linear relationships.

1. Data Linearization Models (HKDSE Core Focus):

- **Exponential Model:** $y = Ab^x$ ($A > 0, b > 0, b \neq 1$)

$$\log_k y = (\log_k b)x + \log_k A$$

The graph of $\log_k y$ against x is a straight line with:

$$\text{Slope } m = \log_k b, \quad \text{Vertical Intercept } c = \log_k A$$

- **Power Model:** $y = Ax^n$ ($A > 0, x > 0$)

$$\log_k y = n \log_k x + \log_k A$$

The graph of $\log_k y$ against $\log_k x$ is a straight line with:

$$\text{Slope } m = n, \quad \text{Vertical Intercept } c = \log_k A$$

2. Standard Real-World Scientific Scales:

- **Sound Intensity Level (Decibels):** $D = 10 \log_{10} \left(\frac{I}{I_0} \right)$, where $I_0 = 10^{-12} \text{ W/m}^2$ is the threshold of hearing.
- **Richter Magnitude Scale (Earthquakes):** $M = \log_{10} \left(\frac{A}{A_0} \right)$, or energy released E given by $\log_{10} E = 4.8 + 1.5M$.
- **pH Scale (Chemistry):** $\text{pH} = -\log_{10}[\text{H}^+]$, where $[\text{H}^+]$ is the hydrogen ion concentration in moles per liter (mol/L).
- **Exponential Growth and Decay:** $N(t) = N_0 e^{kt}$ or $N(t) = N_0 a^t$.

2.5.1 Level 1 Exercise

1. The sound intensity of a vacuum cleaner is 10^{-4} W/m^2 . Calculate its sound intensity level D in decibels (dB) using $D = 10 \log_{10} \left(\frac{I}{I_0} \right)$, where $I_0 = 10^{-12} \text{ W/m}^2$.
2. The hydrogen ion concentration of a fruit juice sample is $[\text{H}^+] = 2.5 \times 10^{-4} \text{ mol/L}$. Find the pH value of the juice correct to 2 decimal places.
3. An earthquake measures $M_1 = 6.2$ on the Richter scale, while a second earthquake measures $M_2 = 4.2$. Find the ratio of the amplitude A_1 of the first earthquake to A_2 of the second earthquake.
4. A sum of \$10,000 is deposited into a bank account earning 4% per annum compounded annually. Find the minimum number of complete years required for the total amount to exceed \$20,000.
5. The population P of a city t years after 2020 is modeled by $P(t) = 500,000 \cdot (1.03)^t$.
 - (a) Find the population in 2025 correct to the nearest integer.

- (b) Find the number of years needed for the population to reach 700,000. Give your answer correct to 1 decimal place.
6. The mass $M(t)$ (in grams) of a radioactive sample remaining after t days is given by $M(t) = 80e^{-0.05t}$.
 - (a) State the initial mass of the sample.
 - (b) Find the mass remaining after 10 days, correct to 3 significant figures.
7. The relation $y = 4(3^x)$ is converted into linear form by plotting $\log_{10} y$ against x . Find the exact slope and vertical intercept of the straight line.
8. The relation $y = 5x^{-2}$ is converted into linear form by plotting $\ln y$ against $\ln x$. Find the exact slope and vertical intercept of the straight line.
9. A cup of hot tea cools according to $T(t) = 20 + 70e^{-0.1t}$, where T is the temperature in $^{\circ}\text{C}$ and t is time in minutes. Find the time taken for the tea to cool down to 40°C , correct to 3 significant figures.
10. The value V (in HKD) of a machine after t years is given by $V(t) = 150,000(0.85)^t$. Find the time t for the machine's value to depreciate to half of its original value. Give your answer correct to 3 significant figures.

2.5.2 Level 2 Exercise

1. Two identical loudspeakers, each producing a sound intensity level of 70 dB, operate simultaneously. Calculate the combined sound intensity level in dB. Give your answer correct to 1 decimal place.
2. A bacterial population grows according to $N(t) = N_0e^{kt}$. If the population grows from 2,000 to 6,000 in 4 hours:
 - (a) Find the exact value of k in terms of natural logarithms.
 - (b) Find the total time required for the population to reach 50,000. Give your answer correct to 3 significant figures.
3. The half-life of Carbon-14 is 5,730 years. The mass remaining is modeled by $N(t) = N_0e^{-\lambda t}$.
 - (a) Find the decay constant λ in terms of $\ln 2$.
 - (b) An ancient wood artifact contains 18% of its original Carbon-14. Estimate the age of the artifact correct to the nearest 100 years.
4. An investment of $\$P$ grows under continuous compounding according to $A(t) = Pe^{rt}$, where $r = 0.05$. A second investment grows under annual compounding $B(t) = P(1 + R)^t$. If both double in value in the exact same time period, find R correct to 4 decimal places.
5. The graph of $\log_{10} y$ against x is a straight line passing through $(1, 0.8)$ and $(3, 1.4)$. Express y explicitly in the form $y = A \cdot b^x$, giving A and b correct to 3 significant figures.
6. The graph of $\ln y$ against $\ln x$ is a straight line with a slope of -1.5 and a vertical intercept of 2.4 . Express y in the form $y = Ax^k$, where A and k are exact constants.
7. The energy E (in Joules) released by an earthquake of Richter magnitude M satisfies $\log_{10} E = 4.8 + 1.5M$.
 - (a) Calculate the energy released by an earthquake of magnitude $M = 7.0$.

- (b) An earthquake releases 1000 times more energy than another. Find the difference in their Richter magnitudes.
8. Solution X has $\text{pH} = 3.0$ and Solution Y has $\text{pH} = 5.0$.
- (a) Find the ratio of $[\text{H}^+]$ concentration of Solution X to Solution Y.
- (b) Equal volumes of Solution X and Solution Y are mixed. Find the pH of the resulting mixture correct to 2 decimal places.
9. A student's score S (out of 100) on a memory test t weeks after learning a vocabulary list is modeled by $S(t) = 85 - 18 \ln(t + 1)$.
- (a) Find the initial score at $t = 0$.
- (b) Find the number of weeks t required for the score to drop to 50. Give your answer correct to 2 decimal places.
10. The cost $C(x)$ (in thousands of dollars) to produce x units of a product is given by $C(x) = C_0 e^{kx}$. It is given that producing 10 units costs \$50,000 and producing 30 units costs \$200,000.
- (a) Find the exact value of k and C_0 .
- (b) Calculate the cost to produce 50 units, correct to the nearest thousand dollars.

2.5.3 Exam-type Problem

Question 1 (11 marks) The population P (in thousands) of a species in a protected reserve at time t (in years) is modeled by $P(t) = Ae^{kt}$, where A and k are positive constants. The graph of $\ln P$ against t is a straight line passing through $(2, 1.8)$ and $(6, 3.0)$.

- (a) Express $\ln P$ as a linear function of t . (2 marks)
- (b) Find the exact value of k and the value of A correct to 3 significant figures. (3 marks)
- (c) Find the minimum number of complete years required for the population to exceed 100,000. (3 marks)
- (d) Using the fundamental limit $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$, evaluate $\lim_{h \rightarrow 0} \frac{P(h) - P(0)}{hA}$. (3 marks)

Question 2 (10 marks) A non-linear relation between x and y is modeled by $y = ab^x$, where a and b are positive constants. The graph of $\log_{10} y$ against x is shown below, passing through the points $(1, 1)$ and $(4, 2.5)$.

- (a) Find the slope and vertical intercept of the straight line of $\log_{10} y$ against x . (2 marks)
- (b) Find the exact values of a and b . (4 marks)
- (c) Find the value of y when $x = 6$, expressing your answer in exact surd form. (2 marks)
- (d) Solve the inequality $y > 1000$ for x . (2 marks)

Question 3 (9 marks) The variables x and y satisfy the power relation $y = ax^n$, where $x > 0$, $a > 0$, and n is a constant. The graph of $\log_2 y$ against $\log_2 x$ is a straight line with a y -intercept of 3 and an x -intercept of -2 .

- (a) Find the value of n and the exact value of a . (4 marks)
- (b) Express y explicitly in terms of x . (2 marks)
- (c) Find the exact value of x when $y = 128$. (3 marks)

Question 4 (10 marks) The sound intensity level D (in decibels) is given by $D = 10 \log_{10} \left(\frac{I}{I_0} \right)$, where $I_0 = 10^{-12} \text{ W/m}^2$.

- (a) Machine A generates a noise level of 80 dB. Find its sound intensity I_A . (2 marks)
- (b) Machine B generates a noise level of 85 dB. Find its sound intensity I_B . (2 marks)
- (c) When Machine A and Machine B operate together, the total sound intensity is $I_{\text{total}} = I_A + I_B$.
 - (i) Calculate I_{total} in W/m^2 .
 - (ii) Find the combined sound intensity level D_{total} correct to 1 decimal place. (4 marks)
- (d) How many identical Machine A's operating simultaneously are required to produce a combined sound intensity level exceeding 95 dB? (2 marks)

Question 5 (11 marks) An investor compares two growth plans for an initial investment of \$100,000:

- **Plan A:** Value $V_1(t) = 100,000(1.06)^t$, compounded annually at 6%.
 - **Plan B:** Value $V_2(t) = 100,000e^{kt}$, continuous growth model.
- (a) If Plan B yields \$135,000 after 5 years, find the exact value of k in terms of natural logarithms. (3 marks)
 - (b) Plotting $\ln V_1$ against t yields a straight line L_1 , and plotting $\ln V_2$ against t yields a straight line L_2 . State the slope of L_1 and L_2 . (3 marks)
 - (c) Determine which plan gives a higher value after 10 years, and calculate the difference to the nearest dollar. (3 marks)
 - (d) Using the limit $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$, evaluate $\lim_{r \rightarrow 0} \frac{\ln(1+r)}{r}$ and interpret its relation to continuous compounding. (2 marks)

Question 6 (12 marks) The spread of an information virus in a computer network is modeled by the logistic function:

$$N(t) = \frac{10000}{1 + 99e^{-0.5t}}$$

where $N(t)$ is the number of infected computers after t hours.

- (a) Find the initial number of infected computers at $t = 0$. (1 mark)
- (b) Express t explicitly as a logarithmic function of N . (4 marks)
- (c) Find the time required for 5000 computers to be infected. Give your answer correct to 2 decimal places. (3 marks)

- (d) Evaluate the limit $\lim_{t \rightarrow \infty} N(t)$ and explain its practical meaning in the context of the problem. (4 marks)

Question 7 (10 marks) According to Newton's Law of Cooling, the temperature $T(t)$ (in $^{\circ}\text{C}$) of an object at time t (in minutes) satisfies:

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

where T_s is the surrounding temperature, T_0 is the initial temperature, and $k > 0$ is a constant. A hot metal bar at 100°C is placed in a room at 20°C . After 5 minutes, its temperature drops to 60°C .

- (a) Show that $k = \frac{1}{5} \ln 2$. (3 marks)
- (b) Find the temperature of the metal bar after 15 minutes. (3 marks)
- (c) Find the time required for the temperature to reach 25°C . Give your answer correct to 1 decimal place. (2 marks)
- (d) Evaluate $\lim_{t \rightarrow \infty} T(t)$. (2 marks)

Question 8 (9 marks) The radioactive decay of Carbon-14 is modeled by $N(t) = N_0 e^{-\lambda t}$, where $\lambda = \frac{\ln 2}{5730} \text{ year}^{-1}$.

- (a) Express $\ln\left(\frac{N(t)}{N_0}\right)$ as a linear function of t . (2 marks)
- (b) A fossil sample is found to have 35% of its original Carbon-14 remaining. Calculate the age of the fossil to the nearest 10 years. (4 marks)
- (c) If the measurement of remaining Carbon-14 has an error margin of $\pm 2\%$, evaluate the range of estimated age for the fossil. (3 marks)

Question 9 (10 marks) The energy E (in Joules) released by an earthquake of magnitude M satisfies the relationship $\log_{10} E = 4.8 + 1.5M$.

- (a) Express E in the form $E = A \cdot 10^{kB}$, where A and k are constants. (3 marks)
- (b) Earthquake X has magnitude 8.0 and Earthquake Y has magnitude 6.0. Show that Earthquake X releases 1000 times as much energy as Earthquake Y. (3 marks)
- (c) If the sum of energies released by two earthquakes of equal magnitude M_0 equals the energy of a single earthquake of magnitude M_1 , show that $M_1 = M_0 + \frac{\log_{10} 2}{1.5}$. (4 marks)

Question 10 (12 marks) A company compares two depreciation models for a piece of heavy machinery with an initial cost of $V_0 = \$500,000$:

- **Model 1 (Reducing Balance):** $V_1(t) = V_0(1 - r)^t$, where $r = 0.15$.
- **Model 2 (Continuous Decay):** $V_2(t) = V_0 e^{-kt}$, where $k = 0.1625$.

- (a) Express $\ln V_1(t)$ and $\ln V_2(t)$ in linear form against t . Write down their slopes. (3 marks)

- (b) Find the time t when $V_1(t) = V_2(t)$. Give your answer correct to 2 decimal places. (3 marks)
- (c) Find the value of each model after 5 years, and find the percentage difference relative to Model 1. (3 marks)
- (d) Using the expansion $\ln(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots$ for $|x| < 1$, show that $\ln(1 - r) \approx -r$ for small r , and deduce why Model 2 closely approximates Model 1 when $k \approx r$. (3 marks)

2.6 The graphs involving logarithmic functions

Definition 2.6.1: Properties of Logarithmic Graphs

For a real constant $a > 0$ and $a \neq 1$, the graph of the logarithmic function $y = \log_a x$ (defined for $x > 0$) has the following fundamental geometric characteristics:

1. Basic Features of $y = \log_a x$:

- **Domain and Range:** Domain is $(0, \infty)$ and Range is $(-\infty, \infty)$.
- **Intercepts:** The x -intercept is $(1, 0)$. There is no y -intercept since $\log_a 0$ is undefined.
- **Vertical Asymptote:** The y -axis (line $x = 0$) serves as a vertical boundary line which the curve approaches as $x \rightarrow 0^+$.
- **Monotonicity:**
 - If $a > 1$, the function is strictly increasing.
 - If $0 < a < 1$, the function is strictly decreasing.
- **Key Reference Points:** The graph always passes through $(1, 0)$, $(a, 1)$, and $(\frac{1}{a}, -1)$.

2. Standard Geometric Transformations of $y = f(x)$ where $f(x) = \log_a x$:

- **Vertical Translation:** $y = \log_a x + k$ shifts the graph vertically by $|k|$ units (upward if $k > 0$, downward if $k < 0$).
- **Horizontal Translation:** $y = \log_a(x - h)$ shifts the graph horizontally by $|h|$ units (rightward if $h > 0$, leftward if $h < 0$). The vertical asymptote becomes $x = h$.
- **Reflections:**
 - $y = -\log_a x$ reflects $y = \log_a x$ across the x -axis.
 - $y = \log_a(-x)$ (defined for $x < 0$) reflects $y = \log_a x$ across the y -axis.
 - $y = a^x$ is the reflection of $y = \log_a x$ across the line $y = x$.
- **Absolute Value Transformations:**
 - $y = |\log_a x|$ reflects the portion of the graph below the x -axis upward above the x -axis.
 - $y = \log_a |x|$ is defined for all $x \neq 0$ and is symmetric about the y -axis (even function).

2.6.1 Level 1 Exercise

1. State the domain, range, x -intercept, and equation of the vertical asymptote for each of the following logarithmic functions:

(a) $y = \log_3(x - 2)$

(b) $y = \ln(x + 4) - 1$

(c) $y = 2 - \log_5 x$

2. Describe the sequence of geometric transformations required to obtain the graph of $y = g(x)$ from the graph of $y = \ln x$:
 - (a) $g(x) = \ln(x + 3) + 5$
 - (b) $g(x) = -\ln(x - 1)$
 - (c) $g(x) = \ln(2x)$
3. Find the exact x -intercept and y -intercept (if any) of the graph of each function:
 - (a) $f(x) = \log_2(x + 8) - 3$
 - (b) $g(x) = \ln(3 - x)$
 - (c) $h(x) = 4 - 2^x$
4. The graph of $y = \log_a(x - b)$ has a vertical asymptote at $x = 3$ and passes through the point $(7, 2)$. Find the exact values of a and b .
5. Let $P(e^2, 2)$ be a point on the graph of $y = \ln x$. Write down the coordinates of the image of P under each of the following transformations:
 - (a) Reflection in the line $y = x$.
 - (b) Reflection in the x -axis followed by a vertical translation upwards by 3 units.
 - (c) Horizontal stretch by a factor of 2.
6. Determine whether each of the following points lies above, below, or on the curve $y = \log_3 x$:
 - (a) $A(9, 2)$
 - (b) $B(3, 1.5)$
 - (c) $C\left(\frac{1}{27}, -2\right)$
7. Sketch the graphs of $y = \log_2 x$ and $y = \log_4 x$ on the same coordinate plane for $x > 0$, clearly indicating:
 - (a) The common intercept.
 - (b) The relative positions of the two curves for $0 < x < 1$ and for $x > 1$.
8. Consider the function $f(x) = |\log_2 x|$.
 - (a) State the domain and range of $f(x)$.
 - (b) Find all x -values such that $f(x) = 3$.
9. The graph of $y = \ln x$ is reflected across the y -axis and then translated to the right by 5 units. Find the equation of the resulting graph and state its domain.
10. State the number of real intersection points between the two graphs $y = \ln x$ and $y = 3 - x$.

2.6.2 Level 2 Exercise

1. Consider the two functions $f(x) = \ln(x^2)$ and $g(x) = 2 \ln x$.
 - (a) State the domains of $f(x)$ and $g(x)$. Explain why the two functions are not identical.
 - (b) Sketch the graphs of $y = f(x)$ and $y = g(x)$ on separate coordinate planes.
2. The graph of $y = a \ln(x + b) + c$ passes through $(0, 0)$ and $(e - 1, 3)$, and has a vertical asymptote at $x = -1$. Find the exact values of a , b , and c .

3. Let $f(x) = \ln|x - 3|$ for $x \neq 3$.
 - (a) State the line of symmetry of the graph $y = f(x)$.
 - (b) Find the x -intercepts and y -intercept of $y = f(x)$.
 - (c) State the domain and range of $f(x)$.
4. Show algebraically that the graph of $y = \ln\left(\frac{3+x}{3-x}\right)$ is symmetric about the origin $(0, 0)$, and state its domain.
5. The graph of $y = 2^{x-1} + 3$ is reflected across the line $y = x$ to obtain the curve C .
 - (a) Find the equation of C in the form $y = f(x)$.
 - (b) State the domain, range, and vertical asymptote of C .
6. Find the exact coordinates of the intersection point(s) of the two curves $y = \log_2(x + 3)$ and $y = 1 + \log_2 x$.
7. Consider the function $h(x) = \ln(x^2 - 4x + 5)$.
 - (a) By completing the square for $x^2 - 4x + 5$, show that $h(x) \geq 0$ for all $x \in \mathbb{R}$.
 - (b) State the minimum value of $h(x)$ and the corresponding value of x .
 - (c) State the range of $h(x)$.
8. The graph of $y = |\ln x - 1|$ intersects the horizontal line $y = k$ at two distinct points. Find the range of values of k .
9. Let $f(x) = \log_3(x + 1)$ and $g(x) = \log_3(x^2 - 1)$.
 - (a) Find the domain of $(g - f)(x)$.
 - (b) Simplify $(g - f)(x)$ and sketch the graph of $y = (g - f)(x)$.
10. A curve is given by $y = \ln(e^x + 2)$.
 - (a) Show that $y > x$ for all $x \in \mathbb{R}$.
 - (b) Find the y -intercept of the curve.
 - (c) Describe the behavior of $y - x$ as $x \rightarrow \infty$.

2.6.3 Exam-type Problem

Question 1 (11 marks) Let $f(x) = \ln(x + 2)$ for $x > -2$.

- (a) Write down the domain, x -intercept, y -intercept, and the equation of the vertical asymptote of the graph $y = f(x)$. (4 marks)
- (b) The graph of $y = f(x)$ is transformed to the graph of $y = g(x)$ by a horizontal stretch by a factor of $\frac{1}{3}$, followed by a vertical translation downwards by $\ln 2$ units. Find the explicit formula for $g(x)$. (3 marks)
- (c) Find the x -intercept of $y = g(x)$. (2 marks)
- (d) Using the limit $\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$, evaluate $\lim_{x \rightarrow 0} \frac{g(x)}{x}$. (2 marks)

Question 2 (10 marks) Let C_1 be the curve $y = e^{x-1} - 2$ and C_2 be the curve $y = \ln(x + 2) + 1$.

- (a) Prove that C_1 and C_2 are reflections of each other across the line $y = x$. (4 marks)
- (b) C_1 intersects the y -axis at point P , and C_2 intersects the x -axis at point Q .
- (i) Find the exact coordinates of P and Q .
- (ii) Calculate the distance PQ . (4 marks)
- (c) State the coordinates of the midpoint of PQ and verify that it lies on the line $y = x$. (2 marks)

Question 3 (9 marks) Consider the function $f(x) = \ln(x^2 - 6x + 10)$ defined for all $x \in \mathbb{R}$.

- (a) Express $x^2 - 6x + 10$ in the form $(x - h)^2 + k$. Hence, explain why $f(x)$ is defined for all real numbers x . (3 marks)
- (b) State the line of symmetry of the graph $y = f(x)$, and find its global minimum value. (3 marks)
- (c) Evaluate the limit $\lim_{x \rightarrow 3} \frac{f(x)}{x - 3}$. (3 marks)

Question 4 (10 marks) Let $g(x) = |\ln(x - 1) - 1|$ for $x > 1$.

- (a) Find the x -intercept of $y = g(x)$. (2 marks)
- (b) Sketch the graph of $y = g(x)$, indicating the vertical asymptote and the x -intercept. (3 marks)
- (c) Determine the number of real roots of the equation $g(x) = k$ for different real values of the constant k . (5 marks)

Question 5 (12 marks) Let $f(x) = \ln(1 + x)$ and $g(x) = x - \frac{x^2}{2}$ for $x > -1$.

- (a) Using the expansion $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ for $|x| < 1$, write down the first three non-zero terms of $f(x) - g(x)$. (2 marks)
- (b) Evaluate $\lim_{x \rightarrow 0} \frac{f(x) - g(x)}{x^3}$. (4 marks)
- (c) Sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same coordinate plane for $x \in (-0.5, 1)$, illustrating their relative positions near $x = 0$. (6 marks)

Question 6 (10 marks) The graph of $y = \log_2(ax + b)$ has an x -intercept at $(3, 0)$ and passes through $(7, 3)$.

- (a) Find the exact values of a and b . (4 marks)
- (b) Find the equation of the vertical asymptote of the graph. (2 marks)
- (c) Find the exact coordinates of the intersection point of $y = \log_2(ax + b)$ and $y = 4 - \log_2(x - 1)$. (4 marks)

Question 7 (11 marks) Let $f(x) = \ln(\sqrt{x^2 + 1} - x)$ for $x \in \mathbb{R}$.

- (a) Show that $\sqrt{x^2 + 1} - x > 0$ for all $x \in \mathbb{R}$. (2 marks)
- (b) Show that $f(x)$ is an odd function. Hence, state the geometric symmetry of the graph $y = f(x)$. (4 marks)
- (c) Show that $f(x) = -\ln(\sqrt{x^2 + 1} + x)$. (2 marks)
- (d) Find $f^{-1}(x)$ and sketch the graph of $y = f^{-1}(x)$. (3 marks)

Question 8 (9 marks) Consider the family of curves $C_k : y = \ln(x + k)$, where $k > 0$ is a real parameter.

- (a) State the vertical asymptote and x -intercept of C_k in terms of k . (2 marks)
- (b) If C_k passes through the origin $(0, 0)$, find the value of k . (2 marks)
- (c) For $k = 1$, show that C_1 intersects the line $y = x$ at $x = 0$. Using the inequality $\ln(1 + x) < x$ for $x > 0$, determine whether C_1 lies above or below the line $y = x$ for $x > 0$. (5 marks)

Question 9 (10 marks) Consider the piecewise defined function:

$$f(x) = \begin{cases} \ln(1 + 2x) & \text{if } x > 0 \\ e^{2x} - 1 & \text{if } x \leq 0 \end{cases}$$

- (a) Find the x -intercept(s) and y -intercept of $y = f(x)$. (2 marks)
- (b) Using the fundamental limits $\lim_{u \rightarrow 0} \frac{\ln(1 + u)}{u} = 1$ and $\lim_{v \rightarrow 0} \frac{e^v - 1}{v} = 1$, evaluate:

$$\lim_{x \rightarrow 0^+} \frac{f(x)}{x} \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{f(x)}{x}$$

- (c) Sketch the graph of $y = f(x)$ for $x \in (-1, 2)$. (4 marks)

Question 10 (12 marks) Let $f(x) = \ln\left(\frac{e^x + 1}{e^x - 1}\right)$ for $x > 0$.

- (a) State the domain of $f(x)$. (1 mark)
- (b) Show that $f(f(x)) = x$ for all $x > 0$. What does this imply about the graph $y = f(x)$ with respect to symmetry? (4 marks)
- (c) Evaluate the limits:

$$\lim_{x \rightarrow 0^+} f(x) \quad \text{and} \quad \lim_{x \rightarrow \infty} f(x)$$

State the equations of the vertical and horizontal asymptotes of $y = f(x)$. (4 marks)

- (d) Sketch the graph of $y = f(x)$, indicating all asymptotes and the line of symmetry. (3 marks)

2.7 Linear Regression on a graph

Definition 2.7.1: Linear Regression on Transformed Graphical Scales

When a non-linear bivariate relationship between x and y is converted into a linear form $Y = mX + c$ via logarithmic transformation, the resulting least-squares regression line can be visually represented on coordinate axes where X and Y represent the transformed variables.

1. Graphical Identification of Model Types:

- **Exponential Model** ($y = Ab^x$): Plotting $Y = \log_k y$ against $X = x$ yields a straight line on a **semi-logarithmic graph**.

$$\text{Slope } m = \log_k b, \quad \text{Vertical Intercept } c = \log_k A$$

- **Power Model** ($y = Ax^n$): Plotting $Y = \log_k y$ against $X = \log_k x$ yields a straight line on a **log-log graph**.

$$\text{Slope } m = n, \quad \text{Vertical Intercept } c = \log_k A$$

2. Extracting Model Parameters directly from Transformed Graphs:

1. **Slope (m)**: Calculated using two distinct points (X_1, Y_1) and (X_2, Y_2) on the regression line:

$$m = \frac{Y_2 - Y_1}{X_2 - X_1}$$

2. **Vertical Intercept (c)**: The value of Y when $X = 0$.
3. **Centroid $(\bar{X}, \bar{Y})$** : The mean point of transformed data lies on the regression line:

$$\bar{Y} = m\bar{X} + c$$

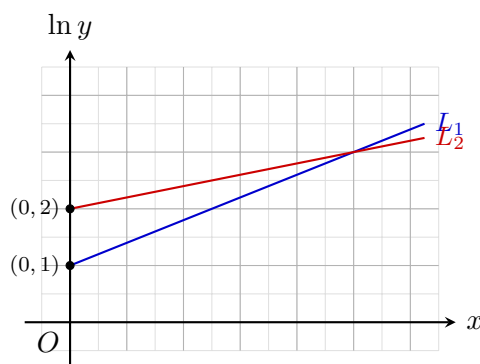
Remark

In HKDSE examination problems, graphs of linear regression lines are often plotted with transformed labels on the axes (e.g., the vertical axis labeled as $\ln y$ or $\log_{10} y$, and the horizontal axis labeled as x or $\ln x$). Students must carefully read axis labels before calculating slopes or intercepts.

2.7.1 Level 1 Exercise

1. The graph of $\log_{10} y$ against x shows a straight line passing through $(0, 1)$ and $(2, 5)$.
 - (a) Find the slope and vertical intercept of the line.
 - (b) Express y in the form $y = A \cdot b^x$.
2. The graph of $\ln y$ against $\ln x$ shows a straight line with a slope of 3 and a vertical intercept of 2. Express y in the form $y = Ax^n$, leaving A in terms of e .
3. A straight line regression graph of $\ln y$ against x passes through the centroid $(\bar{x}, \overline{\ln y}) = (3, 4)$ and has a slope of 0.5.
 - (a) Write down the equation of the regression line in terms of $\ln y$ and x .
 - (b) Find the value of y when $x = 5$.

4. The graph of $\log_2 y$ against x has an x -intercept at $(3, 0)$ and a vertical intercept at $(0, -6)$. Express y in the form $y = A \cdot b^x$.
5. The graph of $\ln y$ against $\ln x$ has an x -intercept at $(\ln 4, 0)$ and a slope of -2 . Express y explicitly as a function of x .
6. Given a scatter diagram of $\log_{10} y$ against x , the line of best fit passes through $(1, 0.3)$ and $(4, 1.2)$. Estimate the value of y when $x = 3$.
7. A non-linear relation $y = 5(2^{3x})$ is plotted as a straight line graph of $\log_2 y$ against x .
 - (a) State the theoretical slope of this line.
 - (b) State the vertical intercept of this line.
8. Two linear regression lines L_1 and L_2 are plotted on the same $\ln y$ against x plane as shown below:

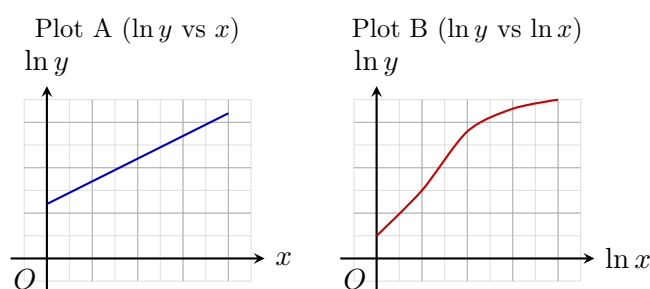


- Line L_1 has y -intercept 1 and slope 0.4. Line L_2 has y -intercept 2 and slope 0.2. Find the exact coordinates $(x, \ln y)$ of the intersection point of L_1 and L_2 , and find the corresponding value of y .
9. A straight line on a graph of $\log_3 y$ against $\log_3 x$ passes through $(1, 2)$ and $(3, 8)$. Find the value of y when $x = 9$.
 10. A set of data points (x, y) produces a straight line when $\ln y$ is plotted against x , but a curved line when y is plotted against x . State whether the underlying relationship is exponential or power law.

2.7.2 Level 2 Exercise

1. The graph of $\log_4 y$ against $\log_2 x$ is a straight line passing through $(2, 3)$ and $(6, 5)$.
 - (a) Find the equation of the straight line in the form $\log_4 y = m \log_2 x + c$.
 - (b) Express y in the form $y = Ax^k$, where A and k are exact constants.
2. The graph of $\frac{1}{\ln y}$ against x is a straight line passing through $(1, 0.5)$ and $(3, 1.5)$. Express y explicitly in terms of x .
3. The graph of $\ln(y - 3)$ against x is a straight line with a slope of -0.2 and a vertical intercept of $\ln 5$.
 - (a) Express y in terms of x .
 - (b) State the horizontal asymptote of the graph of y against x .

4. Data pairs (x, y) are transformed such that the graph of $\log_{10} y$ against x^2 is a straight line passing through $(1, 0.5)$ and $(4, 2.0)$. Express y in the form $y = A \cdot b^{x^2}$.
5. The graph of $\ln y$ against $\ln(x + 1)$ is a straight line with slope 1.5 passing through $(0, \ln 2)$. Express y in terms of x .
6. For a dataset (x, y) , the mean of $\ln x$ is $\overline{\ln x} = 2.0$ and the mean of $\ln y$ is $\overline{\ln y} = 5.0$. The regression line of $\ln y$ on $\ln x$ shown on a graph passes through $(0, 1.0)$. Express y in the form $y = Ax^n$.
7. A regression graph plots $\log_3 \left(\frac{y}{x}\right)$ against x . The line has a slope of 2 and a vertical intercept of 1. Express y explicitly in terms of x .
8. An analyst compares two transformed scatter plots for the same dataset (x, y) :

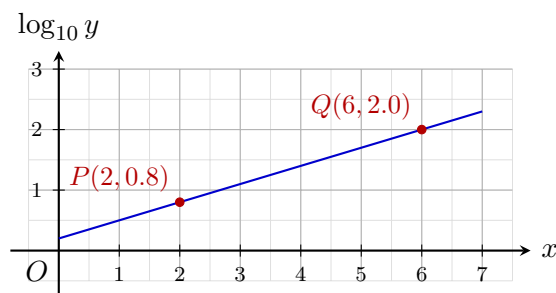


Plot A shows points clustered tightly around a line with slope 0.5 and intercept 1.2. Plot B shows a clear non-linear curve. Write down the most appropriate equation for y in terms of x based on Plot A.

9. The regression line of $\ln y$ against x on a graph is $Y = -0.4X + 2.8$, where $Y = \ln y$ and $X = x$.
 - (a) Express x in terms of y .
 - (b) Find x when $y = 2$, correct to 3 significant figures.
10. The graph of $\ln(y - C)$ against x is a straight line passing through $(2, \ln 8)$ and $(5, \ln 64)$, where $C = 10$ is a known constant. Express y as a function of x .

2.7.3 Exam-type Problem

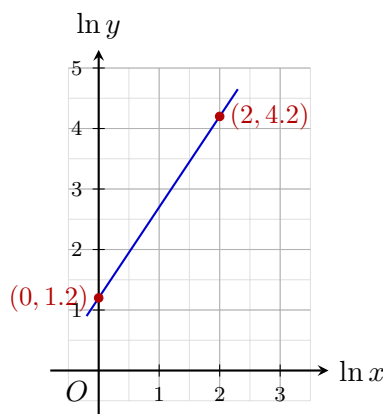
Question 1 (11 marks) The relationship between x and y is modeled by $y = Ab^x$, where A and b are positive constants. The graph of $\log_{10} y$ against x is a straight line shown below, passing through $P(2, 0.8)$ and $Q(6, 2.0)$.



- (a) Find the slope and vertical intercept of the straight line. (2 marks)

- (b) Find the exact value of b and the value of A correct to 3 significant figures. (4 marks)
- (c) Estimate the value of y when $x = 10$. (2 marks)
- (d) Using the limit $\lim_{h \rightarrow 0} \frac{10^h - 1}{h} = \ln 10$, evaluate $\lim_{x \rightarrow 0} \frac{y(x) - A}{xA}$. (3 marks)

Question 2 (10 marks) The graph of $\ln y$ against $\ln x$ is a straight line shown below, passing through $(0, 1.2)$ and $(2, 4.2)$.



- (a) Express $\ln y$ as a linear function of $\ln x$. (2 marks)
- (b) Express y in the form $y = Ax^n$, giving A correct to 3 significant figures and n as an exact decimal. (4 marks)
- (c) Solve the inequality $y > 1000$ for $x > 0$. (2 marks)
- (d) Using the limit $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$, evaluate $\lim_{x \rightarrow 1} \frac{\ln y - 1.2}{x - 1}$. (2 marks)

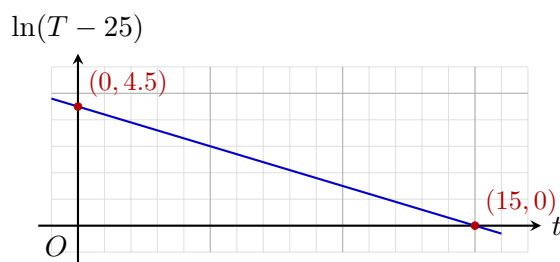
Question 3 (10 marks) A biological population P (in thousands) at time t (in years) is recorded. The scatter plot of $Y = \ln P$ against t yields a linear regression line. The centroid of the transformed data on the graph is $(\bar{t}, \bar{Y}) = (4, 2.5)$, and the slope of the regression line is 0.3.

- (a) Write down the equation of the regression line of Y on t . (2 marks)
- (b) Express P in the form $P(t) = P_0 e^{kt}$, finding the exact value of k and P_0 . (3 marks)
- (c) Find the time required for the population to double its initial value P_0 . (2 marks)
- (d) Using the limit $\lim_{h \rightarrow 0} \frac{e^{kh} - 1}{h} = k$, evaluate:

$$\lim_{h \rightarrow 0} \frac{P(h) - P(0)}{hP_0}$$

(3 marks)

Question 4 (9 marks) The cooling of a liquid is modeled by Newton's Law of Cooling $T(t) = 25 + Ae^{-kt}$. The graph of $\ln(T - 25)$ against t is a straight line cutting the vertical axis at $(0, 4.5)$ and the horizontal axis at $(15, 0)$ as shown below.



- Find the slope and vertical intercept of the straight line on the graph. (2 marks)
- Express $T(t)$ explicitly in terms of t , giving A correct to 1 decimal place and k correct to 3 decimal places. (3 marks)
- Find the temperature of the liquid at $t = 10$. (2 marks)
- Evaluate $\lim_{t \rightarrow \infty} T(t)$ and state its physical meaning. (2 marks)

Question 5 (11 marks) An economic model relates production output Y to capital input X . The graph of $\log_2 Y$ against $\log_4 X$ is a straight line passing through $(1, 2)$ and $(3, 5)$.

- Express $\log_2 Y$ as a linear function of $\log_4 X$. (3 marks)
- Express Y in the form $Y = AX^p$, where A and p are exact constants. (5 marks)
- Find the value of Y when $X = 16$. (3 marks)

Question 6 (10 marks) An Arrhenius plot displays the rate constant k of a chemical reaction against temperature T (in Kelvin) by plotting $\ln k$ against $\frac{1}{T}$. The graph shows a straight line passing through $(0.002, -1.0)$ and $(0.003, -4.0)$.

- Find the slope and vertical intercept of the regression line on the graph. (3 marks)
- Express k in the form $k = Ae^{-E/T}$, giving A and E in exact form or correct to 3 significant figures. (4 marks)
- Calculate k when $T = 400$ K. (3 marks)

Question 7 (10 marks) The graph of $Y = \ln\left(\frac{y}{100}\right)$ against x is a straight line passing through $(0, -0.5)$ and $(5, -1.5)$.

- Express Y as a linear function of x . (2 marks)
- Express y in the form $y = Ae^{-kx}$, where A and k are positive constants. (3 marks)
- Solve $y(x) = 50$ for x . (2 marks)
- Using the limit $\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = 1$, evaluate:

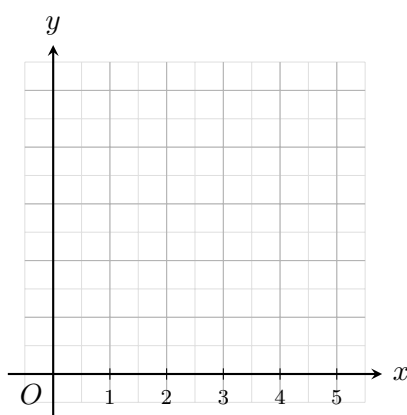
$$\lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x}$$

(3 marks)

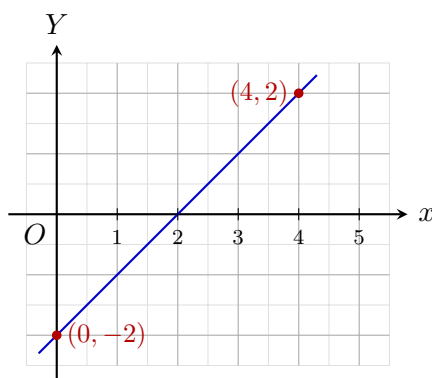
Question 8 (12 marks) Two candidate regression models are fitted to a dataset (x, y) and plotted on transformed axes:

- **Model A** ($\ln y$ against x): Straight line passing through $(1, 2.0)$ and $(4, 3.5)$.
- **Model B** ($\ln y$ against $\ln x$): Straight line passing through $(0, 1.5)$ and $(2, 3.5)$.

- Express y explicitly in terms of x for Model A. (3 marks)
- Express y explicitly in terms of x for Model B. (3 marks)
- Calculate the predicted value of y at $x = 2$ for both models, and find their absolute difference. (3 marks)
- On the coordinate grid provided below, sketch the curves of y against x for Model A and Model B for $x \in (0, 3]$. (3 marks)

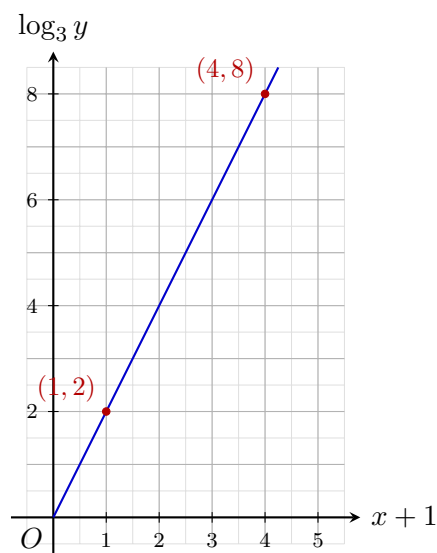


Question 9 (9 marks) The graph of $Y = \ln \left(\frac{y}{100-y} \right)$ against x is a straight line shown below, passing through $(0, -2)$ and $(4, 2)$.



- Find the equation of the line in terms of Y and x . (2 marks)
- Show that y can be written in the logistic form $y = \frac{100}{1+e^{-(x-2)}}$. (4 marks)
- Evaluate $\lim_{x \rightarrow \infty} y(x)$ and state its significance. (3 marks)

Question 10 (11 marks) Data points (x, y) are plotted on a graph of $\log_3 y$ against $(x + 1)$ shown below. The regression line passes through $(1, 2)$ and $(4, 8)$.



- (a) Find the equation of the regression line in terms of $\log_3 y$ and x . (3 marks)
- (b) Express y explicitly as a function of x . (3 marks)
- (c) Solve the inequality $y \geq 81$ for x . (2 marks)
- (d) Using the limit $\lim_{u \rightarrow 0} \frac{\log_3(1+u)}{u} = \frac{1}{\ln 3}$, evaluate:

$$\lim_{x \rightarrow 0} \frac{\log_3(y(x)) - 2}{x}$$

(3 marks)

Solutions to Exercises

Solutions to Linear Relation Between Variables

Level 1 Exercise

1. (a) $2x - 3y + 12 = 0 \implies y = \frac{2}{3}x + 4$.
Slope: $\frac{2}{3}$; y -intercept: 4;
 x -intercept ($y = 0$): -6.
- (b) $4x + 5y - 20 = 0 \implies y = -\frac{4}{5}x + 4$.
Slope: $-\frac{4}{5}$; y -intercept: 4;
 x -intercept ($y = 0$): 5.
- (c) $y = -3x + 7$.
Slope: -3; y -intercept: 7;
 x -intercept ($y = 0$): $\frac{7}{3}$.
2. (a) $m = \frac{10-4}{3-1} = 3$.
 $y - 4 = 3(x - 1) \implies y = 3x + 1$.
- (b) $m = \frac{-1-5}{4-(-2)} = -1$.
 $y - 5 = -(x + 2) \implies y = -x + 3$.
- (c) $m = \frac{2-(-3)}{5-0} = 1$.
 $y = x - 3$.
3. $y - 7 = -3(x - 2) \implies y = -3x + 13$.
4. (a) Rates of change:
 $\frac{5-3}{2-1} = 2$, $\frac{9-5}{4-2} = 2$, $\frac{15-9}{7-4} = 2$.
Since $\frac{\Delta y}{\Delta x}$ is constant ($= 2$), it represents a linear function.
- (b) Rates of change:
 $\frac{2-1}{1-0} = 1$, $\frac{5-2}{3-1} = 1.5$.
Since $\frac{\Delta y}{\Delta x}$ is not constant, it does not represent a linear function.
5. Slope $m = \frac{11-3}{5-a} = 2 \implies 8 = 2(5 - a) \implies 4 = 5 - a \implies a = 1$.
6. (a) Parallel slope $m = 3$.
 $y - 4 = 3(x + 1) \implies 3x - y + 7 = 0$.
- (b) Perpendicular to $2x - y + 4 = 0$ (slope 2) $\implies m = -\frac{1}{2}$.
 $y + 3 = -\frac{1}{2}(x - 2) \implies x + 2y + 4 = 0$.
7. Substitute $y = 2x + 3$ into $3x - 2y + 1 = 0$:
 $3x - 2(2x + 3) + 1 = 0 \implies -x - 5 = 0 \implies x = -5$.
Then $y = 2(-5) + 3 = -7$.
Point of intersection: $(-5, -7)$.
8. $m = \frac{-2-0}{0-4} = \frac{1}{2}$, $c = -2 \implies y = \frac{1}{2}x - 2$.
9. (a) $f(-3) = 5 - 2(-3) = 11$.
 $f(4) = 5 - 2(4) = -3$.
- (b) $y = 5 - 2x \implies 2x = 5 - y \implies x = -\frac{1}{2}y + \frac{5}{2}$.

Level 2 Exercise

1. Let $X = x^2$. Points are $(1, 4)$ and $(9, 10)$.
Slope $m = \frac{10-4}{9-1} = \frac{3}{4}$.
 $y - 4 = \frac{3}{4}(X - 1) \implies y = \frac{3}{4}x^2 + \frac{13}{4}$.

2. $\frac{1}{y} = 3x + 2 \implies y = \frac{1}{3x+2}$.
3. (a) $3(2) + k(3) - 12 = 0 \implies 3k = 6 \implies k = 2$.
 (b) Line $3x + 2y - 12 = 0$.
 x -intercept ($y = 0$): 4.
 y -intercept ($x = 0$): 6.
 Area $= \frac{1}{2} \times 4 \times 6 = 12$.
4. Slope $AC = \frac{14-2}{5-(-1)} = 2$.
 Slope $AB = \frac{k-2}{2-(-1)} = \frac{k-2}{3}$.
 $\frac{k-2}{3} = 2 \implies k = 8$.
5. (a) Passes $(a, 0)$ and $(0, b)$: slope $m = -\frac{b}{a}$.
 $y - b = -\frac{b}{a}x \implies \frac{b}{a}x + y = b \implies \frac{x}{a} + \frac{y}{b} = 1$.
 (b) $\frac{2}{2b} + \frac{3}{b} = 1 \implies \frac{4}{b} = 1 \implies b = 4$.
 Then $a = 2(4) = 8$.
6. Equation: $y^2 = 5\sqrt{x} - 10$.
 For $y \in \mathbb{R}$, $y^2 \geq 0 \implies 5\sqrt{x} - 10 \geq 0 \implies \sqrt{x} \geq 2 \implies x \geq 4$.
7. (a) L_1 passes $(0, 0)$ and $(3, 7) \implies c_1 = 0, m_1 = \frac{7}{3}$.
 $L_1 \perp L_2 \implies m_2 = -\frac{3}{7}$.
 L_2 passes $(3, 7) \implies 7 = -\frac{3}{7}(3) + c_2 \implies c_2 = \frac{58}{7}$.
 (b) x -intercept of L_2 : set $y = 0 \implies 0 = -\frac{3}{7}x + \frac{58}{7} \implies x = \frac{58}{3}$.
8. (a) Slope $m = \frac{75-45}{8-2} = 5$.
 $W - 45 = 5(t - 2) \implies W = 5t + 35$.
 (b) $100 = 5t + 35 \implies 5t = 65 \implies t = 13$.
9. Slope of L_1 : $-\frac{k}{k+2}$. Slope of L_2 : $\frac{3}{2}$.
 Perpendicular $\implies \left(-\frac{k}{k+2}\right)\left(\frac{3}{2}\right) = -1 \implies 3k = 2k + 4 \implies k = 4$.
10. Let $f(x) = ax + b$.
 $f(f(x)) = a(ax + b) + b = a^2x + ab + b = 9x + 8$.
 $a^2 = 9 \implies a = 3$ or $a = -3$.
 If $a = 3$: $3b + b = 8 \implies b = 2 \implies f(x) = 3x + 2$.
 If $a = -3$: $-3b + b = 8 \implies b = -4 \implies f(x) = -3x - 4$.

Exam-type Problem Solutions

Question 1

- (a) Slope $m = \frac{13-5}{6-2} = 2$.
 $y - 5 = 2(x - 2) \implies y = 2x + 1$.
- (b) P (x -int): set $y = 0 \implies x = -\frac{1}{2}$, so $P\left(-\frac{1}{2}, 0\right)$.
 Q (y -int): set $x = 0 \implies y = 1$, so $Q(0, 1)$.
- (c) Area of $\triangle OPQ = \frac{1}{2} \times \left|-\frac{1}{2}\right| \times 1 = \frac{1}{4}$.

Question 2

- (a) $(y - 2x) = 3(x + 1) - 4$.

(b) $y - 2x = 3x - 1 \implies y = 5x - 1.$

(c) x -intercept: set $y = 0 \implies x = \frac{1}{5}.$
 y -intercept: set $x = 0 \implies y = -1.$

Question 3

(a) Points $(0, 32)$ and $(100, 212).$

Slope $m = \frac{212-32}{100-0} = \frac{9}{5}.$

$F = \frac{9}{5}C + 32.$

(b) Set $F = C = x:$

$x = \frac{9}{5}x + 32 \implies -\frac{4}{5}x = 32 \implies x = -40.$

The temperature is $-40^\circ.$

(c) $F - 32 = \frac{9}{5}C \implies C(F) = \frac{5}{9}(F - 32).$

Question 4

(a) Rearrange $L_k:$

$k(2x + y - 5) + (x - y - 1) = 0.$

Solving the system:

$$2x + y - 5 = 0$$

$$x - y - 1 = 0$$

Adding the equations gives $3x - 6 = 0 \implies x = 2.$

Substituting $x = 2$ gives $2 - y - 1 = 0 \implies y = 1.$

Thus, $P = (2, 1).$

(b) Slope of $4x + y - 3 = 0$ is $-4.$

Slope of $L_k: -\frac{2k+1}{k-1}.$

$-\frac{2k+1}{k-1} = -4 \implies 2k + 1 = 4k - 4 \implies k = \frac{5}{2}.$

Question 5

(a) Let $Y = \frac{1}{y}$ and $X = \frac{1}{x}.$

Slope $m = \frac{9-3}{4-1} = 2.$

$Y - 3 = 2(X - 1) \implies \frac{1}{y} = \frac{2}{x} + 1.$

(b) $\frac{1}{y} = \frac{x+2}{x} \implies y = \frac{x}{x+2}.$

(c) As $x \rightarrow \infty, y = \frac{x}{x+2} = \frac{1}{1+\frac{2}{x}} \rightarrow 1.$

Question 6

(a) $100m + c = 5000$ and $300m + c = 11000.$

Subtracting gives $200m = 6000 \implies m = 30.$

Substituting gives $c = 2000.$

$m = 30$: variable cost per unit (\$).

$c = 2000$: fixed cost (\$).

(b) Revenue: $R(x) = 45x.$

Profit: $P(x) = 45x - (30x + 2000) = 15x - 2000.$

- (c) Set $P(x_0) = 0 \implies 15x_0 = 2000 \implies x_0 = \frac{400}{3} \approx 133.33$.
Minimum units to avoid loss: 134.

Question 7

- (a) $(f \circ g)(x) = a(cx + d) + b = (ac)x + (ad + b)$.
Since $a, c \neq 0$, $ac \neq 0$, so it is a linear function with slope ac .
- (b) $(g \circ f)(x) = c(ax + b) + d = (ac)x + (bc + d)$.
 $f \circ g = g \circ f \iff ad + b = bc + d \iff b(c - 1) = d(a - 1)$.
- (c) $(f \circ g)(x) = 2(5x - 1) + 3 = 10x + 1$.
 $\lim_{x \rightarrow \infty} \frac{10x+1}{x} = \lim_{x \rightarrow \infty} \left(10 + \frac{1}{x}\right) = 10$.

Solutions to Linear Regression

Level 1 Exercise Solutions

1. Since the regression line passes through the centroid $(\bar{x}, \bar{y}) = (4, 11)$:

$$11 = 1.8(4) + c \implies 11 = 7.2 + c \implies c = 3.8$$

2. (a) For $x = 4$:

$$\hat{y} = -2.5(4) + 18.4 = -10 + 18.4 = 8.4$$

- (b) For $x = 6.2$:

$$\hat{y} = -2.5(6.2) + 18.4 = -15.5 + 18.4 = 2.9$$

3. (a) Mean values:

$$\bar{x} = \frac{\sum x}{n} = \frac{30}{6} = 5, \quad \bar{y} = \frac{\sum y}{n} = \frac{48}{6} = 8$$

- (b) Slope b and intercept a :

$$\begin{aligned} b &= \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2} \\ &= \frac{6(282) - (30)(48)}{6(190) - 30^2} = \frac{1692 - 1440}{1140 - 900} = \frac{252}{240} = 1.05 \\ a &= \bar{y} - b\bar{x} = 8 - 1.05(5) = 8 - 5.25 = 2.75 \end{aligned}$$

Regression line equation: $y = 2.75 + 1.05x$.

4. (a) The slope 0.75 means that for each additional hour of study, a student's expected exam score increases by 0.75 marks.
- (b) No, it is not appropriate. The regression line of y on x is used exclusively to predict the dependent variable y from x . Predicting x given y would require the regression line of x on y .
5. Since $(\bar{x}, \bar{y})$ lies on the regression line:

$$\bar{y} = 3.4\bar{x} - 5.1 = 3.4(5) - 5.1 = 17 - 5.1 = 11.9$$

6. (a) For $n = 4$:

$$\begin{aligned}\sum x &= 1 + 2 + 3 + 4 = 10 \implies \bar{x} = \frac{10}{4} = 2.5 \\ \sum y &= 3 + 5 + 6 + 8 = 22 \implies \bar{y} = \frac{22}{4} = 5.5\end{aligned}$$

- (b) Summary terms:

$$\begin{aligned}\sum x^2 &= 1^2 + 2^2 + 3^2 + 4^2 = 30 \\ \sum xy &= 1(3) + 2(5) + 3(6) + 4(8) = 63 \\ b &= \frac{4(63) - (10)(22)}{4(30) - 10^2} = \frac{252 - 220}{120 - 100} = \frac{32}{20} = 1.6 \\ a &= 5.5 - 1.6(2.5) = 1.5\end{aligned}$$

Regression line equation: $y = 1.5 + 1.6x$.

Level 2 Exercise Solutions

1. Given $Y = \ln y$ and $X = x$, $Y = 0.45X + 1.30$:

$$\ln y = 0.45x + 1.30 \implies y = e^{0.45x+1.30} = e^{1.30}e^{0.45x}$$

Since $e^{1.30} \approx 3.669 \approx 3.67$:

$$y = 3.67e^{0.45x} \quad (\text{where } A \approx 3.67, k = 0.45)$$

2. (a) No. Since the added point $(10, 25)$ equals $(\bar{x}, \bar{y})$, the new mean values remain:

$$\bar{x}_{\text{new}} = \frac{n(10) + 10}{n + 1} = 10, \quad \bar{y}_{\text{new}} = \frac{n(25) + 25}{n + 1} = 25$$

- (b) No. The new point lies directly on the regression line $y = 2x + 5$ (since $2(10) + 5 = 25$), so its residual is zero and it does not change the least-squares slope.

3. From $\ln y = \ln a + x \ln b$:

$$\text{Slope: } \ln b = 0.5 \implies b = e^{0.5}$$

$$\text{Intercept: } \overline{\ln y} = \ln a + \bar{x} \ln b \implies 2.4 = \ln a + 3.0(0.5)$$

$$\ln a = 2.4 - 1.5 = 0.9 \implies a = e^{0.9}$$

Exact values: $a = e^{0.9}$, $b = e^{0.5}$.

4. Substituting $X = \frac{1}{x}$ and $Y = \frac{1}{y}$:

$$\frac{1}{y} = 2.5 \left(\frac{1}{x} \right) + 0.4 = \frac{2.5 + 0.4x}{x} \implies y = \frac{x}{0.4x + 2.5}$$

Multiplying numerator and denominator by 10:

$$y = \frac{10x}{4x + 25}$$

5. Slope formula:

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{48}{40} = 1.2$$

6. For $(1, 2), (2, 4), (3, 5)$:

$$\bar{x} = 2, \quad \bar{y} = \frac{11}{3}, \quad \sum x^2 = 14, \quad \sum xy = 25$$

$$b = \frac{3(25) - (6)(11)}{3(14) - 6^2} = \frac{75 - 66}{42 - 36} = \frac{9}{6} = 1.5$$

$$a = \frac{11}{3} - 1.5(2) = \frac{11}{3} - 3 = \frac{2}{3} \implies \hat{y} = \frac{2}{3} + 1.5x$$

Residuals $e_i = y_i - \hat{y}_i$:

$$e_1 = 2 - \left(\frac{2}{3} + 1.5(1)\right) = 2 - \frac{13}{6} = -\frac{1}{6}$$

$$e_2 = 4 - \left(\frac{2}{3} + 1.5(2)\right) = 4 - \frac{11}{3} = \frac{1}{3}$$

$$e_3 = 5 - \left(\frac{2}{3} + 1.5(3)\right) = 5 - \frac{31}{6} = -\frac{1}{6}$$

$$\sum e_i = -\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0 \quad (\text{verified})$$

7. (a) Substitute $u = x - 100$ and $v = y - 500$:

$$y - 500 = 3.2(x - 100) + 15 \implies y = 3.2x - 320 + 15 + 500 = 3.2x + 195$$

(b) For $x = 110$: $y = 3.2(110) + 195 = 352 + 195 = 547$.

8. Taking natural logarithms on both sides of $y = Ax^b$:

$$\ln y = \ln(Ax^b) = \ln A + b \ln x$$

Letting $Y = \ln y$ and $X = \ln x$, the equation becomes $Y = bX + \ln A$, which is a linear relation between Y and X with slope b . Thus, the slope of the regression line of $\ln y$ on $\ln x$ equals b .

Exam-type Problem Solutions

Question 1

(a) $\ln P = 0.04t + 1.25$.

(b) $P = e^{1.25+0.04t} = e^{1.25}e^{0.04t}$. Comparing with $P = Ae^{kt}$:

$$k = 0.04, \quad A = e^{1.25} \text{ (exact)}$$

(c) When $t = 10$:

$$\ln P = 0.04(10) + 1.25 = 1.65 \implies P = e^{1.65} \approx 5.21 \text{ thousand}$$

(d) $P(h) = Ae^{kh} = e^{1.25}e^{0.04h}$ and $P(0) = e^{1.25}$:

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{P(h) - P(0)}{h} &= \lim_{h \rightarrow 0} \frac{e^{1.25}(e^{0.04h} - 1)}{h} \\ &= 0.04e^{1.25} \lim_{0.04h \rightarrow 0} \frac{e^{0.04h} - 1}{0.04h} = 0.04e^{1.25}\end{aligned}$$

Question 2

(a) Intercept $a = \bar{Y} - b\bar{X} = 4.2 - (-1.4)(1.5) = 4.2 + 2.1 = 6.3$. Equation: $Y = 6.3 - 1.4X$.

(b) From $\ln P = 6.3 - 1.4 \ln V$:

$$\gamma = 1.4 \text{ (exact)}, \quad \ln C = 6.3 \implies C = e^{6.3} \approx 545 \text{ (3 s.f.)}$$

(c) For $V = 5.0$, $X = \ln 5.0 \approx 1.6094$:

$$Y = 6.3 - 1.4(1.6094) \approx 4.0468 \implies P = e^{4.0468} \approx 57.2$$

Question 3

(a) Mean point:

$$\bar{x} = \frac{1650}{10} = 165, \quad \bar{y} = \frac{580}{10} = 58$$

(b) Slope b :

$$\begin{aligned}b &= \frac{10(96400) - (1650)(580)}{10(273000) - 1650^2} = \frac{964000 - 957000}{2730000 - 2722500} = \frac{7000}{7500} = \frac{14}{15} \\ a &= \bar{y} - b\bar{x} = 58 - \frac{14}{15}(165) = 58 - 154 = -96\end{aligned}$$

Regression line: $y = -96 + \frac{14}{15}x$.

Question 4

(a) The **Exponential Model** is more appropriate because its correlation coefficient $|r| = 0.98$ is much closer to 1 than that of the Linear Model ($r = 0.72$), indicating a stronger linear relationship on transformed variables.

(b) From $\ln y = 0.35x + 0.80$:

$$y = e^{0.80+0.35x} = e^{0.80}(e^{0.35})^x \implies y = 2.23(1.42)^x$$

(where $A = e^{0.80} \approx 2.23$ and $b = e^{0.35} \approx 1.42$).

(c) For $x = 4$:

- Linear Model: $y_1 = 2.1(4) + 3.4 = 11.8$
- Exponential Model: $\ln y_2 = 0.35(4) + 0.80 = 2.20 \implies y_2 = e^{2.20} \approx 9.025$

Absolute difference: $|11.8 - 9.025| = 2.775 \approx 2.78$.

Question 5

- (a) Mean transformed values:

$$\bar{u} = 3\bar{x} - 2, \quad \bar{v} = 2\bar{y} + 5$$

- (b) Deviations $u_i - \bar{u} = 3(x_i - \bar{x})$ and $v_i - \bar{v} = 2(y_i - \bar{y})$:

$$b' = \frac{\sum 3(x_i - \bar{x}) \cdot 2(y_i - \bar{y})}{\sum 9(x_i - \bar{x})^2} = \frac{6 \sum (x_i - \bar{x})(y_i - \bar{y})}{9 \sum (x_i - \bar{x})^2} = \frac{2}{3}b$$

- (c) For $y = 1.5x + 4.0$, $b = 1.5 \implies b' = \frac{2}{3}(1.5) = 1.0$. Since $\bar{y} = 1.5\bar{x} + 4.0$:

$$\bar{v} = 2(1.5\bar{x} + 4.0) + 5 = 3\bar{x} + 13$$

Using $\bar{v} = a' + b'\bar{u}$:

$$3\bar{x} + 13 = a' + 1.0(3\bar{x} - 2) \implies a' = 15$$

Transformed regression equation: $v = u + 15$.

Question 6

- (a) Line passes $(0, 5)$ and $(2, 11)$:

$$a = 5, \quad b = \frac{11 - 5}{2 - 0} = 3$$

- (b) $y(x) = 5 + 3 \ln x$.

- (c) For $x = e^3$: $y = 5 + 3 \ln(e^3) = 5 + 3(3) = 14$.

- (d) Evaluating limit:

$$\lim_{x \rightarrow 1} \frac{y(x) - a}{x - 1} = \lim_{x \rightarrow 1} \frac{3 \ln x}{x - 1} = 3 \lim_{t \rightarrow 0} \frac{\ln(1 + t)}{t} = 3(1) = 3$$

Question 7

- (a) Taking natural logarithms of $T(t) - 20 = Ae^{-kt}$:

$$\ln(T - 20) = \ln A - kt = -0.05t + 3.80$$

- (b) Comparing coefficients:

$$k = 0.05 \text{ (exact)}, \quad \ln A = 3.80 \implies A = e^{3.80} \approx 44.7 \text{ (3 s.f.)}$$

- (c) At $t = 0$: $Y = 3.80 \implies T(0) - 20 = e^{3.80} \implies T(0) = 20 + 44.7 = 64.7^\circ\text{C}$.

- (d) As $t \rightarrow \infty$, $e^{-0.05t} \rightarrow 0$:

$$\lim_{t \rightarrow \infty} T(t) = 20^\circ\text{C}$$

Physical meaning: The temperature of the object approaches the surrounding ambient temperature (20°C) over time.

Question 8

- (a) From $\frac{1}{y} = a + b\left(\frac{1}{x}\right) = 0.2 + 0.5\left(\frac{1}{x}\right)$:

$$a = 0.2, \quad b = 0.5$$

(b) Simplifying:

$$\frac{1}{y} = \frac{0.2x + 0.5}{x} \implies y = \frac{x}{0.2x + 0.5} = \frac{10x}{2x + 5}$$

(c) Evaluating limit:

$$\lim_{x \rightarrow \infty} y(x) = \lim_{x \rightarrow \infty} \frac{10x}{2x + 5} = \frac{10}{2} = 5$$

Physical meaning: As x grows arbitrarily large, y approaches a constant horizontal saturation limit of 5.

Question 9

(a) From $Y = -0.2x + 1.6$:

$$k = 0.2, \quad \ln A = 1.6 \implies A = e^{1.6} \approx 4.95 \text{ (3 s.f.)}$$

(b) For $x = 5$: $Y = -0.2(5) + 1.6 = 0.6$.

$$\frac{100}{y} - 1 = e^{0.6} \implies \frac{100}{y} = 1 + e^{0.6} \approx 2.8221 \implies y \approx 35.4$$

(c) As $x \rightarrow \infty$, $e^{-0.2x} \rightarrow 0$:

$$\lim_{x \rightarrow \infty} y(x) = \lim_{x \rightarrow \infty} \frac{100}{1 + Ae^{-0.2x}} = \frac{100}{1 + 0} = 100$$

Question 10

(a) Setting $\frac{\partial S}{\partial a} = 0$:

$$\begin{aligned} \frac{\partial S}{\partial a} &= -2 \sum_{i=1}^n (y_i - a - bx_i) = 0 \\ \implies \sum y_i - na - b \sum x_i &= 0 \end{aligned}$$

Dividing both sides by n :

$$\frac{1}{n} \sum y_i - a - b \left(\frac{1}{n} \sum x_i \right) = 0 \implies \bar{y} = a + b\bar{x}$$

(b) Using $a = \bar{y} - b\bar{x} = 7 - 3(2) = 1$:

$$\hat{y} = 1 + 3x$$

(c) Since $\hat{y}(x) - a = bx$:

$$\lim_{x \rightarrow 0} \frac{\ln(1 + \hat{y}(x) - a)}{\hat{y}(x) - a} = \lim_{x \rightarrow 0} \frac{\ln(1 + bx)}{bx} = \lim_{t \rightarrow 0} \frac{\ln(1 + t)}{t} = 1$$

Solutions: Logarithmic Function as Inverse Function of Exponential Function

Level 1 Exercise

1. (a) $3^4 = 81 \iff \log_3 81 = 4$
- (b) $e^{2k} = 5 \iff \ln 5 = 2k$

- (c) $10^{-3} = 0.001 \iff \log_{10} 0.001 = -3$
2. (a) $\log_2 64 = 6 \iff 2^6 = 64$
 (b) $\ln(x+1) = 3 \iff x+1 = e^3$
 (c) $\log_{10} y = -2 \iff y = 10^{-2} = 0.01$
3. (a) $\log_5 125 = \log_5(5^3) = 3$
 (b) $\ln(e^{-7}) = -7$
 (c) $10^{\log_{10} 42} = 42$
 (d) $e^{\ln 12 - \ln 4} = e^{\ln(12/4)} = e^{\ln 3} = 3$
4. (a) $x - 4 > 0 \implies x > 4$.
Domain: $(4, \infty)$, **Range:** $(-\infty, \infty)$
 (b) $2x + 6 > 0 \implies x > -3$.
Domain: $(-3, \infty)$, **Range:** $(-\infty, \infty)$
 (c) $x > 0$.
Domain: $(0, \infty)$, **Range:** $(-\infty, \infty)$
5. (a) $\log_7(7^{3x+1}) = 3x + 1$
 (b) $e^{\ln(x^2+1)} = x^2 + 1$
 (c) $10^{\log_{10}(5x)} = 5x$
6. (a) $y = 2^x - 3 \implies 2^x = y + 3 \implies x = \log_2(y + 3)$.
 $f^{-1}(x) = \log_2(x + 3)$
 (b) $y = e^{x+1} \implies x + 1 = \ln y \implies x = \ln y - 1$.
 $f^{-1}(x) = \ln x - 1$
 (c) $y = 4(10^{2x}) \implies 10^{2x} = \frac{y}{4} \implies 2x = \log_{10}(\frac{y}{4})$.
 $f^{-1}(x) = \frac{1}{2} \log_{10}(\frac{x}{4})$
7. (a) $y = \log_3(x + 2) \implies x + 2 = 3^y \implies x = 3^y - 2$.
 $f^{-1}(x) = 3^x - 2$, **Domain:** $(-\infty, \infty)$
 (b) $y = \ln x - 4 \implies \ln x = y + 4 \implies x = e^{y+4}$.
 $f^{-1}(x) = e^{x+4}$, **Domain:** $(-\infty, \infty)$
 (c) $y = 2 \log_5(x - 1) \implies \log_5(x - 1) = \frac{y}{2} \implies x = 5^{y/2} + 1$.
 $f^{-1}(x) = 5^{x/2} + 1$, **Domain:** $(-\infty, \infty)$
8. (a) For $A(1, 0)$: $\ln 1 = 0 \implies$ **Lies on graph.**
 (b) For $B(e^2, 2)$: $\ln(e^2) = 2 \implies$ **Lies on graph.**
 (c) For $C(\frac{1}{e}, -1)$: $\ln(e^{-1}) = -1 \implies$ **Lies on graph.**
9. (a) $\log_2(x - 3) = 4 \implies x - 3 = 2^4 = 16 \implies x = 19$
 (b) $\ln(2x + 1) = 0 \implies 2x + 1 = e^0 = 1 \implies x = 0$
 (c) $\log_{10}(x^2 - 21) = 2 \implies x^2 - 21 = 10^2 = 100 \implies x^2 = 121 \implies x = \pm 11$

Level 2 Exercise

1. (a) $f(-x) = \ln\left(\frac{1+(-x)}{1-(-x)}\right) = \ln\left(\frac{1-x}{1+x}\right) = \ln\left(\left(\frac{1+x}{1-x}\right)^{-1}\right) = -\ln\left(\frac{1+x}{1-x}\right) = -f(x)$.
Thus, $f(x)$ is an odd function.
- (b) $y = \ln\left(\frac{1+x}{1-x}\right) \implies e^y = \frac{1+x}{1-x} \implies e^y - xe^y = 1+x \implies x(1+e^y) = e^y - 1 \implies x = \frac{e^y - 1}{e^y + 1}$.
 $f^{-1}(x) = \frac{e^x - 1}{e^x + 1}$.
Domain: $(-\infty, \infty)$, **Range:** $(-1, 1)$.
2. (a) Since $e^{1-x} > 0 \implies 2e^{1-x} > 0 \implies 3 - 2e^{1-x} < 3$.
Range: $(-\infty, 3)$.
- (b) $y = 3 - 2e^{1-x} \implies e^{1-x} = \frac{3-y}{2} \implies 1-x = \ln\left(\frac{3-y}{2}\right) \implies x = 1 - \ln\left(\frac{3-y}{2}\right)$.
 $g^{-1}(x) = 1 - \ln\left(\frac{3-x}{2}\right)$, **Domain:** $(-\infty, 3)$.
- (c) x -intercept: $g^{-1}(x) = 0 \implies \ln\left(\frac{3-x}{2}\right) = 1 \implies 3-x = 2e \implies x = 3 - 2e$. Point: $(3 - 2e, 0)$.
 y -intercept: $x = 0 \implies y = 1 - \ln\left(\frac{3}{2}\right)$. Point: $\left(0, 1 - \ln\frac{3}{2}\right)$.
3. (a) $(f \circ g)(x) = e^{\ln(x^2+1)} = x^2 + 1$.
 $(g \circ f)(x) = \ln((e^x)^2 + 1) = \ln(e^{2x} + 1)$.
- (b) For $(f \circ g)(x) = x^2 + 1$:
Domain: $(-\infty, \infty)$, **Range:** $[1, \infty)$.
4. (a) $3x + 1 > 0 \implies x > -\frac{1}{3}$.
Domain: $\left(-\frac{1}{3}, \infty\right)$, **Range:** $(-\infty, \infty)$.
- (b) $y = 2 - \ln(3x + 1) \implies \ln(3x + 1) = 2 - y \implies 3x + 1 = e^{2-y} \implies x = \frac{e^{2-y} - 1}{3}$.
 $f^{-1}(x) = \frac{e^{2-x} - 1}{3}$, **Domain:** $(-\infty, \infty)$.
5. Let $u = e^x > 0$:
$$u^2 - 5u + 6 = 0 \implies (u - 2)(u - 3) = 0 \implies u = 2 \text{ or } u = 3$$

Thus, $e^x = 2 \implies x = \ln 2$, or $e^x = 3 \implies x = \ln 3$.
6. (a) Let $y = \sinh(x) \implies y = \frac{e^x - e^{-x}}{2} \implies e^{2x} - 2ye^x - 1 = 0$.
By quadratic formula in e^x : $e^x = \frac{2y + \sqrt{4y^2 + 4}}{2} = y + \sqrt{y^2 + 1}$ (taking positive root since $e^x > 0$).
Since $\sqrt{y^2 + 1} > |y| \geq -y$, $y + \sqrt{y^2 + 1} > 0$ for all $y \in \mathbb{R}$.
For every $y \in \mathbb{R}$, there exists a unique $x = \ln(y + \sqrt{y^2 + 1}) \in \mathbb{R}$. Thus $\sinh(x)$ is bijective.
- (b) $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$.
7. For domain: $x^2 - 4x + 3 > 0 \implies (x - 1)(x - 3) > 0$.
Domain: $(-\infty, 1) \cup (3, \infty)$.
As $x \rightarrow 1^-$, $x^2 - 4x + 3 \rightarrow 0^+ \implies \ln(x^2 - 4x + 3) \rightarrow -\infty$.
As $x \rightarrow \infty$, $x^2 - 4x + 3 \rightarrow \infty \implies \ln(x^2 - 4x + 3) \rightarrow \infty$.
Range: $(-\infty, \infty)$.
8. (a) $y = e^{x-1} + 1 \implies e^{x-1} = y - 1 \implies x = \ln(y - 1) + 1$.
 $f^{-1}(x) = \ln(x - 1) + 1$.

- (b) If (a, b) lies on both graphs, $b = f(a)$ and $a = f(b) \implies b = e^{a-1} + 1$ and $a = e^{b-1} + 1$.
 Subtracting gives $b - a = e^{a-1} - e^{b-1}$.
 If $a > b \implies e^{a-1} > e^{b-1} \implies e^{a-1} - e^{b-1} > 0$, but $b - a < 0$ (contradiction).
 If $a < b \implies e^{a-1} < e^{b-1} \implies e^{a-1} - e^{b-1} < 0$, but $b - a > 0$ (contradiction).
 Hence $a = b$.
- (c) Any intersection point lies on $y = x \implies e^{x-1} + 1 = x \implies e^{x-1} = x - 1$.
 Since $e^u \geq u + 1 > u$ for all $u \in \mathbb{R}$, setting $u = x - 1$ gives $e^{x-1} > x - 1$ for all x .
 Thus, there are **no real solutions**.

9. Domain restriction: $x^2 - 3x > 0 \implies x < 0$ or $x > 3$.
 Inequality: $x^2 - 3x \leq 2^2 = 4 \implies x^2 - 3x - 4 \leq 0 \implies (x-4)(x+1) \leq 0 \implies -1 \leq x \leq 4$.
 Combining with domain: $[-1, 0) \cup (3, 4]$.

Exam-type Problem Solutions

Question 1

- (a) $y = 5 - 2e^{3x} \implies 2e^{3x} = 5 - y \implies 3x = \ln\left(\frac{5-y}{2}\right) \implies x = \frac{1}{3} \ln\left(\frac{5-y}{2}\right)$.
 $f^{-1}(x) = \frac{1}{3} \ln\left(\frac{5-x}{2}\right)$, **Domain:** $(-\infty, 5)$.
- (b) (i) $P = (0, f(0)) = (0, 5 - 2e^0) = (0, 3)$.
 $Q = (f^{-1}(0), 0) = (3, 0)$.
 (ii) $P(0, 3)$ and $Q(3, 0)$ have swapped coordinates $(x, y) \leftrightarrow (y, x)$, which corresponds geometrically to reflection in the line $y = x$.
- (c) At intersection point (k, k) , $k = f(k) \implies k = 5 - 2e^{3k} \implies k + 2e^{3k} - 5 = 0$.
- (d) Let $t = 3f^{-1}(x) = \ln\left(\frac{5-x}{2}\right) \implies e^t = \frac{5-x}{2} \implies x - 3 = -2(e^t - 1)$.
 As $x \rightarrow 3$, $t \rightarrow 0$.
 $\lim_{x \rightarrow 3} \frac{f^{-1}(x)}{x - 3} = \lim_{t \rightarrow 0} \frac{t/3}{-2(e^t - 1)} = -\frac{1}{6} \lim_{t \rightarrow 0} \frac{t}{e^t - 1} = -\frac{1}{6}$.

Question 2

- (a) For any real x , $\sqrt{x^2 + 1} > \sqrt{x^2} = |x| \geq -x \implies x + \sqrt{x^2 + 1} > 0$.
- (c) $y = \ln(x + \sqrt{x^2 + 1}) \implies e^y - x = \sqrt{x^2 + 1}$.
 Squaring both sides: $e^{2y} - 2xe^y + x^2 = x^2 + 1 \implies 2xe^y = e^{2y} - 1 \implies x = \frac{e^y - e^{-y}}{2}$.
 $g^{-1}(x) = \frac{e^x - e^{-x}}{2}$.
- (d) Since the domain of $g^{-1}(x)$ is $(-\infty, \infty)$, the **Range** of $g(x)$ is $(-\infty, \infty)$.

Question 3

- (a) $y = a^x + b$.
 Passes $(0, 3) \implies a^0 + b = 3 \implies b = 2$.
 Passes $(2, 11) \implies a^2 + 2 = 11 \implies a^2 = 9 \implies a = 3$ (since $a > 1$).
- (b) $y = 3^x + 2 \implies 3^x = y - 2 \implies x = \log_3(y - 2)$.
 $f^{-1}(x) = \log_3(x - 2)$.
- (c) **Domain:** $(2, \infty)$, **Range:** $(-\infty, \infty)$.
 x -intercept: $f^{-1}(x) = 0 \implies \log_3(x - 2) = 0 \implies x - 2 = 1 \implies x = 3$, i.e., $(3, 0)$.

Question 4

- (a) $k - x > 0 \implies x < k$. **Domain:** $(-\infty, k)$.
- (b) $y = \ln(k - x) \implies k - x = e^y \implies x = k - e^y$.
 $f^{-1}(x) = k - e^x$. **Range:** $(-\infty, k)$.
- (c) $-1 = k - e^{\ln 4} \implies -1 = k - 4 \implies k = 3$.
- (d) For $k = 3$, $f(x) = f^{-1}(x) \iff \ln(3 - x) = 3 - e^x$.
 Since $f(x)$ and $f^{-1}(x)$ intersect on $y = x$, we have $e^x + x - 3 = 0$.

Question 5

- (a) $h(x) = \frac{e^x - 1}{e^x + 1} = 1 - \frac{2}{e^x + 1}$.
 Since $e^x > 0 \implies e^x + 1 > 1 \implies 0 < \frac{2}{e^x + 1} < 2$.
 Subtracting from 1: $-1 < 1 - \frac{2}{e^x + 1} < 1 \implies -1 < h(x) < 1$.
- (b) $y = \frac{e^x - 1}{e^x + 1} \implies ye^x + y = e^x - 1 \implies e^x(1 - y) = 1 + y \implies e^x = \frac{1 + y}{1 - y}$.
 $h^{-1}(x) = \ln\left(\frac{1 + x}{1 - x}\right)$, **Domain:** $(-1, 1)$.
- (c) $\ln\left(\frac{1 + x}{1 - x}\right) = \ln 3 \implies \frac{1 + x}{1 - x} = 3 \implies 1 + x = 3 - 3x \implies 4x = 2 \implies x = \frac{1}{2}$.

Question 6

- (a) $y = 2^{3x-1} + 4 \implies 2^{3x-1} = y - 4 \implies 3x - 1 = \log_2(y - 4)$.
 $f^{-1}(x) = \frac{\log_2(x-4)+1}{3}$.
- (b) $f^{-1}(x) = \frac{\frac{\ln(x-4)}{\ln 2} + 1}{3} = \frac{\ln(x-4) + \ln 2}{3 \ln 2} = \frac{\ln(2x-8)}{3 \ln 2}$.
- (c) For $f(x)$: y -intercept is $(0, f(0)) = (0, 4.5)$; no x -intercept.
 For $f^{-1}(x)$: x -intercept is $(4.5, 0)$; no y -intercept.
 Line of symmetry: $y = x$.

Question 7

- (a) $x^2 - 2x + 2 = (x - 1)^2 + 1 \geq 1 > 0$ for $x \geq 1$. Thus $f(x)$ is well-defined.
- (b) $y = \ln((x - 1)^2 + 1) \implies (x - 1)^2 = e^y - 1 \implies x - 1 = \sqrt{e^y - 1}$ (since $x \geq 1$).
 $f^{-1}(x) = 1 + \sqrt{e^x - 1}$.
 Since $x \geq 1 \implies f(x) \geq \ln 1 = 0$, **Domain:** $[0, \infty)$.
- (c) Intersections lie on $y = x \implies \ln((x - 1)^2 + 1) = x \implies (x - 1)^2 + 1 = e^x$.
 For $x \geq 1$, $e^x > (x - 1)^2 + 1$, so there are **no intersection points**.

Question 8

- (a) y -intercept ($x = 0$): $y = e^0 - 4e^0 + 3 = 0 \implies (0, 0)$.
 x -intercepts ($y = 0$): $(e^x - 1)(e^x - 3) = 0 \implies x = 0$ or $x = \ln 3$.
 Points: $(0, 0)$ and $(\ln 3, 0)$.
- (b) (i) x -intercept of C_2 : $(0, 0)$.
 y -intercepts of C_2 : $(0, 0)$ and $(0, \ln 3)$.
- (ii) Reflecting C_1 : swap x and $y \implies x = e^{2y} - 4e^y + 3 \implies (e^y - 2)^2 = x + 1$.
 For $y \geq \ln 2 \implies e^y \geq 2 \implies e^y - 2 = \sqrt{x + 1} \implies e^y = 2 + \sqrt{x + 1}$.
 $y = \ln(2 + \sqrt{x + 1})$ for $x \geq -1$.

Question 9

- (a) $p(3) = 0 \implies \log_a(3+b) = 0 \implies 3+b = 1 \implies b = -2$.
 $p^{-1}(2) = 19 \iff p(19) = 2 \implies \log_a(19-2) = 2 \implies a^2 = 17 \implies a = \sqrt{17}$.
- (b) $y = \log_{\sqrt{17}}(x-2) \implies x-2 = (\sqrt{17})^y \implies x = 17^{y/2} + 2$.
 $p^{-1}(x) = 17^{x/2} + 2$, **Domain:** $(-\infty, \infty)$, **Range:** $(2, \infty)$.
- (c) $\log_{\sqrt{17}}(x-2) < 2$.
 Domain restriction: $x > 2$.
 Inequality: $x-2 < 17 \implies x < 19$.
 Solution set: $(2, 19)$.

Question 10

- (a) $f(x) = \frac{e^x(e^x - e^{-x})}{e^x(e^x + e^{-x})} = \frac{e^{2x} - 1}{e^{2x} + 1}$.
- (b) $y = \frac{e^{2x} - 1}{e^{2x} + 1} \implies ye^{2x} + y = e^{2x} - 1 \implies e^{2x}(1-y) = 1+y \implies e^{2x} = \frac{1+y}{1-y}$.
 $f^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$, **Domain:** $(-1, 1)$.
- (c) $\lim_{x \rightarrow 0} \frac{f^{-1}(x)}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2} [\ln(1+x) - \ln(1-x)]}{x} = \frac{1}{2} (1 - (-1)) = 1$.

Solutions to Natural Logarithm

Level 1 Exercise Solutions

- (a) $e^x = 7 \iff x = \ln 7$

(b) $e^{3-x} = 2 \iff 3-x = \ln 2 \iff x = 3 - \ln 2$

(c) $e^{x^2} = 10 \iff x^2 = \ln 10$
- (a) $\ln(\sqrt{e}) = \ln(e^{1/2}) = \frac{1}{2}$

(b) $\ln(e^5) - \ln(e^2) = 5 - 2 = 3$

(c) $e^{2\ln 3} = e^{\ln(3^2)} = 9$

(d) $e^{-\ln 4} = e^{\ln(1/4)} = \frac{1}{4}$
- (a) $x+5 > 0 \implies x > -5$.
Domain: $(-5, \infty)$, **Range:** $(-\infty, \infty)$

(b) $3-2x > 0 \implies x < \frac{3}{2}$.
Domain: $(-\infty, \frac{3}{2})$, **Range:** $(-\infty, \infty)$

(c) $x > 0$.
Domain: $(0, \infty)$, **Range:** $(-\infty, \infty)$
- (a) $e^{\ln(2x-1)} = 2x-1$

(b) $\ln(e^{x^2+3x}) = x^2+3x$

(c) $e^{3\ln x+1} = e \cdot e^{\ln(x^3)} = ex^3$
- (a) $y = 2e^x - 1 \implies e^x = \frac{y+1}{2} \implies x = \ln\left(\frac{y+1}{2}\right)$.
 $f^{-1}(x) = \ln\left(\frac{x+1}{2}\right)$, **Domain:** $(-1, \infty)$

- (b) $y = \ln(x - 3) + 2 \implies x - 3 = e^{y-2} \implies x = e^{y-2} + 3$.
 $f^{-1}(x) = e^{x-2} + 3$, **Domain:** $(-\infty, \infty)$
- (c) $y = \frac{1}{2}e^{4x} \implies e^{4x} = 2y \implies x = \frac{1}{4}\ln(2y)$.
 $f^{-1}(x) = \frac{1}{4}\ln(2x)$, **Domain:** $(0, \infty)$
6. (a) $\ln(3x - 2) = 0 \implies 3x - 2 = e^0 = 1 \implies x = 1$
 (b) $e^{2x+1} = 9 \implies 2x + 1 = \ln 9 \implies x = \frac{\ln 9 - 1}{2} = \ln 3 - \frac{1}{2}$
 (c) $\ln(x^2 - 3) = \ln 6 \implies x^2 - 3 = 6 \implies x^2 = 9 \implies x = \pm 3$
7. (a) $\ln 6 = \ln(2 \cdot 3) = \ln 2 + \ln 3 = a + b$
 (b) $\ln\left(\frac{8}{9}\right) = \ln(2^3) - \ln(3^2) = 3\ln 2 - 2\ln 3 = 3a - 2b$
 (c) $\ln(\sqrt{12}) = \frac{1}{2}\ln(2^2 \cdot 3) = \ln 2 + \frac{1}{2}\ln 3 = a + \frac{1}{2}b$
8. Substitute $x = e^2$ into LHS:

$$\ln(e^2) + \ln(e^2 - e) = 2 + \ln e + \ln(e - 1) = 3 + \ln(e - 1)$$

 Since $\text{LHS} = 3 + \ln(e - 1) \neq 2 + \ln(e - 1) = \text{RHS}$, $x = e^2$ is **not** a solution.
9. • y -intercept ($x = 0$): $y = 3\ln 1 - 2 = -2 \implies (0, -2)$.
 • x -intercept ($y = 0$): $3\ln(x + 1) - 2 = 0 \implies \ln(x + 1) = \frac{2}{3} \implies x = e^{2/3} - 1 \implies (e^{2/3} - 1, 0)$.
10. $e^{\ln(x+2)} = 2x - 5 \implies x + 2 = 2x - 5 \implies x = 7$.
 Check domain: $7 + 2 = 9 > 0$ (valid). Thus, $x = 7$.

Level 2 Exercise Solutions

1. Let $u = \ln x$:

$$u^2 - 3u + 2 = 0 \implies (u - 1)(u - 2) = 0 \implies u = 1 \text{ or } u = 2$$

Thus, $x = e$ or $x = e^2$.

2. Multiply by e^x : $e^{2x} - e^x - 6 = 0$. Let $u = e^x > 0$:

$$u^2 - u - 6 = 0 \implies (u - 3)(u + 2) = 0 \implies u = 3 \quad (\text{since } u > 0)$$

Thus, $e^x = 3 \implies x = \ln 3$.

3. $\frac{e^x + e^{-x}}{2} = 2 \implies e^{2x} - 4e^x + 1 = 0$. Let $u = e^x > 0$:

$$u = \frac{4 \pm \sqrt{16 - 4}}{2} = 2 \pm \sqrt{3}$$

Since $2 \pm \sqrt{3} > 0$, $x = \ln(2 + \sqrt{3})$ or $x = \ln(2 - \sqrt{3})$.

4. • **Domain:** $9 - x^2 > 0 \implies x^2 < 9 \implies (-3, 3)$.
 • **Range:** $0 < 9 - x^2 \leq 9 \implies \ln(9 - x^2) \in (-\infty, \ln 9]$.

5. (a) $(f \circ g)(x) = f(g(x)) = \ln(2e^{x/2} - 1)$.
 $(g \circ f)(x) = g(f(x)) = e^{\frac{1}{2}\ln(2x-1)} = \sqrt{2x-1}$.

- (b) For $(f \circ g)(x)$, require $2e^{x/2} - 1 > 0 \implies e^{x/2} > \frac{1}{2} \implies \frac{x}{2} > -\ln 2 \implies x > -2\ln 2$.
Domain: $(-2\ln 2, \infty)$.

6. Domain: $x^2 - 5x + 6 > 0 \implies (x-2)(x-3) > 0 \implies x < 2$ or $x > 3$. Inequality:
 $x^2 - 5x + 6 \leq 2 \implies x^2 - 5x + 4 \leq 0 \implies 1 \leq x \leq 4$. Intersection with domain:
 $[1, 2) \cup (3, 4]$.

Exam-type Problem Solutions

Question 1

(a) $e^{2x} = 1 + (2x) + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots$

(b) For $x = 0.1$, $e^{0.2} \approx 1 + 2(0.1) + 2(0.1)^2 + \frac{4}{3}(0.1)^3 = 1.221333$.

$$\frac{e^{0.2} - 1}{0.1} \approx \frac{1.221333 - 1}{0.1} = 2.213333 \approx 2.21$$

- (c) Rewrite limit:

$$\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{\ln(1 + 2x)} = \lim_{x \rightarrow 0} \left[\frac{e^{3x} - 1}{3x} \cdot \frac{2x}{\ln(1 + 2x)} \cdot \frac{3}{2} \right] = 1 \cdot 1 \cdot \frac{3}{2} = \frac{3}{2}$$

Question 2

(a) $f(x) = \ln \left(\frac{e^x(1-e^{-x})}{e^x(1+e^{-x})} \right) = \ln \left(\frac{1-e^{-x}}{1+e^{-x}} \right)$.

(b) $y = \ln \left(\frac{e^x - 1}{e^x + 1} \right) \implies e^y = \frac{e^x - 1}{e^x + 1} \implies e^x(1 - e^y) = 1 + e^y \implies e^x = \frac{1+e^y}{1-e^y}$.

$$f^{-1}(x) = \ln \left(\frac{1 + e^x}{1 - e^x} \right)$$

Domain: $1 - e^x > 0 \implies e^x < 1 \implies (-\infty, 0)$. Range: $(0, \infty)$.

Question 3

(a) $P(5) = 2P_0 \implies e^{5k} = 2 \implies k = \frac{\ln 2}{5}$.

(b) $P(t) = 3P_0 \implies e^{kt} = 3 \implies t = \frac{\ln 3}{k} = \frac{5 \ln 3}{\ln 2} \approx 7.92$ years.

(c) $P(0.1) = P_0 e^{0.14 \times 0.1} = P_0 e^{0.014} \approx P_0(1 + 0.014) = 1.014P_0$. Percentage increase $\approx 1.4\%$.

Question 4

(a) $x^2 + 2x + 2 = (x+1)^2 + 1 \geq 1 > 0$ for all $x \in \mathbb{R}$. **Domain:** $(-\infty, \infty)$, **Range:** $[0, \infty)$.

(b) $\ln(x^2 + 2x + 2) = \ln 5 \implies x^2 + 2x - 3 = 0 \implies (x+3)(x-1) = 0$. $x = -3$ or $x = 1$.

- (c) Rewrite limit:

$$\lim_{x \rightarrow 0} \frac{\ln \left(1 + \frac{x^2 + 2x}{2} \right)}{x} = \lim_{x \rightarrow 0} \left[\frac{\ln \left(1 + \frac{x^2 + 2x}{2} \right)}{\frac{x^2 + 2x}{2}} \cdot \frac{x^2 + 2x}{2x} \right] = 1 \cdot 1 = 1$$

Question 5

(a) $e^{2x} - 4e^x + 3 = 0 \implies (e^x - 1)(e^x - 3) = 0 \implies x = 0$ or $x = \ln 3$. x -intercepts: $(0, 0)$ and $(\ln 3, 0)$.

(b) $y = (e^x - 2)^2 - 1 \implies e^x = 2 + \sqrt{y+1}$ (since $x \geq \ln 2 \implies e^x \geq 2$).

$$f^{-1}(x) = \ln(2 + \sqrt{x+1}), \quad \text{Domain: } [-1, \infty)$$

(c) Let $u = \frac{\sqrt{x+1}-2}{4}$. As $x \rightarrow 3$, $u \rightarrow 0$ and $\frac{\sqrt{x+1}-2}{x-3} \rightarrow \frac{1}{4}$.

$$\lim_{x \rightarrow 3} \frac{\ln(1+u)}{4u} \cdot \frac{\sqrt{x+1}-2}{x-3} = 1 \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$

Question 6

(a) $\frac{x+a}{x-a} > 0 \implies x < -a$ or $x > a$. **Domain:** $(-\infty, -a) \cup (a, \infty)$.

(c) $y = \ln\left(\frac{x+a}{x-a}\right) \implies e^y(x-a) = x+a \implies x(e^y-1) = a(e^y+1)$.

$$h^{-1}(x) = a \left(\frac{e^x + 1}{e^x - 1} \right)$$

Question 7

(a) **Domain:** $(-\infty, \infty)$, **Range:** $(\ln k, \infty)$.

(b) $y = \ln(e^x + k) \implies e^x = e^y - k \implies f^{-1}(x) = \ln(e^x - k)$, **Domain:** $(\ln k, \infty)$.

(c) $\ln 2 = \ln(e^{\ln 5} - k) \implies 2 = 5 - k \implies k = 3$.

(d) $\ln(e^x + 3) = 2x \implies e^{2x} - e^x - 3 = 0$. $e^x = \frac{1+\sqrt{13}}{2} \implies x = \ln\left(\frac{1+\sqrt{13}}{2}\right)$.

Question 8

(a) $\lim_{x \rightarrow 0} \frac{\ln(1+3x)}{x} = 3 \lim_{3x \rightarrow 0} \frac{\ln(1+3x)}{3x} = 3(1) = 3$.

(b) $\lim_{x \rightarrow 0} \frac{\ln(1+3x)}{x} - \lim_{x \rightarrow 0} \frac{\ln(1-2x)}{x} = 3 - (-2) = 5$.

(c) $\lim_{x \rightarrow 0} \left[\frac{e^{cx} - 1}{cx} \cdot \frac{3x}{\ln(1+3x)} \cdot \frac{c}{3} \right] = \frac{c}{3} = 2 \implies c = 6$.

Question 9

(a) For $x = e^2 - 1$: $\ln(e^2 - 1 + 1) = \ln(e^2) = 2$.

(b) (i) $\ln\left(\frac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x) = 2x + \frac{2}{3}x^3 + \frac{2}{5}x^5 + \dots$

(ii) Substitute $x = \frac{1}{3}$:

$$\ln 2 \approx 2 \left(\frac{1}{3} + \frac{1}{81} + \frac{1}{1215} \right) = 2 \left(\frac{421}{1215} \right) = \frac{842}{1215}$$

Solutions to Properties of Logarithmic Function

Level 1 Exercise

1. (a) $\ln(x^3y^2) = \ln(x^3) + \ln(y^2) = 3\ln x + 2\ln y$
 (b) $\ln\left(\frac{\sqrt{x}}{y^4}\right) = \ln(x^{1/2}) - \ln(y^4) = \frac{1}{2}\ln x - 4\ln y$
 (c) $\ln\left(\frac{x^2y^3}{z^5}\right) = \ln(x^2) + \ln(y^3) - \ln(z^5) = 2\ln x + 3\ln y - 5\ln z$
2. (a) $2\ln 3 + 3\ln 2 = \ln(3^2) + \ln(2^3) = \ln(9 \cdot 8) = \ln 72$
 (b) $\frac{1}{2}\ln 16 - 2\ln 2 + \ln 5 = \ln(16^{1/2}) - \ln(2^2) + \ln 5 = \ln 4 - \ln 4 + \ln 5 = \ln 5$
 (c) $3\ln x - 2\ln y + \frac{1}{3}\ln z = \ln(x^3) - \ln(y^2) + \ln(\sqrt[3]{z}) = \ln\left(\frac{x^3\sqrt[3]{z}}{y^2}\right)$
3. (a) $\ln(e^4 \cdot \sqrt{e}) = \ln(e^{4.5}) = 4.5 = \frac{9}{2}$
 (b) $\ln 18 - 2\ln 3 + \ln 5 - \ln 10 = \ln\left(\frac{18 \cdot 5}{3^2 \cdot 10}\right) = \ln 1 = 0$
 (c) $e^{3\ln 2 - \ln 4} = e^{\ln 8 - \ln 4} = e^{\ln 2} = 2$
4. (a) $\ln((-4)^2) = 2\ln|-4| = 2\ln 4 = \ln 16$
 (b) For $x < 0$, $|x| = -x \implies \ln((-x)^6) = 6\ln|-x| = 6\ln(-x)$
 (c) $\ln(x^4) - 2\ln|x| = 4\ln|x| - 2\ln|x| = 2\ln|x|$
5. (a) $\ln 30 = \ln(2 \cdot 3 \cdot 5) = \ln 2 + \ln 3 + \ln 5 = p + q + r$
 (b) $\ln\left(\frac{12}{25}\right) = \ln(2^2 \cdot 3) - \ln(5^2) = 2\ln 2 + \ln 3 - 2\ln 5 = 2p + q - 2r$
 (c) $\ln(0.072) = \ln\left(\frac{72}{1000}\right) = \ln\left(\frac{9}{125}\right) = 2\ln 3 - 3\ln 5 = 2q - 3r$
6. (a) $\ln(a^2b^3) = 2\ln a + 3\ln b = 2(3) + 3(5) = 21$
 (b) $\ln\left(\frac{\sqrt{a}}{b^2}\right) = \frac{1}{2}\ln a - 2\ln b = \frac{1}{2}(3) - 2(5) = -\frac{17}{2}$
 (c) $\ln\left(\frac{e^2 \cdot a^4}{\sqrt{b}}\right) = 2 + 4\ln a - \frac{1}{2}\ln b = 2 + 4(3) - \frac{1}{2}(5) = \frac{23}{2}$
7. (a) $\ln(4x) = \ln 12 \implies 4x = 12 \implies x = 3$
 (b) $\ln\left(\frac{x+3}{x}\right) = \ln 2 \implies \frac{x+3}{x} = 2 \implies x = 3$
 (c) $\ln(x^2) = \ln(3x+4) \implies x^2 - 3x - 4 = 0 \implies (x-4)(x+1) = 0$.
 Domain $x > 0 \implies x = 4$.
8. (a) $\log_2 32 = \frac{\ln 32}{\ln 2} = \frac{5\ln 2}{\ln 2} = 5$
 (b) $\log_9 27 = \frac{\ln 27}{\ln 9} = \frac{3\ln 3}{2\ln 3} = \frac{3}{2}$
 (c) $(\log_3 5)(\log_5 81) = \frac{\ln 5}{\ln 3} \cdot \frac{\ln 81}{\ln 5} = \frac{4\ln 3}{\ln 3} = 4$
9. (a) $\ln(3^x) = \ln 7 \implies x\ln 3 = \ln 7 \implies x = \frac{\ln 7}{\ln 3}$
 (b) $(x+1)\ln 2 = x\ln 5 \implies x(\ln 5 - \ln 2) = \ln 2 \implies x = \frac{\ln 2}{\ln(5/2)}$
 (c) $2x - 1 = x\ln 4 \implies x(2 - 2\ln 2) = 1 \implies x = \frac{1}{2(1-\ln 2)}$
10. (a) **False.** Correct: $\ln(a/b) = \ln a - \ln b$.
 Counterexample: $a = e^2, b = e \implies \frac{\ln e^2}{\ln e} = 2 \neq 2 - 1 = 1$.
 (b) **False.** Correct: $\ln(ab) = \ln a + \ln b$.
 Counterexample: $a = 1, b = 1 \implies \ln(1+1) = \ln 2 \neq \ln 1 + \ln 1 = 0$.
 (c) **False.** Correct: $\ln(x^2) = 2\ln x$.
 Counterexample: $x = e \implies (\ln e)^2 = 1 \neq 2\ln e = 2$.

Level 2 Exercise

1. Domain: $x - 2 > 0 \implies x > 2$.

$$\ln((x-2)(x+1)) = \ln(x+7) \implies x^2 - x - 2 = x + 7$$

$$x^2 - 2x - 9 = 0 \implies x = 1 \pm \sqrt{10}$$

Since $x > 2$, $x = 1 + \sqrt{10}$.

2. Let $u = \ln x$:

$$u^2 - 3u + 2 = 0 \implies (u-1)(u-2) = 0 \implies u = 1 \text{ or } u = 2$$

$$x = e \text{ or } x = e^2$$

3. Solving system:

$$\ln x + 2 \ln y = 7 \quad \text{--- (1)}$$

$$2 \ln x - \ln y = 1 \quad \text{--- (2)}$$

$$2 \times (2) + (1) \implies 5 \ln x = 9 \implies \ln x = \frac{9}{5} \implies x = e^{9/5}.$$

$$\text{Substitute into (2)} \implies \ln y = \frac{13}{5} \implies y = e^{13/5}.$$

$$\text{Solution: } (x, y) = (e^{9/5}, e^{13/5}).$$

- 4.

$$\text{LHS} = \ln [(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})]$$

$$= \ln((x+1) - x) = \ln 1 = 0 = \text{RHS}$$

5. For $x < 0$, $|x| = -x$. Using $\ln(x^y) = y \ln |x|$:

$$f(x) = 6 \ln(-x) - 3(2 \ln(-x)) + \ln(-x) = \ln(-x)$$

6. Change base: $\log_x e = \frac{1}{\ln x}$.

$$\frac{1}{\ln x} + 2 \ln x = 3 \implies 2(\ln x)^2 - 3 \ln x + 1 = 0$$

$$(2 \ln x - 1)(\ln x - 1) = 0 \implies \ln x = \frac{1}{2} \text{ or } \ln x = 1$$

$$x = \sqrt{e} \text{ or } x = e$$

7. Take $\ln$ on both sides:

$$(\ln x)^2 = \ln(e^4) + 3 \ln x \implies (\ln x)^2 - 3 \ln x - 4 = 0$$

$$(\ln x - 4)(\ln x + 1) = 0 \implies \ln x = 4 \text{ or } \ln x = -1$$

$$x = e^4 \text{ or } x = \frac{1}{e}$$

8. Domain: $x > 2$.

$$\ln \left(\frac{x^2 - 4}{x - 2} \right) \leq \ln 5 \implies \ln(x+2) \leq \ln 5 \implies x+2 \leq 5 \implies x \leq 3$$

Combining with domain: $(2, 3]$.

$$9. \quad (a) \quad f(x) + f(y) = \ln \left(\frac{(2+x)(2+y)}{(2-x)(2-y)} \right) = \ln \left(\frac{4+2x+2y+xy}{4-2x-2y+xy} \right).$$

Let $u = \frac{4x+4y}{4+xy}$. Then $\frac{2+u}{2-u} = \frac{4+2x+2y+xy}{4-2x-2y+xy}$.

Thus $f(x) + f(y) = f \left(\frac{4x+4y}{4+xy} \right)$.

(b) Using $f(-t) = -f(t)$:

$$f(x) + f(y) + f(-x-y) = \ln \left(\frac{(2+x)(2+y)(2-x-y)}{(2-x)(2-y)(2+x+y)} \right)$$

Exam-type Problem Solutions

Question 1

(a) Domain: $x > 1$.

$$\frac{x^2 - kx + 4}{x - 1} = 3 \implies x^2 - (k+3)x + 7 = 0$$

$$x = \frac{(k+3) \pm \sqrt{(k+3)^2 - 28}}{2}$$

(a) Let $g(x) = x^2 - (k+3)x + 7 = 0$.

• **Case 1** ($\Delta = 0$): $(k+3)^2 = 28 \implies k = -3 + 2\sqrt{7}$.

Root $x = \frac{2\sqrt{7}}{2} = \sqrt{7} > 1$ (valid).

• **Case 2** ($g(1) \leq 0$): $g(1) = 5 - k \leq 0 \implies k \geq 5$.

One root is > 1 and the other is ≤ 1 .

Range of k : $k = -3 + 2\sqrt{7}$ or $k \geq 5$.

Question 2

(a) Slope $m = \frac{7-1}{5-2} = 2$.

$$\ln y - 1 = 2(\ln x - 2) \implies \ln y = 2 \ln x - 3.$$

(b) $y = e^{2 \ln x - 3} = e^{-3} x^2 \implies A = e^{-3}, p = 2$.

(c) When $x = e^2$: $y = e^{-3}(e^2)^2 = e$.

Question 3

$$(a) \quad 1 - \frac{1}{k^2} = \frac{k^2-1}{k^2} = \frac{(k-1)(k+1)}{k^2}.$$

(b) Telescoping product:

$$S_n = \ln \left(\prod_{k=2}^n \frac{(k-1)(k+1)}{k^2} \right) = \ln \left(\frac{1}{2} \cdot \frac{n+1}{n} \right) = \ln \left(\frac{n+1}{2n} \right)$$

$$(c) \quad \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \ln \left(\frac{n+1}{2n} \right) = \ln \left(\frac{1}{2} \right) = -\ln 2.$$

Question 4

(a) $\ln(xy) = \ln(e^5) \implies \ln x + \ln y = 5$.

Since $\ln x + \ln y = 5$ and $\ln x \cdot \ln y = 6$, $\ln x$ and $\ln y$ are roots of $t^2 - 5t + 6 = 0$.

(b) $t^2 - 5t + 6 = 0 \implies t = 2 \text{ or } t = 3.$

Ordered pairs: $(x, y) = (e^2, e^3)$ or $(e^3, e^2).$

Question 5

(a) $\ln N(t) = \ln N_0 - \lambda t.$

(b) $N(T) = \frac{1}{2}N_0 \implies e^{-\lambda T} = \frac{1}{2} \implies T = \frac{\ln 2}{\lambda}.$

(c) $N(12) = \frac{1}{4}N_0 = (1/2)^2 N_0 \implies 2T = 12 \implies T = 6 \text{ years}.$
 $\lambda = \frac{\ln 2}{6}.$

For 10%: $e^{-\lambda t} = 0.1 \implies t = \frac{\ln 10}{\lambda} = \frac{6 \ln 10}{\ln 2} \approx 19.9 \text{ years}.$

Question 6

(a) Let $u = 3x + 2x^2:$

$$\lim_{x \rightarrow 0} \frac{\ln(1+u)}{u} \cdot \frac{3x+2x^2}{x} = 1 \cdot 3 = 3$$

(b) $\lim_{x \rightarrow 0} \frac{4 \ln(1+x)}{2x} = 2 \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 2(1) = 2.$

Question 7

(a) $x^4 - 16 > 0 \implies x^2 > 4 \implies D = (-\infty, -2) \cup (2, \infty).$

(b) For $x \in D:$

$$f(x) - \ln(x^2 + 4) = \ln\left(\frac{x^4 - 16}{x^2 + 4}\right) = \ln(x^2 - 4)$$

(c) $\ln(x^2 - 4) = \ln 5 \implies x^2 - 4 = 5 \implies x = \pm 3.$

Question 8

(a) $1 - \frac{2}{k(k+1)} = \frac{k^2+k-2}{k(k+1)} = \frac{(k-1)(k+2)}{k(k+1)}.$

(b) Telescoping sum:

$$\begin{aligned} \ln P_n &= \sum_{k=2}^n \ln\left(\frac{k-1}{k}\right) + \sum_{k=2}^n \ln\left(\frac{k+2}{k+1}\right) \\ &= \ln\left(\frac{1}{n}\right) + \ln\left(\frac{n+2}{3}\right) = \ln\left(\frac{n+2}{3n}\right) \end{aligned}$$

(c) $P_n = \frac{n+2}{3n}.$

$$\lim_{n \rightarrow \infty} P_n = \lim_{n \rightarrow \infty} \frac{n+2}{3n} = \frac{1}{3}.$$

Question 9

(a) Divide numerator and denominator by $e^x:$

$$g(x) = \ln\left(\frac{1+e^{-x}}{1-e^{-x}}\right) = \ln(1+e^{-x}) - \ln(1-e^{-x})$$

(b) Let $y = g(x) \implies e^y = \frac{e^x+1}{e^x-1} \implies e^x = \frac{e^y+1}{e^y-1} \implies x = g(y).$

Thus $g^{-1}(x) = g(x).$

(c) Let $t = e^{-x} \rightarrow 0$ as $x \rightarrow \infty:$

$$\lim_{x \rightarrow \infty} e^x g(x) = \lim_{t \rightarrow 0} \frac{\ln(1+t) - \ln(1-t)}{t} = 1 - (-1) = 2$$

Solutions to Change of Base

Level 1 Exercise

1. (a) $\log_2 7 = \frac{\ln 7}{\ln 2}$
 (b) $\log_5 100 = \frac{\ln 100}{\ln 5}$
 (c) $\log_{10} e = \frac{\ln e}{\ln 10} = \frac{1}{\ln 10}$
2. (a) $\log_2 8 = \log_2(2^3) = 3$
 (b) $\log_9 27 = \frac{\ln 27}{\ln 9} = \frac{3 \ln 3}{2 \ln 3} = \frac{3}{2}$
 (c) $\log_{16} 64 = \frac{\ln 64}{\ln 16} = \frac{6 \ln 2}{4 \ln 2} = \frac{3}{2}$
3. (a) $\log_3 5 \cdot \log_5 9 = \frac{\ln 5}{\ln 3} \cdot \frac{\ln 9}{\ln 5} = \frac{2 \ln 3}{\ln 3} = 2$
 (b) $\log_2 3 \cdot \log_3 4 \cdot \log_4 8 = \frac{\ln 3}{\ln 2} \cdot \frac{\ln 4}{\ln 3} \cdot \frac{\ln 8}{\ln 4} = \frac{\ln 8}{\ln 2} = 3$
 (c) $\frac{\log_5 16}{\log_5 2} = \log_2 16 = 4$
4. (a) $\log_2 3 = \frac{\ln 3}{\ln 2} = \frac{b}{a}$
 (b) $\log_3 12 = \frac{\ln 12}{\ln 3} = \frac{2 \ln 2 + \ln 3}{\ln 3} = \frac{2a+b}{b}$
 (c) $\log_6 18 = \frac{\ln 18}{\ln 6} = \frac{\ln 2 + 2 \ln 3}{\ln 2 + \ln 3} = \frac{a+2b}{a+b}$
5. (a) $\log_x 81 = 4 \implies x^4 = 81 \implies x = 3$
 (b) $\log_x 5 = \frac{1}{2} \implies x^{1/2} = 5 \implies x = 25$
 (c) $\log_x e = 2 \implies x^2 = e \implies x = \sqrt{e}$

6. Proof:

$$\frac{\log_{10} A}{\log_{10} B} = \frac{\frac{\ln A}{\ln 10}}{\frac{\ln B}{\ln 10}} = \frac{\ln A}{\ln B}$$

7. (a) $2^{\log_2 5} = 5$
 (b) $3^{\log_9 16} = 3^{\frac{1}{2} \log_3 16} = 3^{\log_3 4} = 4$
 (c) $4^{\log_2 3} = (2^2)^{\log_2 3} = (2^{\log_2 3})^2 = 3^2 = 9$
8. $\log_4 x = \log_2 3 \implies \frac{\log_2 x}{2} = \log_2 3 \implies \log_2 x = \log_2 9 \implies x = 9.$
9. $\log_a \sqrt{b} \cdot \log_b(a^3) = \left(\frac{1}{2} \log_a b\right) (3 \log_b a) = \frac{3}{2} \left(\frac{\ln b}{\ln a} \cdot \frac{\ln a}{\ln b}\right) = \frac{3}{2}.$
10. $\log_a b = \frac{\ln b}{\ln a} = \frac{1}{\frac{\ln a}{\ln b}} = \frac{1}{\log_b a}.$

Level 2 Exercise

1. $\log_2 x + \frac{\log_2 x}{2} = 6 \implies \frac{3}{2} \log_2 x = 6 \implies \log_2 x = 4 \implies x = 16.$
2. Let $u = \log_4 x$:

$$\frac{1}{u} + u = \frac{5}{2} \implies 2u^2 - 5u + 2 = 0$$

$$(2u - 1)(u - 2) = 0 \implies u = \frac{1}{2} \text{ or } u = 2$$

$$x = 4^{1/2} = 2 \text{ or } x = 4^2 = 16.$$

3. Change base to 2:

$$\begin{cases} \log_2 x + \frac{1}{2} \log_2 y = 3 \\ \frac{1}{2} \log_2 x + \log_2 y = 3 \end{cases}$$

Adding equations: $\frac{3}{2} \log_2(xy) = 6 \implies xy = 16$.

Subtracting equations: $\frac{1}{2} \log_2\left(\frac{x}{y}\right) = 0 \implies x = y$.

Since $x, y > 0$, $x = 4, y = 4$.

4. Taking $\log_2$ on both sides:

$$(\log_2 x)^2 = 4 + \log_2 x \implies (\log_2 x)^2 - \log_2 x - 4 = 0$$

$$\log_2 x = \frac{1 \pm \sqrt{17}}{2} \implies x = 2^{\frac{1 \pm \sqrt{17}}{2}}$$

5. Using $\frac{1}{\log_x y} = \log_y x$:

$$\text{LHS} = \log_{abc} a + \log_{abc} b + \log_{abc} c = \log_{abc}(abc) = 1$$

6. Change base to 2:

$$\log_2(x+1) - \frac{1}{3} \log_2(x+1) = \frac{2}{3}$$

$$\frac{2}{3} \log_2(x+1) = \frac{2}{3} \implies x+1 = 2 \implies x = 1$$

7. $a^x = b^y = c^z = d \implies \frac{1}{x} = \log_d a, \frac{1}{y} = \log_d b, \frac{1}{z} = \log_d c$.

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \log_d a + \log_d b + \log_d c = \log_d(abc)$$

8. $\frac{\ln 2}{\ln x} < 1$:

- For $x > 1$: $\ln 2 < \ln x \implies x > 2$.
- For $0 < x < 1$: $\ln 2 > \ln x \implies x < 2 \implies 0 < x < 1$.

Solution: $(0, 1) \cup (2, \infty)$.

9. (a) $\log_{a^2}(b^3) = \frac{3 \ln b}{2 \ln a} = \frac{3}{2} \log_a b = \frac{3}{2} k$

(b) $\log_{\sqrt{a}}\left(\frac{1}{b}\right) = \frac{-\ln b}{\frac{1}{2} \ln a} = -2 \log_a b = -2k$

10. $\log_3 x - \frac{2}{\log_3 x} + 1 = 0 \implies (\log_3 x)^2 + \log_3 x - 2 = 0$.

$$(\log_3 x + 2)(\log_3 x - 1) = 0 \implies x = \frac{1}{9} \text{ or } x = 3$$

Exam-type Problem Solutions

Question 1

$$\log_3 x + \frac{1}{2} \log_3 x + \frac{1}{3} \log_3 x = \frac{11}{2}$$

$$\frac{11}{6} \log_3 x = \frac{11}{2} \implies \log_3 x = 3 \implies x = 27$$

Question 2

$$(a) \log_x 16 = \frac{\log_4 16}{\log_4 x} = \frac{2}{\log_4 x}.$$

$$\frac{2}{\log_4 x} - \log_4 x = 3 \implies (\log_4 x)^2 + 3 \log_4 x - 2 = 0$$

$$(b) \log_4 x = \frac{-3 \pm \sqrt{17}}{2} \implies x = 4^{\frac{-3 \pm \sqrt{17}}{2}} = 2^{-3 \pm \sqrt{17}}.$$

Question 3

$$(a) \text{ Slope } m = \frac{7-3}{3-1} = 2.$$

$$\log_2 y - 3 = 2(\log_4 x - 1) \implies \log_2 y = 2 \log_4 x + 1$$

$$(b) \text{ Since } \log_4 x = \frac{1}{2} \log_2 x:$$

$$\log_2 y = \log_2 x + 1 = \log_2(2x) \implies y = 2x$$

Here $A = 2, p = 1$.

$$(c) \text{ When } x = 16: y = 2(16) = 32.$$

Question 4

$$(a) 2 \log_b x = \log_a x + \log_c x \implies \frac{2 \ln x}{\ln b} = \frac{\ln x}{\ln a} + \frac{\ln x}{\ln c}. \text{ Since } x \neq 1, \frac{2}{\ln b} = \frac{1}{\ln a} + \frac{1}{\ln c}.$$

$$(b) \frac{2}{\ln b} = \frac{\ln(ac)}{\ln a \ln c} \implies \ln b = \frac{2 \ln a \ln c}{\ln(ac)} \implies b = \exp\left(\frac{2 \ln a \ln c}{\ln(ac)}\right).$$

$$(c) \text{ For } a = 2, c = 8:$$

$$\frac{2}{\ln b} = \frac{1}{\ln 2} + \frac{1}{3 \ln 2} = \frac{4}{3 \ln 2} \implies \ln b = \frac{3}{2} \ln 2 \implies b = 2^{3/2} = 2\sqrt{2}$$

$$\textbf{Question 5} \text{ Let } u = \log_x y: u + \frac{1}{u} = \frac{5}{2} \implies 2u^2 - 5u + 2 = 0 \implies u = 2 \text{ or } u = \frac{1}{2}.$$

- If $u = 2 \implies y = x^2 \implies x^3 = 64 \implies x = 4, y = 16$.
- If $u = \frac{1}{2} \implies x = y^2 \implies y^3 = 64 \implies x = 16, y = 4$.

Solutions: $(4, 16)$ or $(16, 4)$.

Question 6

$$(a) \log_a x = \frac{\ln x}{\ln a}.$$

$$(b) \text{ Let } u = x - 1 \rightarrow 0 \text{ as } x \rightarrow 1:$$

$$\lim_{x \rightarrow 1} \frac{\log_a x}{x - 1} = \frac{1}{\ln a} \lim_{u \rightarrow 0} \frac{\ln(1 + u)}{u} = \frac{1}{\ln a}$$

$$(c) \text{ For } a = 2: \lim_{x \rightarrow 1} \frac{\log_2 x}{x - 1} = \frac{1}{\ln 2}.$$

Question 7

$$(a) \frac{1}{\log_k(n!)} = \frac{1}{\frac{\ln(n!)}{\ln k}} = \frac{\ln k}{\ln(n!)}.$$

$$(b) P_n = \sum_{k=2}^n \frac{\ln k}{\ln(n!)} = \frac{\ln(2 \cdot 3 \cdots n)}{\ln(n!)} = \frac{\ln(n!)}{\ln(n!)} = 1.$$

$$(c) \lim_{n \rightarrow \infty} P_n = \lim_{n \rightarrow \infty} 1 = 1.$$

Question 8

(a) Take $\ln$ on both sides:

$$(2x + 1) \ln 2 + x \ln 3 = (1 - x) \ln 5$$

$$x(2 \ln 2 + \ln 3 + \ln 5) = \ln 5 - \ln 2 \implies x = \frac{\ln 5 - \ln 2}{\ln 12 + \ln 5}$$

$$(b) x = \frac{\ln(5/2)}{\ln 60} = \log_{60}(2.5).$$

Question 9

(a) Base: $x > 0, x \neq 1$. Argument: $3x - 2 > 0 \implies x > \frac{2}{3}$.

Domain: $(\frac{2}{3}, 1) \cup (1, \infty)$.

(b) • Case $x > 1$: $3x - 2 > x^2 \implies x^2 - 3x + 2 < 0 \implies 1 < x < 2$.

• Case $\frac{2}{3} < x < 1$: $3x - 2 < x^2 \implies x^2 - 3x + 2 > 0 \implies \frac{2}{3} < x < 1$.

Solution: $(\frac{2}{3}, 1) \cup (1, 2)$.

Question 10

$$(a) \log_2(1 + 4x) = \frac{\ln(1+4x)}{\ln 2}, \log_3(1 + 2x) = \frac{\ln(1+2x)}{\ln 3}.$$

$$(b) (i) \lim_{x \rightarrow 0} \frac{\log_2(1 + 4x)}{x} = \frac{4}{\ln 2} \lim_{4x \rightarrow 0} \frac{\ln(1 + 4x)}{4x} = \frac{4}{\ln 2}.$$

$$(ii) \lim_{x \rightarrow 0} \frac{\log_2(1 + 4x)}{\log_3(1 + 2x)} = \frac{\ln 3}{\ln 2} \lim_{x \rightarrow 0} \frac{\frac{\ln(1+4x)}{x}}{\frac{\ln(1+2x)}{x}} = \frac{\ln 3}{\ln 2} \cdot \frac{4}{2} = \frac{2 \ln 3}{\ln 2}.$$

Solutions to Application Problems of Logarithmic Function

Level 1 Exercise Solutions

$$1. D = 10 \log_{10} \left(\frac{10^{-4}}{10^{-12}} \right) = 10 \log_{10}(10^8) = 10 \times 8 = 80 \text{ dB}.$$

$$2. \text{pH} = -\log_{10}(2.5 \times 10^{-4}) = 4 - \log_{10}(2.5) \approx 4 - 0.3979 = 3.60.$$

$$3. M_1 - M_2 = 6.2 - 4.2 = 2.0.$$

$$\text{Since } M = \log_{10} \left(\frac{A}{A_0} \right), \log_{10} \left(\frac{A_1}{A_2} \right) = 2.0 \implies \frac{A_1}{A_2} = 10^2 = 100.$$

$$4. 10000(1.04)^n > 20000 \implies (1.04)^n > 2.$$

$$n > \frac{\ln 2}{\ln 1.04} \approx 17.67 \implies \text{Minimum complete years} = 18.$$

$$5. (a) t = 2025 - 2020 = 5.$$

$$P(5) = 500000 \cdot (1.03)^5 \approx 579637.$$

$$(b) 500000 \cdot (1.03)^t = 700000 \implies (1.03)^t = 1.4.$$

$$t = \frac{\ln 1.4}{\ln 1.03} \approx 11.4 \text{ years}.$$

$$6. (a) \text{Initial mass } M(0) = 80 \text{ g}.$$

$$(b) M(10) = 80e^{-0.5} \approx 48.5 \text{ g}.$$

7. $\log_{10} y = \log_{10}(4 \cdot 3^x) = (\log_{10} 3)x + \log_{10} 4$.
Slope $m = \log_{10} 3$, vertical intercept $c = \log_{10} 4$.
8. $\ln y = \ln(5x^{-2}) = -2\ln x + \ln 5$.
Slope $m = -2$, vertical intercept $c = \ln 5$.
9. $20 + 70e^{-0.1t} = 40 \implies e^{-0.1t} = \frac{20}{70} = \frac{2}{7}$.
 $t = \frac{-\ln(2/7)}{0.1} = \frac{\ln 3.5}{0.1} \approx 12.5$ minutes.
10. $150000(0.85)^t = 75000 \implies (0.85)^t = 0.5$.
 $t = \frac{\ln 0.5}{\ln 0.85} \approx 4.26$ years.

Level 2 Exercise Solutions

1. $70 = 10 \log_{10} \left(\frac{I}{I_0} \right) \implies I = 10^7 I_0$.
Combined intensity $I_{\text{total}} = 2 \times 10^7 I_0$.
 $D_{\text{total}} = 10 \log_{10}(2 \times 10^7) = 10(7 + \log_{10} 2) \approx 73.0$ dB.
2. (a) $2000e^{4k} = 6000 \implies e^{4k} = 3 \implies k = \frac{\ln 3}{4}$.
(b) $2000e^{kt} = 50000 \implies e^{kt} = 25 \implies t = \frac{\ln 25}{k} = \frac{8 \ln 5}{\ln 3} \approx 11.7$ hours.
3. (a) $e^{-\lambda(5730)} = 0.5 \implies 5730\lambda = \ln 2 \implies \lambda = \frac{\ln 2}{5730}$.
(b) $e^{-\lambda t} = 0.18 \implies t = \frac{-\ln 0.18}{\lambda} = 5730 \times \frac{\ln(50/9)}{\ln 2} \approx 14200$ years.
4. $e^{0.05t} = 2 \implies t = \frac{\ln 2}{0.05}$.
 $(1 + R)^t = 2 \implies t \ln(1 + R) = \ln 2 \implies \ln(1 + R) = 0.05$.
 $R = e^{0.05} - 1 \approx 0.0513$.
5. Slope $m = \frac{1.4 - 0.8}{3 - 1} = 0.3$.
Intercept $c = 0.8 - 0.3(1) = 0.5$.
 $\log_{10} y = 0.3x + 0.5 \implies y = 10^{0.5} \cdot (10^{0.3})^x \approx 3.16 \cdot (2.00)^x$.
6. $\ln y = -1.5 \ln x + 2.4 \implies y = e^{2.4} x^{-1.5}$.
 $A = e^{2.4}$, $k = -1.5$.
7. (a) $\log_{10} E = 4.8 + 1.5(7.0) = 15.3 \implies E = 10^{15.3} \approx 2.00 \times 10^{15}$ J.
(b) $\frac{E_1}{E_2} = 1000 \implies 1.5(M_1 - M_2) = \log_{10} 1000 = 3 \implies M_1 - M_2 = 2.0$.
8. (a) $[\text{H}^+]_X = 10^{-3}$, $[\text{H}^+]_Y = 10^{-5}$. Ratio $= \frac{10^{-3}}{10^{-5}} = 100$.
(b) $[\text{H}^+]_{\text{mix}} = \frac{10^{-3} + 10^{-5}}{2} = 5.05 \times 10^{-4}$ mol/L.
 $\text{pH} = -\log_{10}(5.05 \times 10^{-4}) \approx 3.30$.
9. (a) $S(0) = 85 - 18 \ln 1 = 85$.
(b) $85 - 18 \ln(t + 1) = 50 \implies \ln(t + 1) = \frac{35}{18}$.
 $t = e^{35/18} - 1 \approx 5.99$ weeks.
10. (a) $C_0 e^{10k} = 50$ and $C_0 e^{30k} = 200 \implies e^{20k} = 4 \implies k = \frac{\ln 2}{10}$.
 $C_0 e^{\ln 2} = 50 \implies 2C_0 = 50 \implies C_0 = 25$.
(b) $C(50) = 25e^{50(\frac{\ln 2}{10})} = 25(2^5) = 800 \implies \$800,000$.

Exam-type Problem Solutions

Question 1

- (a) Slope $m = \frac{3.0-1.8}{6-2} = 0.3$.
 Intercept $c = 1.8 - 0.3(2) = 1.2 \implies \ln P = 0.3t + 1.2$.
- (b) $k = 0.3$ (exact).
 $\ln A = 1.2 \implies A = e^{1.2} \approx 3.32$.
- (c) $P > 100 \implies \ln P > \ln 100 \approx 4.605 \implies 0.3t + 1.2 > 4.605$.
 $t > 11.35 \implies$ Minimum complete years = 12.
- (d) $\lim_{h \rightarrow 0} \frac{P(h) - P(0)}{hA} = \lim_{h \rightarrow 0} \frac{Ae^{0.3h} - A}{hA} = \lim_{h \rightarrow 0} \frac{e^{0.3h} - 1}{h} = 0.3$.

Question 2

- (a) Slope $m = \frac{2.5-1}{4-1} = 0.5$.
 Vertical intercept $c = 1 - 0.5(1) = 0.5$.
- (b) $\log_{10} y = 0.5x + 0.5 \implies y = 10^{0.5x+0.5} = \sqrt{10} \cdot (\sqrt{10})^x$.
 $a = \sqrt{10}, b = \sqrt{10}$.
- (c) For $x = 6$: $y = \sqrt{10} \cdot (\sqrt{10})^6 = 1000\sqrt{10}$.
- (d) $\sqrt{10} \cdot 10^{0.5x} > 1000 = 10^3 \implies 10^{0.5x+0.5} > 10^3 \implies 0.5x + 0.5 > 3 \implies x > 5$.

Question 3

- (a) Points $(0, 3)$ and $(-2, 0)$.
 Slope $n = \frac{3-0}{0-(-2)} = \frac{3}{2}$.
 Vertical intercept $\log_2 a = 3 \implies a = 2^3 = 8$.
- (b) $y = 8x^{3/2}$.
- (c) $128 = 8x^{3/2} \implies x^{3/2} = 16 \implies x = 16^{2/3} = 4\sqrt[3]{4}$.

Question 4

- (a) $80 = 10 \log_{10}(I_A/10^{-12}) \implies I_A = 10^{-4} \text{ W/m}^2$.
- (b) $85 = 10 \log_{10}(I_B/10^{-12}) \implies I_B = 10^{-3.5} \text{ W/m}^2$.
- (c) (i) $I_{\text{total}} = 10^{-4} + 10^{-3.5} \approx 4.16 \times 10^{-4} \text{ W/m}^2$.
 (ii) $D_{\text{total}} = 10 \log_{10}(4.16 \times 10^8) \approx 86.2 \text{ dB}$.
- (d) $10 \log_{10}(N \cdot 10^{-4}/10^{-12}) > 95 \implies \log_{10} N + 8 > 9.5 \implies N > 10^{1.5} \approx 31.6 \implies 32$ machines.

Question 5

- (a) $100000e^{5k} = 135000 \implies e^{5k} = 1.35 \implies k = \frac{\ln(1.35)}{5} = \frac{\ln(27/20)}{5}$.
- (b) Slope of $L_1 = \ln(1.06)$.
 Slope of $L_2 = k = \frac{\ln(1.35)}{5}$.

- (c) $V_1(10) = 100000(1.06)^{10} \approx \$179,085$.
 $V_2(10) = 100000e^{10k} = 100000(1.35)^2 = \$182,250$.
 Plan B gives higher value by $\$182,250 - \$179,085 = \$3,165$.
- (d) $\lim_{r \rightarrow 0} \frac{\ln(1+r)}{r} = 1$. This shows that as discrete compounding periods approach infinity ($r \rightarrow 0$), annual nominal rate equals the continuous rate.

Question 6

- (a) $N(0) = \frac{10000}{1+99} = 100$.
- (b) $1 + 99e^{-0.5t} = \frac{10000}{N} \implies e^{-0.5t} = \frac{10000-N}{99N} \implies t = 2 \ln \left(\frac{99N}{10000-N} \right)$.
- (c) For $N = 5000$: $t = 2 \ln \left(\frac{99 \times 5000}{5000} \right) = 2 \ln 99 \approx 9.19$ hours.
- (d) $\lim_{t \rightarrow \infty} N(t) = \frac{10000}{1+0} = 10000$.
 The number of infected computers eventually reaches the maximum capacity of 10,000.

Question 7

- (a) $T(t) = 20 + 80e^{-kt}$.
 $T(5) = 20 + 80e^{-5k} = 60 \implies e^{-5k} = \frac{1}{2} \implies 5k = \ln 2 \implies k = \frac{1}{5} \ln 2$.
- (b) $T(15) = 20 + 80e^{-15(\frac{1}{5} \ln 2)} = 20 + 80(2^{-3}) = 20 + 10 = 30^\circ\text{C}$.
- (c) $20 + 80e^{-kt} = 25 \implies e^{-kt} = \frac{1}{16} = 2^{-4} \implies kt = 4 \ln 2 \implies t = 20.0$ minutes.
- (d) $\lim_{t \rightarrow \infty} T(t) = 20 + 80(0) = 20^\circ\text{C}$.

Question 8

- (a) $\ln \left(\frac{N(t)}{N_0} \right) = -\lambda t$.
- (b) $-\lambda t = \ln(0.35) \implies t = \frac{-\ln 0.35}{\ln 2 / 5730} = 5730 \times \frac{\ln(20/7)}{\ln 2} \approx 8680$ years.
- (c) Range 33% to 37%:
 $t_{33\%} = 5730 \times \frac{-\ln 0.33}{\ln 2} \approx 9170$ years.
 $t_{37\%} = 5730 \times \frac{-\ln 0.37}{\ln 2} \approx 8220$ years.
 Estimated age range: 8220 to 9170 years.

Question 9

- (a) $E = 10^{4.8+1.5M} = 10^{4.8} \cdot 10^{1.5M}$.
 $A = 10^{4.8}$, $k = 1.5$.
- (b) $E_X = 10^{4.8+12} = 10^{16.8}$, $E_Y = 10^{4.8+9} = 10^{13.8}$.
 $\frac{E_X}{E_Y} = 10^{16.8-13.8} = 10^3 = 1000$.
- (c) $2 \cdot 10^{4.8+1.5M_0} = 10^{4.8+1.5M_1} \implies \log_{10} 2 + 4.8 + 1.5M_0 = 4.8 + 1.5M_1 \implies M_1 = M_0 + \frac{\log_{10} 2}{1.5}$.

Question 10

- (a) Model 1: $\ln V_1(t) = \ln 500000 + t \ln(0.85)$, slope $= \ln(0.85) \approx -0.1625$.
 Model 2: $\ln V_2(t) = \ln 500000 - 0.1625t$, slope $= -0.1625$.
- (b) $V_1(t) = V_2(t) \implies t(\ln(0.85) + 0.1625) = 0 \implies t = 0.00$ years.
- (c) $V_1(5) = 500000(0.85)^5 \approx \$221,853$.
 $V_2(5) = 500000e^{-0.1625(5)} \approx \$221,874$.
 Percentage difference $= \frac{221874 - 221853}{221853} \times 100\% \approx 0.01\%$.
- (d) $\ln(1 - r) = -r - \frac{r^2}{2} - \dots \approx -r$ for small r .
 Thus $\ln V_1(t) = \ln V_0 + t \ln(1 - r) \approx \ln V_0 - rt$, which matches Model 2 when $k \approx r$.

Solutions to The Graphs Involving Logarithmic Functions

Level 1 Exercise Solutions

1. (a) For $y = \log_3(x - 2)$:
 - **Domain:** $x - 2 > 0 \implies (2, \infty)$
 - **Range:** $(-\infty, \infty)$
 - **x -intercept:** $\log_3(x - 2) = 0 \implies x - 2 = 1 \implies x = 3$, point $(3, 0)$
 - **Vertical Asymptote:** $x = 2$
- (b) For $y = \ln(x + 4) - 1$:
 - **Domain:** $x + 4 > 0 \implies (-4, \infty)$
 - **Range:** $(-\infty, \infty)$
 - **x -intercept:** $\ln(x + 4) = 1 \implies x + 4 = e \implies x = e - 4$, point $(e - 4, 0)$
 - **Vertical Asymptote:** $x = -4$
- (c) For $y = 2 - \log_5 x$:
 - **Domain:** $x > 0 \implies (0, \infty)$
 - **Range:** $(-\infty, \infty)$
 - **x -intercept:** $\log_5 x = 2 \implies x = 25$, point $(25, 0)$
 - **Vertical Asymptote:** $x = 0$ (y -axis)
2. (a) $g(x) = \ln(x + 3) + 5$: Translate the graph of $y = \ln x$ leftward by 3 units, then upward by 5 units.
- (b) $g(x) = -\ln(x - 1)$: Translate the graph of $y = \ln x$ rightward by 1 unit, then reflect it across the x -axis.
- (c) $g(x) = \ln(2x)$: Compress the graph of $y = \ln x$ horizontally by a factor of $\frac{1}{2}$ (or translate upward by $\ln 2$).
3. (a) $f(x) = \log_2(x + 8) - 3$:
 - x -intercept: $\log_2(x + 8) = 3 \implies x + 8 = 8 \implies x = 0$, point $(0, 0)$
 - y -intercept: $f(0) = \log_2(8) - 3 = 0$, point $(0, 0)$
- (b) $g(x) = \ln(3 - x)$:
 - x -intercept: $\ln(3 - x) = 0 \implies 3 - x = 1 \implies x = 2$, point $(2, 0)$
 - y -intercept: $g(0) = \ln 3$, point $(0, \ln 3)$
- (c) $h(x) = 4 - 2^x$:
 - x -intercept: $2^x = 4 \implies x = 2$, point $(2, 0)$
 - y -intercept: $h(0) = 4 - 2^0 = 3$, point $(0, 3)$

4. Vertical asymptote at $x = 3 \implies b = 3$.
 Passing through $(7, 2) \implies 2 = \log_a(7 - 3) = \log_a 4 \implies a^2 = 4 \implies a = 2$ (since $a > 0$).
 Exact values: $a = 2, b = 3$.
5. (a) Reflection in $y = x$: Swap coordinates $\implies (2, e^2)$.
 (b) Reflection in x -axis gives $(e^2, -2)$; upward translation by 3 units gives $(e^2, 1)$.
 (c) Horizontal stretch by a factor of 2: Multiply x -coordinate by 2 $\implies (2e^2, 2)$.
6. (a) $A(9, 2)$: $\log_3 9 = 2 \implies$ Point lies ****on**** the curve.
 (b) $B(3, 1.5)$: $\log_3 3 = 1 < 1.5 \implies$ Point lies ****above**** the curve.
 (c) $C\left(\frac{1}{27}, -2\right)$: $\log_3\left(\frac{1}{27}\right) = -3 < -2 \implies$ Point lies ****above**** the curve.
7. (a) Both curves pass through the common x -intercept $(1, 0)$.
 (b) For $0 < x < 1$: $\log_2 x < \log_4 x < 0$, so $y = \log_2 x$ lies below $y = \log_4 x$.
 For $x > 1$: $\log_2 x > \log_4 x > 0$, so $y = \log_2 x$ lies above $y = \log_4 x$.
8. (a) **Domain:** $(0, \infty)$, **Range:** $[0, \infty)$.
 (b) $f(x) = 3 \implies |\log_2 x| = 3 \implies \log_2 x = 3$ or $-3 \implies x = 8$ or $\frac{1}{8}$.
9. Reflection across y -axis gives $y = \ln(-x)$ for $x < 0$.
 Rightward translation by 5 units gives $y = \ln(-(x - 5)) = \ln(5 - x)$.
Domain: $5 - x > 0 \implies (-\infty, 5)$.
10. $y = \ln x$ is strictly increasing from $-\infty$ to ∞ for $x > 0$, while $y = 3 - x$ is strictly decreasing.
 At $x = 1$, $\ln 1 = 0 < 2 = 3 - 1$; at $x = e^2$, $\ln(e^2) = 2 > 3 - e^2$.
 By the Intermediate Value Theorem, there is ****1**** real intersection point.

Level 2 Exercise Solutions

1. (a) Domain of $f(x)$: $x^2 > 0 \implies x \neq 0$, i.e., $(-\infty, 0) \cup (0, \infty)$.
 Domain of $g(x)$: $x > 0$, i.e., $(0, \infty)$.
 The functions are not identical because their domains differ: $f(x) = 2 \ln |x|$ is defined for negative numbers, whereas $g(x)$ is not.
 (b) $y = f(x)$ has two symmetric branches about the y -axis, while $y = g(x)$ consists only of the right branch ($x > 0$).
2. Vertical asymptote at $x = -1 \implies b = 1$.
 Passes $(0, 0) \implies 0 = a \ln(1) + c = c \implies c = 0$.
 Passes $(e - 1, 3) \implies 3 = a \ln(e - 1 + 1) = a \ln e = a \implies a = 3$.
 Exact values: $a = 3, b = 1, c = 0$.
3. (a) Line of symmetry: $x = 3$.
 (b) x -intercepts: $|x - 3| = 1 \implies x - 3 = \pm 1 \implies x = 2$ or 4 , points $(2, 0)$ and $(4, 0)$.
 y -intercept: $f(0) = \ln |0 - 3| = \ln 3$, point $(0, \ln 3)$.
 (c) **Domain:** $(-\infty, 3) \cup (3, \infty)$, **Range:** $(-\infty, \infty)$.
4. Let $f(x) = \ln\left(\frac{3+x}{3-x}\right)$.
 Domain: $\frac{3+x}{3-x} > 0 \implies (x + 3)(x - 3) < 0 \implies (-3, 3)$.
 $f(-x) = \ln\left(\frac{3-x}{3+x}\right) = \ln\left(\left(\frac{3+x}{3-x}\right)^{-1}\right) = -\ln\left(\frac{3+x}{3-x}\right) = -f(x)$.
 Since $f(-x) = -f(x)$, the graph is symmetric about the origin $(0, 0)$.

5. (a) Swap x and y : $x = 2^{y-1} + 3 \implies 2^{y-1} = x - 3 \implies y - 1 = \log_2(x - 3)$.
Equation of C : $y = \log_2(x - 3) + 1$.
(b) **Domain:** $(3, \infty)$, **Range:** $(-\infty, \infty)$, **Vertical Asymptote:** $x = 3$.
6. Set $\log_2(x + 3) = 1 + \log_2 x = \log_2(2x)$ for $x > 0$:
 $x + 3 = 2x \implies x = 3$.
 $y = \log_2(3 + 3) = \log_2 6$.
Intersection point: $(3, \log_2 6)$.
7. (a) $x^2 - 4x + 5 = (x - 2)^2 + 1 \geq 1$ for all $x \in \mathbb{R}$.
Thus $h(x) = \ln((x - 2)^2 + 1) \geq \ln 1 = 0$.
(b) Minimum value: 0, attained at $x = 2$.
(c) **Range:** $[0, \infty)$.
8. The graph $y = |\ln x - 1|$ has a vertex at $(e, 0)$ where $y = 0$.
For $y = k$ to intersect the graph at two distinct points, we require $k > 0$.
Range of k : $(0, \infty)$.
9. (a) Domain of $f(x)$: $x > -1$. Domain of $g(x)$: $x^2 - 1 > 0 \implies x < -1$ or $x > 1$.
Domain of $(g - f)(x)$: Intersection is $(1, \infty)$.
(b) $(g - f)(x) = \log_3(x^2 - 1) - \log_3(x + 1) = \log_3\left(\frac{x^2 - 1}{x + 1}\right) = \log_3(x - 1)$ for $x > 1$.
10. (a) Since $e^x + 2 > e^x > 0$ for all $x \in \mathbb{R}$, taking natural logarithms yields $\ln(e^x + 2) > \ln(e^x) = x \implies y > x$.
(b) y -intercept ($x = 0$): $y = \ln(e^0 + 2) = \ln 3$, point $(0, \ln 3)$.
(c) $y - x = \ln(e^x + 2) - \ln(e^x) = \ln(1 + 2e^{-x}) \rightarrow \ln 1 = 0$ as $x \rightarrow \infty$.

Exam-type Problem Solutions

Question 1

- (a) **Domain:** $(-2, \infty)$, **x -intercept:** $(-1, 0)$, **y -intercept:** $(0, \ln 2)$, **Vertical Asymptote:** $x = -2$.
- (b) Horizontal stretch by factor $\frac{1}{3} \implies \ln(3x + 2)$.
Vertical translation down by $\ln 2 \implies g(x) = \ln(3x + 2) - \ln 2 = \ln\left(\frac{3}{2}x + 1\right)$.
- (c) $g(x) = 0 \implies \frac{3}{2}x + 1 = 1 \implies x = 0$, point $(0, 0)$.
- (d) $\lim_{x \rightarrow 0} \frac{g(x)}{x} = \lim_{x \rightarrow 0} \frac{\ln\left(1 + \frac{3}{2}x\right)}{x} = \frac{3}{2} \lim_{\frac{3}{2}x \rightarrow 0} \frac{\ln\left(1 + \frac{3}{2}x\right)}{\frac{3}{2}x} = \frac{3}{2}(1) = \frac{3}{2}$.

Question 2

- (a) Swap x and y in C_1 : $x = e^{y-1} - 2 \implies x + 2 = e^{y-1} \implies y - 1 = \ln(x + 2) \implies y = \ln(x + 2) + 1$, which is C_2 .
Since C_2 is the inverse function of C_1 , they are reflections across $y = x$.
- (b) (i) P : $x = 0 \implies y = e^{-1} - 2$, so $P(0, e^{-1} - 2)$.
 Q : $y = 0 \implies \ln(x + 2) = -1 \implies x = e^{-1} - 2$, so $Q(e^{-1} - 2, 0)$.
(ii) $PQ = \sqrt{(e^{-1} - 2 - 0)^2 + (0 - (e^{-1} - 2))^2} = \sqrt{2(e^{-1} - 2)^2} = \sqrt{2}(2 - e^{-1})$.

- (c) Midpoint of PQ : $\left(\frac{e^{-1}-2}{2}, \frac{e^{-1}-2}{2}\right)$.

Since x -coordinate equals y -coordinate, it lies on the line $y = x$.

Question 3

- (a) $x^2 - 6x + 10 = (x - 3)^2 + 1$.

Since $(x - 3)^2 + 1 \geq 1 > 0$ for all $x \in \mathbb{R}$, $f(x)$ is defined for all real numbers.

- (b) Line of symmetry: $x = 3$.

Global minimum value: $f(3) = \ln(1) = 0$.

- (c) $\lim_{x \rightarrow 3} \frac{f(x)}{x - 3} = \lim_{x \rightarrow 3} \frac{\ln(1 + (x - 3)^2)}{x - 3} = \lim_{t \rightarrow 0} \left[\frac{\ln(1 + t^2)}{t^2} \cdot t \right] = 1 \cdot 0 = 0$.

Question 4

- (a) $g(x) = 0 \implies \ln(x - 1) = 1 \implies x - 1 = e \implies x = e + 1$, point $(e + 1, 0)$.

- (b) Graph features: Vertical asymptote $x = 1$, x -intercept $(e + 1, 0)$, V-shape curve reflecting below x -axis portion.

- (c)
- For $k < 0$: 0 real roots.
 - For $k = 0$: 1 real root ($x = e + 1$).
 - For $k > 0$: 2 real roots ($x = 1 + e^{1+k}$ and $x = 1 + e^{1-k}$).

Question 5

- (a) $f(x) - g(x) = \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots\right) - \left(x - \frac{x^2}{2}\right) = \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$

First three non-zero terms: $\frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5}$.

- (b) $\lim_{x \rightarrow 0} \frac{f(x) - g(x)}{x^3} = \lim_{x \rightarrow 0} \left(\frac{1}{3} - \frac{x}{4} + \frac{x^2}{5} - \dots\right) = \frac{1}{3}$.

- (c) Near $x = 0$: For $x > 0$, $f(x) > g(x)$ (graph of f lies above g); for $x < 0$, $f(x) < g(x)$ (graph of f lies below g); they intersect at $(0, 0)$.

Question 6

- (a) Intercept $(3, 0) \implies 3a + b = 1$.

Passes $(7, 3) \implies 7a + b = 2^3 = 8$.

Subtracting gives $4a = 7 \implies a = \frac{7}{4}$.

$b = 1 - 3\left(\frac{7}{4}\right) = -\frac{17}{4}$.

- (b) Vertical asymptote: $\frac{7}{4}x - \frac{17}{4} = 0 \implies x = \frac{17}{7}$.

- (c) $\log_2\left(\frac{7x-17}{4}\right) + \log_2(x-1) = 4 \implies (7x-17)(x-1) = 64 \implies 7x^2 - 24x - 47 = 0$.

Since $x > \frac{17}{7}$, $x = \frac{12+\sqrt{473}}{7}$.

Intersection point: $\left(\frac{12+\sqrt{473}}{7}, 4 - \log_2\left(\frac{5+\sqrt{473}}{7}\right)\right)$.

Question 7

- (a) $\sqrt{x^2 + 1} > \sqrt{x^2} = |x| \geq x \implies \sqrt{x^2 + 1} - x > 0$ for all $x \in \mathbb{R}$.

(b) $f(-x) = \ln(\sqrt{x^2+1} + x) = \ln\left(\frac{1}{\sqrt{x^2+1}-x}\right) = -f(x)$.
 Thus $f(x)$ is an odd function, symmetric about the origin $(0, 0)$.

(c) $f(x) = \ln\left(\frac{1}{\sqrt{x^2+1}+x}\right) = -\ln(\sqrt{x^2+1} + x)$.

(d) $e^y = \sqrt{x^2+1} - x$ and $e^{-y} = \sqrt{x^2+1} + x \implies x = \frac{e^{-y}-e^y}{2}$.
 $f^{-1}(x) = \frac{e^{-x}-e^x}{2} = -\sinh x$.

Question 8

- (a) Vertical asymptote: $x = -k$. x -intercept: $(1 - k, 0)$.
 (b) Passes $(0, 0) \implies \ln(k) = 0 \implies k = 1$.
 (c) At $x = 0$, $\ln(1 + 0) = 0$, so C_1 intersects $y = x$ at $x = 0$.
 Since $\ln(1 + x) < x$ for $x > 0$, C_1 lies **below** the line $y = x$ for $x > 0$.

Question 9

- (a) x -intercept: $(0, 0)$, y -intercept: $(0, 0)$.
 (b) $\lim_{x \rightarrow 0^+} \frac{f(x)}{x} = \lim_{x \rightarrow 0^+} \frac{\ln(1 + 2x)}{x} = 2$.
 $\lim_{x \rightarrow 0^-} \frac{f(x)}{x} = \lim_{x \rightarrow 0^-} \frac{e^{2x} - 1}{x} = 2$.
 (c) Continuous and smooth curve passing through origin with slope 2.

Question 10

- (a) **Domain:** $e^x - 1 > 0 \implies x > 0$, i.e., $(0, \infty)$.
 (b) Let $y = \ln\left(\frac{e^x+1}{e^x-1}\right) \implies e^y = \frac{e^x+1}{e^x-1} \implies e^x = \frac{e^y+1}{e^y-1} \implies x = f(y)$.
 Thus $f(f(x)) = x$, meaning the graph is symmetric about the line $y = x$.
 (c) $\lim_{x \rightarrow 0^+} f(x) = \infty \implies$ Vertical asymptote $x = 0$.
 $\lim_{x \rightarrow \infty} f(x) = \ln 1 = 0 \implies$ Horizontal asymptote $y = 0$.
 (d) Strictly decreasing curve in quadrant I, asymptotic to $x = 0$ and $y = 0$, symmetric about $y = x$.

Solutions to Linear Regression on a Graph

Level 1 Exercise

- (a) Slope $m = \frac{5-1}{2-0} = 2$, vertical intercept $c = 1$.
 (b) $\log_{10} y = 2x + 1 \implies y = 10^{2x+1} = 10 \cdot 100^x$.
 Thus, $A = 10, b = 100$.
- $\ln y = 3 \ln x + 2 \implies y = e^{3 \ln x + 2} = e^2 x^3$.
 Here $A = e^2, n = 3$.
- (a) $\ln y - 4 = 0.5(x - 3) \implies \ln y = 0.5x + 2.5$.
 (b) When $x = 5$: $\ln y = 0.5(5) + 2.5 = 5 \implies y = e^5$.

4. Slope $m = \frac{0-(-6)}{3-0} = 2$, vertical intercept $c = -6$.
 $\log_2 y = 2x - 6 \implies y = 2^{2x-6} = \frac{1}{64} \cdot 4^x$.
 Thus, $A = \frac{1}{64}, b = 4$.
5. $\ln y = -2(\ln x - \ln 4) = \ln(16x^{-2}) \implies y = \frac{16}{x^2}$.
6. Slope $m = \frac{1.2-0.3}{4-1} = 0.3$.
 $\log_{10} y - 0.3 = 0.3(x - 1) \implies \log_{10} y = 0.3x$.
 When $x = 3$: $\log_{10} y = 0.9 \implies y = 10^{0.9} \approx 7.94$.
7. $\log_2 y = \log_2(5 \cdot 2^{3x}) = 3x + \log_2 5$.
 (a) Theoretical slope $m = 3$.
 (b) Vertical intercept $c = \log_2 5$.
8. $L_1 : \ln y = 0.4x + 1$ and $L_2 : \ln y = 0.2x + 2$.
 $0.4x + 1 = 0.2x + 2 \implies 0.2x = 1 \implies x = 5$.
 $\ln y = 0.4(5) + 1 = 3 \implies y = e^3$.
 Intersection point: $(5, 3)$, corresponding $y = e^3$.
9. Slope $m = \frac{8-2}{3-1} = 3$.
 $\log_3 y - 2 = 3(\log_3 x - 1) \implies \log_3 y = 3\log_3 x - 1$.
 When $x = 9$, $\log_3 x = 2 \implies \log_3 y = 3(2) - 1 = 5 \implies y = 3^5 = 243$.
10. Exponential model.

Level 2 Exercise

1. (a) Slope $m = \frac{5-3}{6-2} = 0.5$.
 $\log_4 y - 3 = 0.5(\log_2 x - 2) \implies \log_4 y = 0.5\log_2 x + 2$.
 (b) $\frac{\ln y}{\ln 4} = 0.5\frac{\ln x}{\ln 2} + 2 \implies \ln y = \ln x + 4\ln 2 \implies y = 16x$.
 Thus $A = 16, k = 1$.
2. Slope $m = \frac{1.5-0.5}{3-1} = 0.5$.
 $\frac{1}{\ln y} - 0.5 = 0.5(x - 1) \implies \frac{1}{\ln y} = 0.5x \implies y = e^{2/x}$.
3. (a) $\ln(y - 3) = -0.2x + \ln 5 \implies y = 5e^{-0.2x} + 3$.
 (b) Horizontal asymptote: $y = 3$.
4. Slope $m = \frac{2.0-0.5}{4-1} = 0.5$.
 $\log_{10} y - 0.5 = 0.5(x^2 - 1) \implies \log_{10} y = 0.5x^2 \implies y = (\sqrt{10})^{x^2}$.
 Form $y = A \cdot b^{x^2} \implies A = 1, b = \sqrt{10} \approx 3.16$.
5. $\ln y = 1.5\ln(x + 1) + \ln 2 \implies y = 2(x + 1)^{1.5}$.
6. Slope $m = \frac{5.0-1.0}{2.0-0} = 2.0$.
 $\ln y = 2\ln x + 1.0 \implies y = e^{\ln(x^2)+1} = ex^2$.
 Form $y = Ax^n \implies A = e, n = 2$.
7. $\log_3\left(\frac{y}{x}\right) = 2x + 1 \implies \frac{y}{x} = 3^{2x+1} \implies y = 3x \cdot 9^x$.
8. Based on Plot A: $\ln y = 0.5x + 1.2 \implies y = e^{1.2}e^{0.5x} \approx 3.32e^{0.5x}$.
9. (a) $\ln y = -0.4x + 2.8 \implies 0.4x = 2.8 - \ln y \implies x = 7 - 2.5\ln y$.
 (b) When $y = 2$: $x = 7 - 2.5\ln 2 \approx 5.27$.
10. Slope $m = \frac{\ln 64 - \ln 8}{5-2} = \frac{\ln 8}{3} = \ln 2$.
 $\ln(y - 10) - \ln 8 = \ln 2(x - 2) \implies \ln(y - 10) = \ln(2^{x+1}) \implies y = 2^{x+1} + 10$.

Exam-type Problem Solutions

Question 1

- (a) Slope $m = \frac{2.0-0.8}{6-2} = 0.3$.
Vertical intercept $c = 0.8 - 0.3(2) = 0.2$.
- (b) $\log_{10} y = 0.3x + 0.2 \implies y = 10^{0.2} \cdot (10^{0.3})^x$.
 $b = 10^{0.3} \approx 1.995$ (exact $10^{0.3}$), $A = 10^{0.2} \approx 1.58$.
- (c) For $x = 10$: $\log_{10} y = 0.3(10) + 0.2 = 3.2 \implies y = 10^{3.2} \approx 1580$.
- (d) $\lim_{x \rightarrow 0} \frac{y(x) - A}{xA} = \lim_{x \rightarrow 0} \frac{10^{0.3x} - 1}{x} = 0.3 \lim_{0.3x \rightarrow 0} \frac{10^{0.3x} - 1}{0.3x} = 0.3 \ln 10$.

Question 2

- (a) Slope $m = \frac{4.2-1.2}{2-0} = 1.5$.
 $\ln y = 1.5 \ln x + 1.2$.
- (b) $y = e^{1.2} x^{1.5} \approx 3.32x^{1.5}$.
 $A = e^{1.2} \approx 3.32, n = 1.5$.
- (c) $3.3201x^{1.5} > 1000 \implies 1.5 \ln x > \ln 1000 - 1.2 \implies x > 44.9$.
- (d) $\lim_{x \rightarrow 1} \frac{\ln y - 1.2}{x - 1} = \lim_{x \rightarrow 1} \frac{1.5 \ln x}{x - 1} = 1.5$.

Question 3

- (a) $Y - 2.5 = 0.3(t - 4) \implies Y = 0.3t + 1.3$.
- (b) $\ln P = 0.3t + 1.3 \implies P(t) = e^{1.3} e^{0.3t}$.
 $k = 0.3, P_0 = e^{1.3}$.
- (c) $P(t) = 2P_0 \implies e^{0.3t} = 2 \implies t = \frac{\ln 2}{0.3} \approx 2.31$ years.
- (d) $\lim_{h \rightarrow 0} \frac{P(h) - P(0)}{hP_0} = \lim_{h \rightarrow 0} \frac{e^{kh} - 1}{h} = k = 0.3$.

Question 4

- (a) Slope $m = \frac{0-4.5}{15-0} = -0.3$.
Vertical intercept $c = 4.5$.
- (b) $\ln(T - 25) = -0.3t + 4.5 \implies T(t) = 25 + e^{4.5} e^{-0.3t}$.
 $A = e^{4.5} \approx 90.0, k = 0.300$.
- (c) $T(10) = 25 + e^{1.5} \approx 29.5^\circ\text{C}$.
- (d) $\lim_{t \rightarrow \infty} T(t) = 25^\circ\text{C}$.
Physical meaning: The temperature of the liquid approaches the ambient temperature of 25°C over time.

Question 5

- (a) Slope $m = \frac{5-2}{3-1} = 1.5$.
 $\log_2 Y - 2 = 1.5(\log_4 X - 1) \implies \log_2 Y = 1.5 \log_4 X + 0.5$.

- (b) Since $\log_4 X = \frac{1}{2} \log_2 X$:
 $\log_2 Y = 0.75 \log_2 X + 0.5 \implies Y = 2^{0.5} X^{0.75} = \sqrt{2} X^{3/4}$.
 $A = \sqrt{2}, p = \frac{3}{4} = 0.75$.

- (c) When $X = 16$: $Y = \sqrt{2}(16)^{3/4} = 8\sqrt{2} \approx 11.31$.

Question 6

- (a) Slope $m = \frac{-4.0 - (-1.0)}{0.003 - 0.002} = -3000$.
 Vertical intercept $c = -1.0 - (-3000)(0.002) = 5.0$.
- (b) $\ln k = 5.0 - \frac{3000}{T} \implies k = e^5 e^{-3000/T}$.
 $A = e^5 \approx 148, E = 3000$.
- (c) For $T = 400$ K: $k = e^{5.0 - 7.5} = e^{-2.5} \approx 0.0821$.

Question 7

- (a) Slope $m = \frac{-1.5 - (-0.5)}{5 - 0} = -0.2$.
 $Y = -0.2x - 0.5$.
- (b) $\frac{y}{100} = e^{-0.5 - 0.2x} \implies y = 100e^{-0.5} e^{-0.2x}$.
 $A = 100e^{-0.5} \approx 60.7, k = 0.2$.
- (c) $100e^{-0.5 - 0.2x} = 50 \implies -0.5 - 0.2x = -\ln 2 \implies x = 5 \ln 2 - 2.5 \approx 0.966$.
- (d) $\lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x} = 100e^{-0.5} \lim_{x \rightarrow 0} \frac{e^{-0.2x} - 1}{x} = -20e^{-0.5} \approx -12.1$.

Question 8

- (a) Model A: Slope $m = \frac{3.5 - 2.0}{4 - 1} = 0.5$.
 $\ln y = 0.5x + 1.5 \implies y = e^{1.5} e^{0.5x}$.
- (b) Model B: Slope $m = \frac{3.5 - 1.5}{2 - 0} = 1.0$.
 $\ln y = \ln x + 1.5 \implies y = e^{1.5} x$.
- (c) At $x = 2$: $y_A = e^{2.5} \approx 12.18, y_B = 2e^{1.5} \approx 8.96$.
 Absolute difference: $|12.18 - 8.96| = 3.22$.
- (d) Both curves start at $x \rightarrow 0$ with $y_B(0) = 0$ and $y_A(0) = e^{1.5} \approx 4.48$, both increasing for $x > 0$.

Question 9

- (a) Slope $m = \frac{2 - (-2)}{4 - 0} = 1$.
 $Y = x - 2$.
- (b) $\frac{y}{100 - y} = e^{x-2} \implies y = (100 - y)e^{x-2} \implies y = \frac{100}{1 + e^{-(x-2)}}$.
- (c) $\lim_{x \rightarrow \infty} y(x) = 100$.
 Significance: The value of y approaches a horizontal saturation limit of 100 as x grows large.

Question 10

(a) Slope $m = \frac{8-2}{4-1} = 2$.
 $\log_3 y - 2 = 2((x+1) - 1) \implies \log_3 y = 2x + 2$.

(b) $y = 3^{2x+2} = 9 \cdot 9^x$.

(c) $9 \cdot 9^x \geq 81 \implies 9^x \geq 9 \implies x \geq 1$.

(d) $\lim_{x \rightarrow 0} \frac{\log_3(y(x)) - 2}{x} = \lim_{x \rightarrow 0} \frac{2x}{x} = 2$.